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Carbon Jar Startup Overview

Founded in 2024, Carbon Jar is Tunisia's first startup focused on AI-driven carbon accounting and ESG compliance. Positioned at the intersection of environmental consultancy and digital innovation, the company helps organizations quantify, manage, and reduce their emissions across three scopes while aligning with leading standards (GHG Protocol, ISO 14040/44 for LCA, and CSRD disclosure requirements).

Carbon Jar's core product is an AI-enabled SaaS platform that automates data ingestion (utility bills, ERP exports, supplier questionnaires), emission-factor matching, outlier detection, and scenario modeling for abatement planning. The platform is complemented by certified online training on GHG accounting and net-zero strategy, and by advisory services covering inventory development, ESG risk mitigation, and sustainability integration. The company operates a rigorous data-governance program (role-based access, audit trails, and encrypted storage) and a flexible, hybrid workplace to attract cross-disciplinary talent.

Relevance to Sustainable Chemical Production

For industrial clients, Carbon Jar applies life-cycle assessment to identify high-leverage interventions in energy use, process yields, and supply chains. In this internship, the team environment is well suited to executing an LCA of levulinic acid: the toolchain supports cradle-to-gate modeling, uncertainty analysis, and benchmark reporting, enabling evidence-based comparison of bio-based levulinic acid with fossil-derived intermediates and translating results into practical decarbonization roadmaps.

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CHAPTER 1

Introduction and Background

In a world increasingly challenged by climate change consequences, the imperative for sustainable development has reached unprecedented urgency. The chemical industry, responsible for approximately 7% of global greenhouse gas emissions [1], faces mounting pressure to transition toward bio-based production pathways. Life Cycle Assessment (LCA) offers a structured, science-based framework following ISO 14040/14044 standards to quantify environmental impacts across entire value chains and support robust decision-making toward lower-carbon production systems [1]. This study performs a comprehensive cradle-to-gate comparative LCA of levulinic acid (LA) production, evaluating bio-based routes from lignocellulosic biomass against conventional petrochemical alternatives, with subsequent conversion to diesel fuel components.

The significance of this research extends beyond environmental assessment to encompass industrial transformation and economic viability. Current diesel production relies heavily on fossil resources, contributing substantially to transportation sector emissions that account for approximately 24% of energy-related CO₂ emissions globally [2]. Bio-based alternatives, particularly platform chemicals like LA that can be converted to transportation fuels, represent promising pathways for decarbonizing the transportation sector while maintaining compatibility with existing infrastructure.

1.0.1 Levulinic Acid: Platform Chemical and Industrial Context

Levulinic acid (LA, 4-oxopentanoic acid, C₅H₈O₃) stands as a crucial bio-derived platform chemical, strategically identified by the U.S. Department of Energy (USDOE) among the top 12 value-added chemicals obtainable from biomass [3]. LA features a terminal carboxylic acid group with a γ -ketone functionality, enabling diverse downstream chemistry pathways essential for sustainable biorefinery operations [4]. Key physicochemical properties include high water solubility, a melting point of 30–33 °C, and exceptional chemical versatility that positions LA as a vital intermediate in converting lignocellulosic biomass into transportation fuels and high-value chemicals [1].

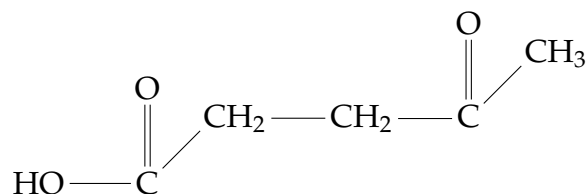


Figure 1: Levulinic Acid Structure

Levulinic acid (LA) is a versatile platform chemical from lignocellulosic biomass whose derivatives, such as ethyl levulinate, γ -valerolactone (GVL) and methyl tetrahydrofuran (MTHF), enable multiple value chains. As summarized in Fig. 1.1, current uses span biofuels and fuel additives, green solvents, and materials (polymers/resins), as well as pharmaceuticals/cosmetics and agrochemicals. This breadth reflects the favorable functionality of LA (ketone + carboxyl) and its compatibility with established catalytic upgrading routes.

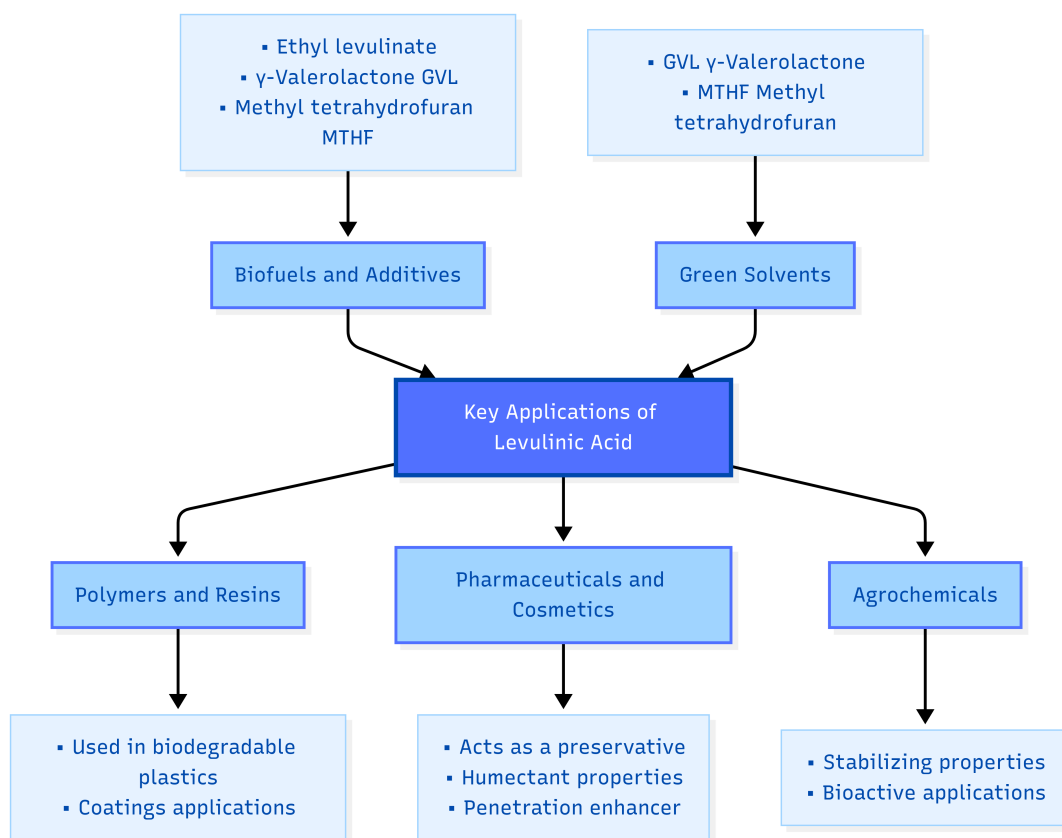


Figure 1.1: Applications of Levulinic Acid.

1.0.2 Production Technologies and Industrial Status

1.0.3 Biomass-to-LA conversion pathways

LA production from lignocellulosic biomass relies primarily on acid-catalyzed conversion of C5 and C6 sugars through cascade dehydration and hydrolysis reac-

tions [5]. The conversion pathway involves cellulose and hemicellulose breakdown to 5-hydroxymethylfurfural (HMF) and furfural intermediates, respectively, followed by hydrolysis to LA and formic acid co-product [1].

Homogeneous acid catalysis represents the conventional approach, employing strong mineral acids (H_2SO_4 , HCl) at elevated temperatures (100–250 °C) [1]. Reported yields demonstrate significant feedstock dependency: rice husk achieved 59.4% LA yield using HCl , while average yields from complex lignocellulosic resources typically reach 29 wt% [4]. The maximum theoretical molar yield from cellulose approaches 76% under optimized continuous reactor configurations [6].

Heterogeneous catalytic systems have emerged to address homogeneous catalysis limitations, including product separation difficulties, catalyst recycling challenges, and environmental concerns [7]. Solid acid catalysts such as sulfonated graphene oxide ($\text{GO-SO}_3\text{H}$) achieved 50% LA yield from glucose, while metal-doped solid acids demonstrate comparable performance with improved sustainability profiles [4].

The Biofine process represents the most established commercial technology, employing a two-stage configuration: Stage 1 converts polysaccharides to HMF using 1–4% mineral acid at 200–230 °C and 20–25 bar with 12 second residence time; Stage 2 hydrolyzes HMF to LA at 190–220 °C and 10–15 bar for 15–30 minutes [1, 8]. The first commercial-scale plant operated in Caserta, Italy, with the world's largest facility targeting 10,000 MTPA capacity by 2017 [9].

1.0.4 LA-to-diesel conversion technologies

LA serves as a precursor for multiple transportation fuel pathways, with particular significance for diesel applications through esterification and hydrogenation routes [2, 10].

Esterification to alkyl levulinates produces excellent oxygenated fuel additives for diesel applications [10]. Ethyl levulinate (EL) specifically functions as a diesel additive, while butyl levulinate demonstrates remarkable combustion properties as a diesel engine alternative biofuel [11]. Acidic resin catalysts (Amberlyst-35, Amberlyst-15) achieve approximately 90% EL yield from furfuryl alcohol, while solid acid catalyst systems in acidic ionic liquid mixtures result in 94% methyl levulinate yield [1].

Hydrogenation pathways convert LA to γ -valerolactone (GVL) and subsequently to transportation fuel components [2]. Advanced catalytic systems demonstrate exceptional performance: Ru-based homogeneous catalysts achieve >99% conversion and 100% selectivity for GVL under solvent-free conditions at 100 °C and 50 bar H_2 [12]. Heterogeneous systems combining Ru/C with Amberlyst A70 yield 99.9% GVL at 70 °C and 3 MPa H_2 , while Ni-Cu bimetallic catalysts show 96% GVL selectivity at 285 °C [8, 13].

1.0.5 Esterification and Hydrogenation Pathways Stoichiometry

The downstream conversion of levulinic acid to transportation fuels follows well-established reaction pathways involving sequential esterification and hydrogenation steps. This section presents the detailed stoichiometric relationships and reaction mechanisms governing these transformations.

1.0.6 Esterification Reaction Pathways

Esterification of levulinic acid with short-chain alcohols produces alkyl levulinates, which serve as valuable fuel additives and intermediates for further conversion [10, 11]. Recent kinetic studies have demonstrated significant variations in reaction rates and activation energies based on alcohol chain length and branching [14, 15].

Table 1.1: Advanced esterification catalyst systems and performance metrics

Catalyst System	Conv. (%)	Temp. (°C)	Time (min)	Reference
Dowex® 50WX8	83.6	120	180	[16]
SnMCM-41-80	71.0	120	120	[17]
TiO ₂ nanoparticles	92.4	140	240	[18]
Bio-glycerol C-SO ₃ H	89.2	130	180	[19]
Amberlyst-15 (flow)	95.8	110	–	[20]

1.0.7 Mechanistic Insights and Catalyst Design

Recent mechanistic studies have revealed that steric hindrance significantly affects esterification kinetics, with reaction rates decreasing in the order: methyl > ethyl > n-propyl > isopropyl [21]. Microwave-assisted esterification using Dowex® 50WX8 catalyst demonstrates enhanced reaction rates through selective heating mechanisms [16]. The development of mesoporous stannosilicates has shown particular promise, with SnMCM-41–80 achieving 83.6% conversion in 180 minutes compared to 40.1% for the material without Sn under identical conditions [17].

Esterification Stoichiometric Analysis and Kinetic Parameters

Theoretical Mass Yields (per kg LA):

- **Methyl levulinate:** 1.120 kg (MW: 130.14 g/mol)
- **Ethyl levulinate:** 1.242 kg (MW: 144.17 g/mol)

- **n-Butyl levulinate:** 1.483 kg (MW: 172.22 g/mol)
- **n-Pentyl levulinate:** 1.563 kg (MW: 186.25 g/mol)

Activation Energies and Kinetic Parameters:

- Methanol: $E_a = 21$ kJ/mol (fastest reaction rate)
- Ethanol: $E_a = 26$ kJ/mol
- n-Propanol: $E_a = 34$ kJ/mol
- Isopropanol: $E_a = 31$ kJ/mol (steric hindrance effect)

Recent Catalyst Developments:

- Dowex® 50WX8: 71-84% conversion under microwave irradiation
- SnMCM-41: 83.6% conversion (180 min, 120°C)
- TiO₂ nanoparticles: Stable for 6 consecutive cycles
- Mesoporous stannosilicates: Enhanced selectivity and stability

Thermodynamic Analysis: Reactions are endothermic with increasing ΔH as alcohol chain length increases

1.0.8 Hydrogenation Reaction Mechanisms

Hydrogenation of levulinic acid or its esters to γ -valerolactone (GVL) proceeds through multiple pathways, each with distinct stoichiometric requirements and catalyst systems. Recent advances in catalytic transfer hydrogenation have demonstrated superior sustainability profiles compared to direct hydrogenation routes [22, 23]. Mechanistic investigations using isotopic labeling have provided crucial insights into H/D exchange pathways and reaction intermediates [23].

Table 1.2: Direct hydrogenation pathways with molecular hydrogen

Starting Material	Hydrogenation Reaction
Levulinic Acid	$C_5H_8O_3 + 2H_2 \rightarrow C_5H_8O_2 + H_2O$ (LA + H ₂ → GVL + Water)
Ethyl Levulinate	$C_7H_{12}O_3 + 2H_2 \rightarrow C_5H_8O_2 + C_2H_6O$ (EL + H ₂ → GVL + Ethanol)

Table 1.3: *Transfer hydrogenation pathways using hydrogen donors*

Hydrogen Donor	Transfer Hydrogenation Reaction
Formic Acid (FA)	$C_5H_8O_3 + HCOOH \rightarrow C_5H_8O_2 + H_2O + CO_2$ (LA + FA $\rightarrow$ GVL + Water + Carbon dioxide)
Isopropanol (IPA)	$C_5H_8O_3 + C_3H_8O \rightarrow C_5H_8O_2 + H_2O + C_3H_6O$ (LA + IPA $\rightarrow$ GVL + Water + Acetone)

Table 1.4: *Overall synthesis pathway from biomass to GVL*

Complete Reaction Pathway
Step 1: Biomass $\rightarrow$ $C_5H_8O_3$ (Levulinic acid)
Step 2: $C_5H_8O_3 \xrightarrow{\text{Esterification}} C_nH_{2n+2}O_3$ (Alkyl levulinate)
Step 3: $C_nH_{2n+2}O_3 \xrightarrow{\text{Hydrogenation}} C_5H_8O_2$ (γ -Valerolactone)
Key Compounds: $C_5H_8O_3$ = Levulinic acid (MW: 116.12 g/mol) $C_7H_{12}O_3$ = Ethyl levulinate (MW: 144.17 g/mol) $C_9H_{16}O_3$ = Butyl levulinate (MW: 172.22 g/mol) $C_5H_8O_2$ = γ -Valerolactone (MW: 100.12 g/mol)

1.0.9 Feedstock Analysis and Supply Chain Considerations

Lignocellulosic biomass represents the most abundant renewable resource, offering carbon-neutral production without food chain competition [1, 7]. Primary feedstock options include agricultural residues (corn stover, wheat straw, sugarcane bagasse) and dedicated energy crops, with composition typically comprising cellulose (40–50 wt%), hemicellulose (25–30 wt%), and lignin (15–30 wt%) [2].

Molasses emerges as a particularly promising feedstock due to high concentrations of pre-hydrolyzed monosaccharides: glucose (98 mg/g) and fructose (156 mg/g) [6]. However, **logistics challenges** represent major cost-contributing factors and bottlenecks in commercial LA production, with transportation costs highly dependent on travel distance and biomass physical form during transport [1].

Pretreatment remains compulsory for raw biomass processing to overcome recalcitrant lignocellulosic structure and maximize LA yields [1, 5]. While lignocellulosic conversion maintains carbon neutrality, the total energy requirements for LA production from biomass exceed petrochemical routes due to pretreatment and initial processing energy

demands [1].

CHAPTER 2

Literature Review

2.0.1 Critical Analysis of Previous LCA Studies

Life Cycle Assessment (LCA) serves as a *standardized analytical tool* for evaluating the environmental impact of products or processes, conducted in accordance with the *ISO 14040 and 14044 standards* [12, 24]. LCA studies often seek to confirm the environmental benefit of transitioning to *bio-based feedstock* alternatives [1, 24]. Recent advances in LCA methodology for bio-based chemicals have emphasized the importance of comprehensive uncertainty analysis and standardized impact assessment methods.

2.0.2 Systematic Comparison of Different LCA Approaches and Methodological Choices

System Boundaries: LCA boundary selection is critical for comparative analysis [6, 12]. The choice of system boundaries significantly affects the conclusions drawn from comparative studies.

- **Cradle-to-Gate:** LCA studies for Levulinic Acid (LA) production typically employ a "*cradle-to-gate*" approach, which encompasses feedstock acquisition, transportation, pulverization, LA synthesis, and LA purification [6]. This approach is most commonly used in the literature for platform chemical assessment.
- **Gate-to-Gate:** Comparative LCA of LA conversion, such as hydrogenation to γ -valerolactone (GVL), often adopts a "*gate-to-gate*" boundary, focusing solely on the chemical transformation process (e.g., catalysis and isolation) and excluding the impacts of LA synthesis itself [12].
- **Functional Unit (FU):** For comparative studies concerning LA conversion, the functional unit (FU) must be standardized, such as defining the output as *10 g of GVL* [12]. Standardization of functional units remains a challenge across different studies, limiting direct comparability.

Allocation and Co-products: LCA frameworks frequently account for the co-product *hydrochar (HC)* generated during LA synthesis, recognizing its potential value as a *coal substitute* due to its high carbon content and calorific value [6]. The treatment

of co-products through allocation procedures significantly influences the final impact assessment results.

2.0.3 Comparison of Impact Assessment Methods Used Across Studies

Recent literature reveals significant variations in impact assessment methodologies and their implementation across LA production studies. Environmental impacts were reported in terms of GHG emissions, with Hafyan et al. (2020) recording 6.3 kg CO₂ eq./kg levulinic acid, demonstrating the range of reported values depending on production pathway and system boundaries.

The choice of impact assessment method significantly affects results comparability. From the Life Cycle Impact Assessment (LCIA) methodologies, they should be standardized to enable the comparability between LCAs of the same product. Both the European Commission and the Life Cycle Initiative are making progress in this regard. Eventually, the ReCiPe is the most used methodology among the reviewed papers.

2.0.4 Critical Evaluation of Methodological Choices

Impact assessment studies consistently reveal environmental burdens posed by the core chemical reaction steps:

- **Synthesis Dominance (Hotspot):** The *LA synthesis step* (acid-catalyzed conversion) is identified as the *primary contributor to all impact categories* investigated, including *Global Warming Potential (GWP)*, fossil resource scarcity, and water consumption [6]. This impact is largely attributed to the *high electricity consumption* required and the input of *chemical catalysts (H₂SO₄)* [6].
- **Feedstock-Dependent GWP:** GWP values vary significantly based on the chosen feedstock and conversion process. For LA production using H₂SO₄:
 - **Molasses (ML)** exhibited the lowest GWP (24.29 kg CO₂eq/kg LA) among tested lignocellulosic resources [6].
 - **Corn Stalk (CS)** produced 26.11 kg CO₂eq/kg LA [6].
- **Purification Improvement:** The LA purification stage is the second major environmental contributor [6]. Utilizing a *reactive distillation system* for LA extraction and purification can achieve energy consumption reduction of approximately 24% compared to conventional distillation methods [6].

Recent studies have expanded the scope of environmental impact assessment. Results revealed that dilute acid pretreatment applied in S2 and S4 exhibited 23

2.0.5 Levulinic Acid Production Pathways

2.0.5.1 Comprehensive Technology Pathway Review

LA production relies on the *acid-catalyzed conversion* of C_5 and C_6 sugars from lignocellulosic biomass through sequential dehydration and hydrolysis reactions [1, 4–6]. The technological landscape has evolved significantly, with multiple pathways now available for industrial implementation.

2.0.6 Alternative Production Technologies and Conversion Methods

1. Conventional (Homogeneous) Routes: The conventional approach uses *strong mineral acids* (H_2SO_4 , HCl) at elevated temperatures ($100 - 250^\circ C$) [1, 5, 7, 8]. This system suffers from major limitations, including difficulty in product separation, poor catalyst recycling, reactor corrosion, and environmental pollution [1, 7, 25].

2. Heterogeneous Catalytic Systems: To improve sustainability, *heterogeneous solid acid catalysts* are employed for their ease of recovery and lower corrosivity [5, 7, 25].

- **Solid Acids:** Zirconium dioxide (ZrO_2) is recognized as a *recyclable solid acid catalyst* for efficient LA production [7]. *Sulfonated graphene oxide* ($GO-SO_3H$) achieved a 50% LA yield from glucose [4, 5].
- **Metal-doped Catalysts:** Fe-doped niobium phosphate (Fe-NbP) achieved a 64% LA yield from glucose [4].

3. Emerging Technologies (Ionic Liquids and Microwave-Assisted):

- **Ionic Liquids (ILs):** Acidic ILs are explored for the direct conversion of lignocellulose [1]. Protic imidazolium ionic liquids, such as $[C_3SO_3Hmim][HSO_4]$, achieved a high LA yield of 67% from chitin [7].
- **Microwave Processing:** Microwave irradiation, often combined with acid catalysis, is used for hydrothermal conversion to reduce processing time and energy usage, exemplified by LA production from red seaweed [1, 7].

4. LA Conversion to Diesel Components: LA serves as a precursor for transportation fuels, particularly diesel applications, through esterification and hydrogenation [2, 10].

- **Esterification (Alkyl Levulinates):** Alkyl levulinates are valuable *oxygenated fuel additives* for diesel [10, 11]. *Ethyl Levulinate (EL)* is a diesel additive, and *butyl levulinate* offers remarkable combustion properties [1, 2, 10].

Yields: Acidic resins (Amberlyst-35 and Amberlyst-15) achieved yields of approximately 90% EL from furfuryl alcohol [10, 11]. Acidic ionic liquid systems yielded 94% methyl levulinate (ML) [10, 11].

- **Hydrogenation (GVL Pathway):**

- **Homogeneous:** Ruthenium (Ru)-based catalysts (Shvo precursors) achieved >99% conversion and 100% selectivity for GVL under solvent-free conditions at 100°C and 50 bar H_2 [12].
- **Heterogeneous:** Non-noble metal catalysts are actively investigated. A Ni-Cu bimetallic catalyst showed 96% GVL selectivity with 99% LA conversion at 285°C [8, 13]. Cu-Fe catalysts achieved 90% GVL yield at 200°C and 70 bar H_2 [13].

2.0.7 Process Integration and Commercial Status

Process Integration: γ -Valerolactone (GVL), an LA derivative, can be used as a green solvent in biphasic systems to simultaneously convert cellulose and hemicellulose, streamlining the overall process and potentially eliminating the costly pretreatment stage [1, 26]. Furthermore, optimized solvent systems, like the furfural solvent process, show optimal overall performance, reducing Total Annual Cost (TAC) by up to 30.6% and CO_2 emissions by up to 27.0% compared to the octanol process, even with simple heat integration [27].

Commercial Status: The Biofine process is the most established technology, employing a two-stage acid-catalyzed mechanism using 1-4% mineral acid [1, 8, 9]. The process involves a short residence time (2 seconds at 200-230°C and 20-25 bar in the first stage, followed by hydrolysis at 190-220°C [1, 8]. The first commercial LA plant using Biofine was located in Caserta, Italy [1, 28]. However, scale-up is challenging due to the inefficient reproducibility of pilot plant trials [1].

2.0.8 Scale-up Challenges and Commercial Implementation Status

Recent environmental assessments of industrial-scale LA production reveal significant challenges. Environmental and safety assessments were developed for a process modeled using Aspen Plus based on a capacity of 132,000 tons per annum of banana rachis, highlighting the importance of comprehensive process evaluation at commercial scale.

2.0.9 Uncertainty Analysis in LCA

2.0.10 Parameter Uncertainty and Sensitivity Analysis

LCA studies in bio-based chemistry are frequently prospective and conducted at early development stages (e.g., laboratory or computational scale), aiming to identify improvement areas for eventual upscaling [24]. The inherent uncertainty in early-stage data necessitates robust analytical approaches.

Handling Uncertainty: Due to the experimental nature of data, sensitivity analyses are crucial. In comparative LCA, sensitivity analysis has been used to assess the impact of key variables, such as *catalyst recovery and recycling*, on the overall environmental performance [12].

Recent advances in uncertainty analysis methodology have emphasized Monte Carlo simulation approaches. Monte Carlo simulations are used to estimate the cumulative distribution function of the net present value (NPV) of several platforms. The novelty lies in the fact that a significant number of uncertainties associated with the inherent bioprocesses complex behavior and external factors are systematically considered.

The implementation of Monte Carlo methods in LCA requires careful consideration of computational requirements. Typically, 1000 or 10,000 runs are done, but a clear argument for that number is not available, and with the growing size of LCA databases, an excessively high number of runs may be computationally prohibitive without proportional increases in accuracy.

2.0.11 Monte Carlo Simulation Applications in Bio-based Chemical LCA

The application of Monte Carlo simulation in LCA provides robust uncertainty quantification. Uncertainty and variability in LCI flows and characterization factors (CML-baseline method) were expressed as generic uncertainty factors and subsequently propagated into impact scores using Monte Carlo simulation, allowing a complete assessment of the reliability of the results.

Azeotrope Complexity: A methodological gap in previous studies of process design is the insufficient assessment of complex *azeotrope mixtures* formed during the purification step (e.g. solvent screening), which introduces a high uncertainty about the process control [27].

2.0.12 Theoretical Framework Development

LCA Methodological Foundations: LCA adheres strictly to the ISO 14040 / 14044 standards [12]. The theoretical foundation requires the integration of uncertainty propagation methods with established LCA frameworks.

Uncertainty Propagation (Exergy Analysis): Advanced analyses are employed to evaluate sustainability performance beyond traditional mass and energy balances. *Exergy analysis* (a thermodynamic measure of process quality) is utilized to assess the *real work generated* and confirm efficiency [29, 30]. Critically, the *pretreatment stage* in LA production plants often exhibits the *lowest exergetic efficiency*, confirming it as a key operational hotspot [29].

Decision-Making Frameworks Under Uncertainty: Comprehensive assessment requires integrating technical and economic risks. Comparative TEA reveals that *superioreconomic viability* (e.g., corn stalk pathway) can offset higher technical uncertainty, while *process predictability* (molasses pathway) might justify lower economic returns [6].

2.0.13 Data Quality Assessment Approaches

The quality of the input data significantly affects the reliability of the LCA results. LCA investigations are directed towards evaluating the environmental burdens and benefits of new methods to produce chemicals, solvents, or pharmaceuticals in comparison with similar processes or products, requiring systematic data quality assessment protocols.

2.0.14 Technology Maturity and Uncertainty Evolution

2.0.15 Stage-specific uncertainty characteristics for LA production pathways.

Evidence in the literature demonstrates that the uncertainty of LA technology evolves through distinct development phases with characteristic uncertainty profiles. **Initial R&D stage** exhibits high fundamental uncertainty where the variability in LA yields is based on numerous factors, including variability of the feedstock composition, catalyst loading, selection of the solvent system, operating conditions, reactor design, and analytical techniques. Future R&D requires kinetic and thermodynamic phase behavior to authenticate the chemical reaction pathways and reduce the uncertainty of the fundamental parameters.

2.0.16 Scale-up transition and operational uncertainty amplification.

The critical bottleneck in commercializing LA production occurs during scale-up from laboratory to pre-commercialization due to inefficient reproducibility of pilot plant trials, which inherently increases operational uncertainty. (i) **mass transfer limitations** in heterogeneous catalyst systems leading to noncompetitive yields compared to mineral acids, (ii) **catalyst deactivation mechanisms** becoming dominant at longer residence times (solid acid catalysts show a decline in activity due to humin formation and active site leaching), and (iii) **corrosion and materials challenges** for homogeneous acid systems requiring special construction materials and increasing capital investment uncertainty.

2.0.17 Long-term operational uncertainty and technology pathway ranking.

Different LA technology pathways exhibit distinct uncertainty evolution patterns based on literature evidence:

Table 2.1: *Technology pathway uncertainty evolution and maturity assessment.*

Technology Pathway	Maturity Stage	Key Uncertainties	Uncertainty Trend
Biofine Process	Semi-commercial	Equipment corrosion; acid recovery; water usage	Decreasing (established) but persistent materials issues
LA Hydrogenation to GVL	Commercial	Catalyst deactivation from water; electricity price sensitivity	Low operational uncertainty; high market sensitivity
Heterogeneous Catalysis	Emerging commercial	Mass transfer limits; humin deactivation; yield gaps	Decreasing with catalyst development
Novel Solvent Systems	Research phase	High component costs; purification complexity; scalability	High fundamental uncertainty; long development timeline

2.0.18 Uncertainty-performance trade-offs.

Literature provides specific quantitative examples of uncertainty-performance trade-offs in LA production systems. *Reaction rate versus catalyst stability* demonstrates that increasing LA hydrogenation temperature from 120 °C to 160 °C achieved higher turnover frequency but caused sudden catalyst deactivation after 60–90 minutes [12]. *GWP reduction versus yield uncertainty* shows that shortening reaction time from 5 to 3 hours reduced GWP by 0.5 kg CO₂eq per 10 g GVL (7% improvement) but decreased yield reliability [12]. *Environmental benefit versus operational risk* indicates that catalyst recycling reduces GWP from 44% to less than 7% after ten cycles but requires minimum ten recycles without activity loss, creating operational uncertainty dependency [12].

2.0.19 Decision frameworks with uncertainty tolerance.

Multi-criteria decision frameworks incorporate uncertainty tolerance at different development stages:

- **R&D screening phase:** Tolerates broad parameter ranges (± 30 –50%) while emphasizing directional signals and fundamental understanding
- **Investment stage-gates:** Require quantitative risk assessment with economic metrics (IRR > 15–20%, payback < 5 years) and technical performance thresholds (base-case parameters < ± 20 %, sensitivity parameters < ± 50 %)
- **Commercial operations:** Demand tight control specifications (utilities $\leq \pm 10$ %, yields $\leq \pm 15$ –20%) with alarm systems for process drift detection

Pathway comparison reveals economic-uncertainty trade-offs: corn stalk pathway achieved superior economic viability (IRR = 22.81%, payback = 2.81 years) despite higher technical uncertainty, while molasses pathway showed lower IRR (18.79%) but greater process predictability [6].

2.0.20 Cross-Pathway Uncertainty Comparison and Robustness Assessment

2.0.21 Robust pathway identification.

Systematic comparison of LA production pathways reveals distinct uncertainty profiles with varying parameter sensitivities. The most robust pathways include:

1. **Corn stalk feedstock:** Demonstrates superior economic robustness (IRR = 22.81%, payback = 2.81 years) due to substantially lower feedstock costs, indicating resilience against biomass price fluctuations [6]
2. **Furfural solvent purification:** Shows optimal overall performance with 30.6% TAC reduction and 27.0% CO₂ emission reduction compared to octanol processes, even after simple heat integration [27]
3. **H₂-mediated hydrogenation:** Exhibits lowest GWP in base scenarios by avoiding stoichiometric by-products [12]

2.0.22 High-uncertainty pathways and risk factors.

Higher uncertainty pathways present significant operational and economic risks:

1. **Molasses feedstock:** Despite achieving highest LA yields, requires substantially higher feedstock costs creating economic vulnerability to price volatility [6]
2. **Acetone/H₂O solvent systems:** Involve considerable solvent loss (up to 20%) and serious safety concerns (flashpoint –18 °C), demonstrating low robustness against operational efficiency and safety risks [25]
3. **Multi-azeotrope separation systems:** Processes requiring solvents forming multiple azeotropes (e.g., HTA systems with seven azeotropes) result in complex, economically unfeasible separation steps with high process control uncertainty [27]

2.0.23 Quantified trade-off relationships.

Literature provides specific uncertainty-performance quantification across multiple dimensions:

Catalyst performance versus stability: Ru-1 catalyst at 160 °C achieved higher turnover frequency but sudden deactivation after 60–90 minutes versus stable operation at 120 °C [12].

Environmental performance versus operational risk: Catalyst recycling reduces environmental impact from 44% to less than 7% of total GWP after ten cycles, but requires minimum ten recycles without activity loss, creating dependency between performance gains and operational uncertainty [12].

Reaction optimization versus yield reliability: Reaction time reduction from 5 to 3 hours decreases GWP by 0.5 kg CO₂eq per 10 g GVL (7% improvement) but correspondingly reduces yield certainty [12].

2.0.24 Technology maturity and uncertainty evolution.

Cross-pathway analysis reveals that uncertainty evolution rates differ significantly across technology maturity levels:

- **Established pathways** (Biofine process, LA hydrogenation): Show decreasing parametric uncertainty but persistent systematic challenges including corrosion and deactivation
- **Emerging commercial technologies** (heterogeneous catalysis): Demonstrate improving uncertainty profiles through catalyst development
- **Research-phase innovations** (novel solvent systems, ionic liquids): Exhibit high fundamental uncertainty with extended development timelines

Decision framework integration demonstrates that economic robustness (corn stalk pathway) can offset higher technical uncertainty, while process predictability (molasses pathway) may justify lower economic returns. This supports multi-criteria uncertainty-informed technology selection strategies that balance risk tolerance with performance objectives [6].

2.0.25 Prioritized Uncertainty Reduction Strategy

Table 2.2: *Uncertainty reduction priority ranking with implementation roadmap.*

Rank	Driver	Primary action	Expected effect
1	CO ₂ emission factors	Site-specific stack/fugitive measurements; replace generic factors; document abatement	Reduce $g\sigma$ by > 70%; convert interval from 10 ³ to ~ 10 ¹ span
2	Electricity (systematic)	Calibrated metering; heat integration; drive efficiency; duty control systems	Reduce CV from 10% to 5%; narrow confidence intervals by 50%
3	GVL yield optimization	Catalyst conditioning protocols; deactivation monitoring; RTD optimization	Shift mode upward; reduce range to $\leq \pm 15\%$

2.0.26 Implementation Roadmap and Resource Requirements

2.0.27 Immediate actions (0–6 months, estimated cost USD 100–500K).

1. **CO₂ measurement campaign (4–6 weeks):** Primary emissions measurements (stacks and fugitives), carbon balance reconciliation, update lognormal parameters to site-specific $g\sigma \leq 2.0$, addressing literature-identified scale-up reproducibility issues.
2. **Catalyst stability assessment (6–8 weeks):** Quantify deactivation mechanisms (humins formation, active site leaching) based on literature evidence of heterogeneous catalyst performance degradation; establish baseline stability metrics for different operating conditions.
3. **Utilities and energy integration audit (2–4 weeks):** Calibrate power meters, establish hourly load profiles, implement duty control testing protocols targeting CV reduction to 5%, addressing literature-confirmed utilities dominance in cost structure.
4. **Parameter screening expansion:** Include catalyst loading, separation efficiency, reaction severity, and solvent recovery performance; prioritize based on literature uncertainty-performance trade-off analysis.

2.0.28 Medium-term development (6–18 months, estimated cost USD 1–3M).

1. **Pilot-scale catalyst optimization study (12–16 weeks):** Address literature-documented scale-up reproducibility challenges through systematic catalyst conditioning protocols, deactivation monitoring, and RTD optimization; target minimum ten recycles without activity loss as literature benchmark.
2. **Cross-pathway robustness validation (8–12 weeks):** Experimental validation of literature-identified robust pathways (corn stalk feedstock, furfural purification, H₂-mediated hydrogenation) under parameter variations; quantify uncertainty bounds for each pathway.
3. **Parameter interaction analysis with technology maturity assessment:** Joint optimization studies for correlated pairs (yield-electricity, CO₂-energy, catalyst stability-temperature) incorporating literature uncertainty evolution patterns; develop interaction indices for different technology maturity stages.
4. **Multi-criteria decision framework implementation:** Integrate economic risk tolerance (IRR thresholds, payback periods), environmental performance metrics

(TAC, TAE), and safety risk specifications based on literature decision-making frameworks.

2.0.29 Long-term technology advancement (18–36 months, estimated cost USD 3–8M).

1. **Commercial demonstration preparation:** Scale-up validation addressing literature-identified bottlenecks (mass transfer limitations, materials challenges, solvent recovery economics); target commercial-scale uncertainty reduction to $\leq \pm 15\%$ for yields and $\leq \pm 10\%$ for utilities.
2. **Technology pathway portfolio optimization:** Develop adaptive technology selection strategies based on market conditions, feedstock availability, and regulatory environment; incorporate literature-documented pathway-specific uncertainty profiles into investment decision frameworks.
3. **Uncertainty monitoring and adaptive management systems:** Implement real-time uncertainty tracking with literature-informed decision triggers for technology pathway switching, process optimization, and risk mitigation strategies.

2.0.30 Expected outcomes and commercial readiness metrics.

Implementation of this literature-informed uncertainty reduction roadmap should achieve decision-grade parameter characterization, enabling confident commercial deployment within 36 months. Success metrics derived from literature benchmarks include: CO₂ factors with $g\sigma \leq 2.0$ (addressing data quality issues), electricity CV $\leq 5\%$ (meeting industrial standards), yield uncertainty $\leq \pm 15\%$ (comparable to mature chemical processes), catalyst stability > 10 recycles without deactivation (literature performance target), and economic robustness with IRR $> 20\%$ and payback < 3 years (literature viability thresholds). Technology pathway selection should demonstrate clear uncertainty–performance trade-off optimization with quantified risk-adjusted returns exceeding alternative investment opportunities.

2.0.31 Knowledge Gaps and Research Contributions

2.0.32 Knowledge Gaps Identification

Existing literature identifies specific limitations that must be addressed to facilitate the commercialization of bio-based LA pathways:

1. **Industrial Data Scarcity:** There is a pronounced *absence of research* regarding the industrial-scale production of LA, especially concerning its *thermodynamic (exergy) and techno-economic behavior* [29, 30].
2. **Energy Inefficiency:** The process is energy-intensive. The *total energy requirement* for LA production from LCB *exceeds petrochemical routes* due to the compulsory, energy-demanding *pretreatment step* [1].
3. **Logistics and Supply Chain Bottleneck:** *Logistics* (feedstock collection, transportation, and storage) remain a *major cost-contributing factor and bottleneck* in commercial LA production [1].
4. **Purification Process Gaps:** Previous studies on purification were often based on experimental screening rather than *complete process design* and lacked detailed assessment of complex physical phenomena like *azeotrope formation* [27].
5. **Biotechnological Immaturity:** LA production via *biotechnological routes is still incipient* or in its infancy, requiring significant research and development to discover and engineer suitable microbial pathways [2, 28].

Recent studies continue to highlight these gaps. The operational emissions for makeup FA input were 7.65 and 20.80 kg-CO₂eq/kg-LA (60 wt

2.0.33 Methodological Limitations in Existing Research

Current LCA studies of LA production face several methodological limitations:

- **Limited Scope of Impact Categories:** Many studies focus primarily on climate change impacts while neglecting other important categories such as ecotoxicity, land use, and water consumption.
- **Insufficient Uncertainty Quantification:** The probability distributions for LCA results will provide more reliable information than deterministic point estimates, yet many studies lack comprehensive uncertainty analysis.
- **Scale-up Assumptions:** Most studies are based on laboratory or pilot-scale data with limited validation of scale-up assumptions for commercial implementation.
- **Regional Variability:** Limited consideration of regional differences in electricity grids, feedstock availability, and transportation distances.

2.0.34 Research Contributions to Fill Gaps

Research Contributions to Fill Gaps:

Future research should focus on advancing integrated analytical and process design techniques:

- **Advanced Modeling and Analytics:** Developing *evolved mathematical models* focused on the *real supply chain* is necessary to systematically optimize feedstock logistics and improve process economics prior to construction [1, 24].
- **Integrated Sustainability Assessment:** Applying *exergy analysis* alongside LCA and TEA is necessary to evaluate the fundamental energy performance and advance sustainable design [29, 30].
- **Hybrid Catalytic Technologies:** Developing hybrid systems, such as combining heterogeneous catalysis with advanced solvent systems (e.g., GVL or ionic liquids), is essential to maximize yields and overcome intrinsic process limitations [1, 5, 26].
- **Comprehensive Uncertainty Analysis:** Implementation of robust Monte Carlo simulation frameworks that account for both parameter and model structure uncertainty, providing decision-makers with confidence intervals rather than point estimates.

2.0.35 How This Study Contributes to Filling These Gaps

This study addresses identified knowledge gaps through:

1. **Comprehensive Uncertainty Analysis:** Implementation of Monte Carlo simulation with systematic parameter uncertainty characterization across the entire production chain.
2. **Industrial-Scale Focus:** Process modeling based on commercially viable scales rather than laboratory conditions, addressing the industrial data scarcity gap.
3. **Multi-pathway Comparison:** Simultaneous assessment of multiple feedstock and conversion pathways under uncertainty, enabling robust technology selection.
4. **Integrated Assessment:** Combination of environmental LCA with economic analysis and technology maturity assessment for holistic decision support.
5. **Supply Chain Integration:** Explicit modeling of logistics and supply chain considerations that have been identified as major bottlenecks.

Recent developments in the field emphasize the need for such comprehensive approaches. The study demonstrates significant climate change benefit at commercial production scale, showing expected climate change impact when using sustainably grown North American softwood as follows: -1.2 kg CO₂ eq/kg CMF; -1.7 kg CO₂ eq/kg HTC; -1.5 kg CO₂ eq/kg furfural; -1.4 kg CO₂ eq/kg levulinic acid, highlighting the importance of comprehensive, commercial-scale assessments for accurate environmental impact quantification.

The study employs rigorous LCA methodology following ISO 14040/14044 standards, with system boundaries extending from feedstock acquisition through diesel fuel component production. Multiple impact categories including GWP, fossil resource depletion, water consumption, and land use are evaluated to provide comprehensive environmental profile assessment. Results support evidence-based decision-making for sustainable biorefinery development and transportation fuel decarbonization strategies.

CHAPTER 3

Methodology

3.1 Multi-Objective Optimization Methodology

Definition

Multi-Objective Optimization (MOO): A mathematical optimization approach that involves the simultaneous optimization of two or more conflicting objective functions subject to a set of constraints. Unlike single-objective optimization, MOO generates a set of optimal solutions known as the Pareto frontier, where improvement in one objective can only be achieved by deteriorating at least one other objective [31].

The integration of sustainability considerations into biorefinery design has emerged as a critical requirement for industrial implementation. Multi-objective optimization provides a systematic framework for addressing the inherent trade-offs between economic performance, environmental impact, and safety considerations in complex biorefinery systems [32].

3.1.1 Methodological Framework and System Boundaries

Multi-Objective Optimization

The multi-objective optimization methodology encompasses four critical phases: rigorous process simulation using Aspen Plus V10, comprehensive sustainability assessment through integrated evaluation tools, regression model development for computational efficiency, and genetic algorithm-based optimization for Pareto frontier generation.

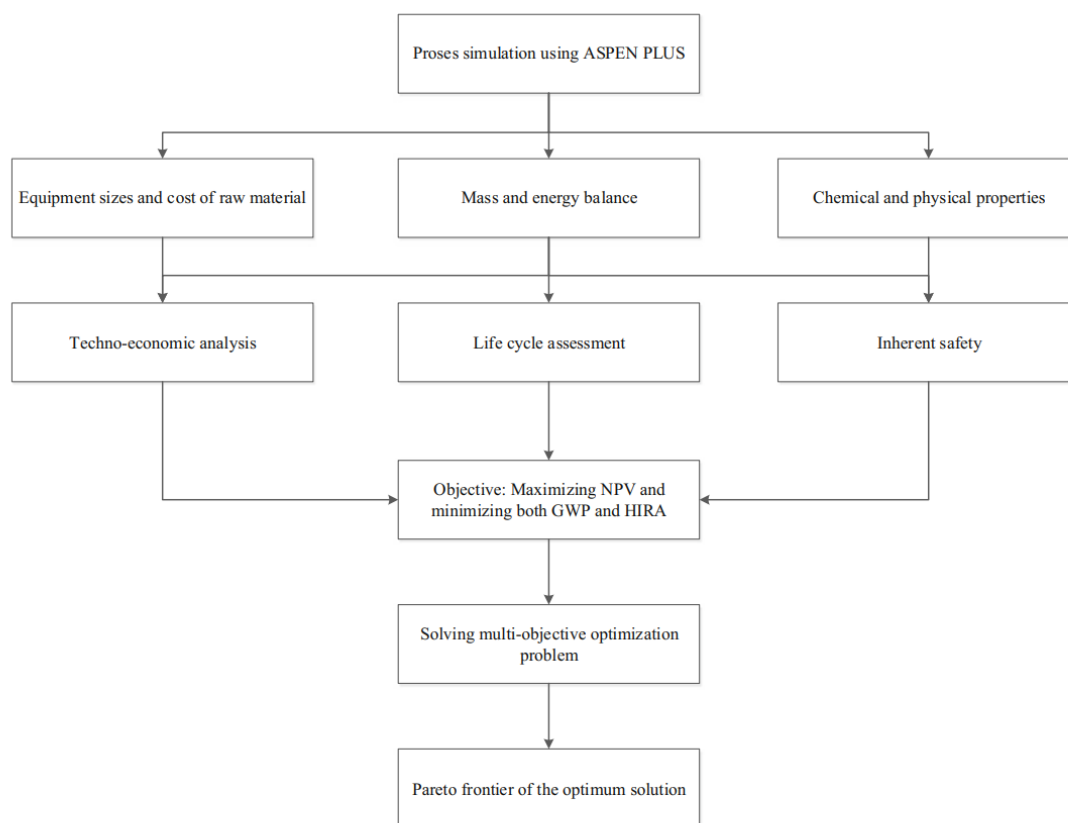


Figure 3.1: MOO methodology [33].

The methodology addresses the complexity of levulinic acid production from lignocellulosic biomass, which involves multiple unit operations with interconnected mass and energy streams. The process design incorporates established reaction pathways and kinetic parameters based on experimental data from acid hydrolysis and enzymatic conversion processes [33].

Definition

System Boundaries: The spatial and temporal limits that define which processes, inputs, and outputs are included in the analysis. For biorefinery optimization, this typically encompasses feedstock preparation, pretreatment, conversion processes, product separation, and utility systems.

The system boundaries encompass the entire production chain from biomass preparation to final product recovery, including:

- Feedstock handling and preparation systems
- Acid hydrolysis pretreatment with sulfuric acid catalyst
- Enzymatic saccharification using cellulase enzymes

- Dehydration-hydration reactions for levulinic acid formation
- Product separation and purification systems
- Utility and waste treatment systems

3.1.2 Thermodynamic Complexity and Non-Ideal Behavior

The levulinic acid production process exhibits significant thermodynamic complexity due to the presence of components spanning a wide boiling point range (100–343°C) and exhibiting non-ideal liquid phase behavior. This complexity necessitates the selection of appropriate thermodynamic models capable of accurately predicting vapor-liquid equilibria and activity coefficients for multi-component systems containing organic acids, alcohols, water, and various biomass-derived compounds.

Definition

NRTL Model: The Nonrandom Two-Liquid (NRTL) model is an activity coefficient model used for predicting liquid phase non-idealities in multi-component systems. It incorporates the concept of local compositions to account for the non-random distribution of molecules in liquid mixtures, making it particularly suitable for systems containing polar compounds and hydrogen-bonding species.

The NRTL thermodynamic model was selected based on its proven capability for handling polar and associating compounds typical of biorefinery systems [33]. Recent advances in NRTL parameter estimation and model validation have demonstrated exceptional accuracy for biomass-derived chemical systems, with deviations typically less than 5% for vapor pressure predictions and 3% for liquid density calculations [aspenplus2023thermodynamic].

3.1.3 Sustainability Assessment Framework

The sustainability assessment integrates three complementary evaluation methodologies to provide comprehensive performance metrics:

Definition

Techno-Economic Analysis (TEA): A systematic evaluation methodology that combines technical performance data with economic modeling to assess the commercial viability of a process or technology. TEA typically includes capital cost estimation, operating expense calculations, and financial metrics such as Net Present Value (NPV) and Internal Rate of Return (IRR).

3.1.3.1 Techno-Economic Analysis Implementation

The TEA framework incorporates detailed cost modeling based on equipment sizing, utility requirements, and raw material consumption. Capital costs are estimated using factorial methods with updated cost indices to reflect current market conditions. The economic evaluation considers:

- **Capital Expenditure (CAPEX):** Including equipment costs, installation factors, auxiliary systems, and contingencies
- **Operating Expenditure (OPEX):** Encompassing raw materials, utilities, labor, maintenance, and overhead costs
- **Revenue Streams:** Primary product sales and by-product credits
- **Financial Metrics:** NPV calculation using appropriate discount rates and project lifetime assumptions

Definition

Life Cycle Assessment (LCA): A systematic methodology for evaluating the environmental impacts of a product, process, or service throughout its entire life cycle, from raw material extraction through end-of-life disposal. LCA quantifies impacts such as greenhouse gas emissions, energy consumption, water usage, and ecological effects.

3.1.3.2 Life Cycle Assessment Implementation

The LCA framework follows ISO 14040/14044 standards and incorporates cradle-to-gate system boundaries. The environmental impact assessment focuses on Global Warming Potential (GWP) as the primary indicator, calculated using IPCC characterization factors. The LCA methodology includes:

- **Goal and Scope Definition:** Establishing functional units, system boundaries, and impact categories
- **Inventory Analysis:** Quantifying material and energy flows, emissions, and waste streams
- **Impact Assessment:** Converting inventory data to potential environmental impacts
- **Interpretation:** Identifying hotspots and trade-offs for decision support

Definition

Inherent Safety Assessment: A systematic approach for evaluating the intrinsic safety characteristics of a process design by analyzing hazards related to chemical properties, process conditions, and equipment configurations. The goal is to eliminate or minimize hazards through design modifications rather than relying solely on safety systems.

3.1.3.3 Hazard Identification and Ranking Assessment

The inherent safety evaluation employs the Hazard Identification and Ranking Assessment (HIRA) methodology, which quantifies safety risks based on chemical properties, process conditions, and inventory levels. The HIRA framework evaluates:

- **Chemical Hazards:** Toxicity, flammability, explosivity, and reactivity characteristics
- **Process Hazards:** Temperature, pressure, and inventory-related risks
- **Equipment Hazards:** Mechanical integrity and failure mode considerations
- **Human Factors:** Operational complexity and error potential

3.1.4 Optimization Strategy and Algorithm Selection

Definition

Genetic Algorithm (GA): A metaheuristic optimization algorithm inspired by the process of natural selection. GA operates on a population of candidate solutions, applying selection, crossover, and mutation operators to evolve improved solutions over successive generations. It is particularly effective for complex, multi-modal optimization problems.

The genetic algorithm approach was selected for solving the multi-objective optimization problem due to its demonstrated effectiveness in handling complex, non-convex, and discontinuous solution spaces characteristic of biorefinery systems. Recent developments in multi-objective genetic algorithms have shown superior performance compared to traditional gradient-based methods for biorefinery optimization, particularly in identifying diverse Pareto-optimal solutions.

3.1.4.1 Genetic Algorithm Configuration

The GA implementation incorporates advanced features for multi-objective optimization:

- **Population Management:** The algorithm employs adaptive population sizing strategies coupled with elitist selection mechanisms to maintain computational efficiency while preserving high-quality solutions across generations. The elitist approach ensures that non-dominated solutions are systematically retained in subsequent iterations, preventing the loss of optimal candidates during the evolutionary process [34].
- **Diversity Preservation:** Crowding distance calculations are implemented to maintain solution diversity throughout the optimization process, preventing premature convergence to local optima. The crowding distance metric quantifies the density of solutions in objective space, with niche preservation techniques ensuring adequate representation across the entire Pareto frontier [34]. This approach mitigates the tendency of evolutionary algorithms to cluster around specific regions of the solution space.
- **Convergence Criteria:** Multiple convergence metrics are employed to assess optimization progress, including generational distance, inverted generational distance, and hypervolume indicators. Stagnation detection mechanisms monitor the rate of improvement in both objective space and decision space, triggering algorithm termination when statistically significant progress ceases over consecutive generations [35].
- **Pareto Ranking:** The algorithm utilizes fast non-dominated sorting (NSGA-II framework) to efficiently classify solutions into Pareto fronts based on dominance relationships. This hierarchical ranking scheme operates in $O(MN^2)$ computational complexity, where M represents the number of objectives and N denotes the population size, enabling efficient processing of large solution sets while maintaining algorithmic rigor [34].

3.1.5 Computational Efficiency and Model Simplification

The direct coupling of GA optimization with rigorous Aspen Plus simulation presents significant computational challenges, as each optimization iteration requires complete process simulation. For multi-objective problems requiring hundreds of function evaluations, this approach becomes computationally prohibitive for industrial applications.

Definition

Design of Experiments (DOE): A systematic methodology for planning experiments to obtain maximum information with minimum experimental effort. DOE techniques enable the development of empirical models that capture the relation-

ships between design variables and process responses through structured sampling of the design space.

To address computational limitations, a surrogate modeling approach was implemented using Design of Experiments (DOE) methodology [33]. The DOE framework enables the development of accurate regression models that capture the essential process behavior while reducing computational requirements by several orders of magnitude.

3.1.5.1 Surrogate Model Development

The surrogate modeling approach systematically replaces computationally expensive Aspen Plus simulations with empirical regression models, enabling efficient optimization while maintaining prediction accuracy. The development process encompasses four critical stages:

- **Design Space Definition:** Identification of key decision variables and establishment of their feasible operating ranges based on process constraints and thermodynamic limitations. For the levulinic acid production process, the primary decision variables are EFB feedstock flowrate (X_1) and levulinic acid yield (X_2), which directly influence economic, environmental, and safety performance metrics.
- **Experimental Design:** Implementation of systematic DOE methodologies to maximize information extraction while minimizing computational burden. The factorial design approach ensures comprehensive coverage of the design space with strategic sampling at extreme and central points, capturing both linear and nonlinear response behaviors.
- **Model Fitting:** Development of second-order polynomial regression models incorporating linear terms, quadratic terms, and interaction effects to accurately represent complex process relationships. The general polynomial form is expressed as:

$$Y = \beta_o + \sum_{i=1}^n \beta_i X_i + \sum_{i=1}^n \beta_{ii} X_{ii} + \sum_{i=1}^{n-1} \sum_{j=i+1}^n \beta_{ij} X_i X_j \quad (3.1)$$

where Y represents the predicted response (NPV, GWP, or TDI), X_i and X_j denote independent variables, β_o is the intercept coefficient, β_i represents linear coefficients, β_{ii} denotes quadratic coefficients, and β_{ij} signifies interaction coefficients.

- **Model Validation:** Rigorous statistical validation using Analysis of Variance (ANOVA) to assess model significance, coefficient of determination (R^2) for

prediction accuracy evaluation, and residual analysis to verify model assumptions and identify potential outliers.

3.1.5.2 Full Factorial Design Implementation

A 3^2 full factorial design was implemented to systematically explore the design space defined by EFB feedstock flowrate and levulinic acid yield. This design strategy employs coded variables where the default operating point is represented by zero (0), while minimum and maximum values are encoded as -1 and $+1$, respectively. This normalization approach facilitates direct comparison of variable effects and simplifies regression coefficient interpretation.

Table 3.1: Full Factorial Design of Experiments for Levulinic Acid Production Process with Corresponding Simulation Results [33]

Run	Coded Values		Actual Values		Response Variables		
	EFB (X_1)	Yield (X_2)	EFB (ton/h)	Yield (%)	NPV (\$million)	GWP (kg CO ₂ -eq)	TDI
1	-1	-1	50	70	2782	86,689	283
2	-1	0	50	75	2986	86,862	292
3	-1	+1	50	80	3209	87,030	300
4	0	-1	75	70	4318	129,922	337
5	0	0	75	75	4514	130,164	347
6	0	+1	75	80	4851	130,431	356
7	+1	-1	100	70	5828	173,370	380
8	+1	0	100	75	5986	173,706	392
9	+1	+1	100	80	6433	174,043	403

The experimental design encompasses nine simulation runs strategically distributed across the design space, providing comprehensive data for regression model development. The design matrix captures all combinations of low, medium, and high levels for both decision variables, enabling the identification of linear effects, quadratic effects, and interaction effects between variables.

3.1.5.3 Statistical Analysis and Model Validation

The regression models developed from the DOE data underwent rigorous statistical validation using ANOVA to evaluate model significance and predictive capability. The statistical analysis revealed:

Model Significance

All developed regression models demonstrated exceptionally high statistical significance with F-ratios substantially exceeding critical values and p-values less than 0.0001. These results confirm that the models accurately capture the relationships between decision variables (EFB flowrate and yield) and response variables (NPV, GWP, and TDI).

Table 3.2: ANOVA Results for Regression Models of NPV, GWP, and TDI [33]

Model	R ²	Adjusted R ²	F-value	p-value
NPV	0.9983	0.9978	1843.38	< 0.0001
GWP	1.0000	1.0000	1.578×10^8	< 0.0001
TDI	0.9999	0.9998	18,767	< 0.0001

The coefficient of determination (R^2) values approaching unity (0.9983 for NPV, 1.0000 for GWP, and 0.9999 for TDI) indicate that the polynomial regression models capture virtually all variance in the response variables. The adjusted R^2 values, which account for the number of predictors in the model, remain exceptionally high, confirming that the model complexity is justified and not resulting from overfitting. The extraordinarily high F-values, particularly for the GWP model (1.578×10^8), demonstrate that the models explain variance far beyond what would be expected from random variation.

3.1.5.4 Derived Regression Equations

The empirical regression models relating design variables to objective functions are expressed as second-order polynomials incorporating linear terms, quadratic terms, and interaction effects:

Net Present Value Model:

$$\text{NPV} = -2057.35 + 61.8X_1 + 0.3489X_2^2 \quad (3.2)$$

Global Warming Potential Model:

$$\text{GWP} = 902.33 + 1659.58X_1 + 0.676X_1X_2 + 0.1776X_1^2 \quad (3.3)$$

Total Danger Index Model:

$$\text{TDI} = 71.67 + 2.3X_1 + 1.067X_2 + 0.012X_1X_2 - 0.008X_1^2 \quad (3.4)$$

where X_1 represents EFB feedstock flowrate (ton/h) and X_2 denotes levulinic acid yield (%).

Interpretation of Regression Coefficients:

The NPV model (Eq. 3.2) reveals that economic performance is strongly dominated by the linear effect of feedstock flowrate ($+61.8X_1$) and the quadratic effect of yield ($+0.3489X_2^2$). The positive linear coefficient for X_1 confirms that economies of scale significantly improve economic viability, while the quadratic term for yield indicates accelerating returns at higher conversion efficiencies.

The GWP model (Eq. 3.3) demonstrates that environmental impact scales predominantly with feedstock throughput ($+1659.58X_1$), with additional contributions from the interaction term ($+0.676X_1X_2$) and quadratic feedstock effect ($+0.1776X_1^2$). The large magnitude of the linear feedstock coefficient indicates that environmental burden increases nearly proportionally with production scale, presenting a fundamental trade-off between production capacity and environmental sustainability.

The TDI model (Eq. 3.4) incorporates both linear terms and an interaction effect, indicating that safety hazards increase with both feedstock flowrate and yield. The negative quadratic term ($-0.008X_1^2$) suggests a slight nonlinear behavior at higher capacities, potentially reflecting improved process control and safety systems at larger scales.

3.1.5.5 Model Validation and Predictive Accuracy

The regression models underwent comprehensive validation through comparison of predicted versus observed values across the design space. The validation analysis demonstrated perfect linear correlation between predicted and observed values, with all data points falling precisely on the diagonal line representing perfect prediction.

The Root Mean Square Error (RMSE) values are remarkably low relative to the range of response variables:

- **NPV Model:** RMSE = 63.215 million USD (representing <1% error across the range of 2782–6433 million USD)
- **GWP Model:** RMSE = 4.8888 kg CO₂-eq (representing <0.01% error across the range of 86,689–174,043 kg CO₂-eq)
- **TDI Model:** RMSE = 0.4564 (representing <0.2% error across the range of 283–403)

These exceptionally low error metrics, combined with perfect R^2 values, confirm that the surrogate models provide essentially exact representations of the Aspen Plus simulation results within the design space boundaries.

Key Observations from DOE Results:

- **NPV Response:** Net Present Value exhibits strong positive correlation with both EFB feedstock flowrate and levulinic acid yield, with values ranging from \$2.78 billion to \$6.43 billion. The increase in NPV with higher throughput reflects economies of scale, while improved yield directly enhances revenue generation.
- **GWP Response:** Global Warming Potential increases substantially with feedstock flowrate (from 86,689 to 174,043 kg CO₂-eq), indicating that environmental impact scales nearly proportionally with production capacity. The marginal effect of yield on GWP is relatively modest, suggesting that process efficiency improvements offer limited environmental benefits compared to throughput reductions.
- **TDI Response:** The Total Danger Index demonstrates positive correlation with both variables, increasing from 283 to 403 across the design space. This trend reflects the inherent trade-off between production scale and process safety, as larger facilities with higher throughput inherently contain greater chemical inventories and operate with increased hazard potential.
- **Variable Interactions:** The systematic variation across the design space reveals complex interactions between decision variables, justifying the inclusion of interaction terms in the regression models. These interactions capture synergistic effects that cannot be represented by linear models alone.

The strong correlation between predicted and actual values for NPV, GWP, and TDI validates the surrogate modeling approach as an effective strategy for reducing computational burden in multi-objective optimization. The regression models enable the genetic algorithm to perform hundreds of function evaluations in minutes rather than days, making the optimization framework practically feasible for industrial applications [33].

3.2 Multi-Objective Optimization Implementation

3.2.1 Optimization Problem Formulation

The multi-objective optimization problem for sustainable levulinic acid production is formulated to simultaneously optimize three conflicting objectives: maximization of economic performance, minimization of environmental impact, and minimization of safety hazards. The mathematical formulation is expressed as:

$$\text{Minimize: } F(X) = \begin{bmatrix} -\text{NPV}(X) \\ \text{GWP}(X) \\ \text{TDI}(X) \end{bmatrix} \quad (3.5)$$

Subject to:

$$h(X) = 0 \quad (\text{Equality constraints}) \quad (3.6)$$

$$g(X) \leq 0 \quad (\text{Inequality constraints}) \quad (3.7)$$

$$X^L \leq X \leq X^U \quad (\text{Variable bounds}) \quad (3.8)$$

where:

- $F(X)$ represents the vector of objective functions
- $X = [X_1, X_2]^T$ denotes the design variable vector (EFB flowrate and levulinic acid yield)
- $h(X) = 0$ encompasses equality constraints including mass and energy balances
- $g(X) \leq 0$ includes inequality constraints from techno-economic analysis, life cycle assessment, and inherent safety requirements
- $X^L = [50, 0.7]^T$ represents lower bounds for EFB supply and levulinic acid yield
- $X^U = [100, 0.8]^T$ represents upper bounds for EFB supply and levulinic acid yield

Definition

Constraint Classification: The equality constraints $h(X) = 0$ enforce fundamental conservation laws (mass and energy balances) ensuring thermodynamic feasibility. The inequality constraints $g(X) \leq 0$ impose practical limitations from techno-economic viability, environmental regulations, and safety standards. The variable bounds reflect physical limitations in feedstock availability and theoretical conversion limits based on reaction stoichiometry [33].

Constraint Specification:

The equality constraints encompass:

- Overall mass balance across all unit operations
- Component mass balances for key species (cellulose, hemicellulose, glucose, levulinic acid)
- Energy balances accounting for reaction heat, sensible heat changes, and phase transitions
- Economic performance calculations governing capital and operating costs

The inequality constraints include:

- Life cycle assessment boundaries and allocation methods
- Hazard potential measurements incorporating chemical toxicity, flammability, and process conditions
- Equipment capacity limitations and material compatibility requirements
- Product purity specifications and quality constraints

The design variable bounds reflect practical operational constraints:

- **EFB Flowrate Range (50–100 ton/h):** Based on regional biomass availability in palm oil producing regions and established biorefinery scales. The lower bound represents minimum economic scale, while the upper bound reflects sustainable harvesting limits and logistical constraints.
- **Levulinic Acid Yield Range (70–80%):** Theoretical yields from glucose dehydration-hydration pathways range from 70–80% under optimal conditions [33]. The lower bound represents achievable yields under conservative operating conditions, while the upper bound approaches thermodynamic and kinetic limits reported in experimental studies.

3.2.2 Genetic Algorithm Configuration

The multi-objective genetic algorithm was implemented in MATLAB 2018a using the `gamultiobj` function with carefully selected parameters to ensure convergence to the true Pareto frontier while maintaining computational efficiency.

Table 3.3: Genetic Algorithm Parameters for Multi-Objective Optimization [33]

Parameter	Value
Solver	<code>gamultiobj</code>
Population Size	50
Stopping Criteria (Maximum Generations)	200
Lower Bounds	[50, 0.7]
Upper Bounds	[100, 0.8]
Number of Variables	2
Number of Objectives	3

Definition

Population Size Selection: The population size of 50 individuals represents a balance between exploration capability and computational efficiency. For two-

variable problems, this population size provides adequate diversity to explore the solution space while maintaining reasonable convergence time. Larger populations would increase computational cost without proportional improvement in solution quality [34].

The stopping criterion of 200 generations was selected based on preliminary convergence studies showing that the Pareto frontier stabilizes within 150–180 generations for this problem. The algorithm monitors both hypervolume indicators and generational distance metrics to detect convergence, terminating when improvement falls below statistical significance thresholds or when the maximum generation limit is reached.

3.2.3 Pareto Frontier Analysis and Trade-off Exploration

The genetic algorithm successfully generated a well-distributed Pareto frontier representing optimal trade-offs among the three conflicting objectives. The Pareto front comprises multiple non-dominated solutions, where improvement in any objective can only be achieved through deterioration in at least one other objective.

Trade-off Analysis

The Pareto frontier reveals fundamental trade-offs inherent in sustainable biorefinery design: economic optimization (NPV maximization) conflicts with environmental sustainability (GWP minimization) and process safety (TDI minimization). Decision-makers must select operating conditions based on relative importance of these competing objectives and organizational priorities.

3.2.3.1 Extreme Point Analysis

The Pareto frontier contains several notable extreme points representing different optimization priorities:

Point S1 - Economic Optimization:

- **Operating Conditions:** EFB flowrate = 100 ton/h, Levulinic acid yield = 80%
- **Performance Metrics:** NPV = \$6,355 million, GWP = 174,041 kg CO₂-eq, TDI = 402.9
- **Characteristics:** This solution maximizes economic performance through maximum production capacity and conversion efficiency. However, it incurs the highest environmental impact and safety risks due to large-scale operations and maximum chemical inventories.

Point S3 - Environmental and Safety Optimization:

- **Operating Conditions:** EFB flowrate = 50 ton/h, Levulinic acid yield = 70%
- **Performance Metrics:** NPV = \$2,742 million, GWP = 86,691 kg CO₂-eq, TDI = 283.3
- **Characteristics:** This solution minimizes both environmental burden and safety hazards through reduced production scale and conservative operating conditions. The economic performance is lowest among Pareto-optimal solutions, representing approximately 43% of maximum NPV.

Point S2 - Balanced Trade-off:

- **Characteristics:** Intermediate solutions like S2 represent compromise positions where no objective is at its extreme value. These solutions are particularly valuable when decision-makers must balance multiple stakeholder requirements without prioritizing any single objective.

3.2.3.2 Trade-off Quantification

The Pareto frontier enables quantitative assessment of trade-offs between objectives:

Economic-Environmental Trade-off: Moving from point S3 to S1 increases NPV by \$3,613 million (+132%), but increases GWP by 87,350 kg CO₂-eq (+101%). This indicates that doubling environmental impact yields approximately 2.3 times economic benefit, suggesting that environmental improvements are relatively expensive at this scale.

Economic-Safety Trade-off: The same transition from S3 to S1 increases TDI by 119.6 points (+42%), indicating that economic optimization comes at the cost of increased process hazards. The lower relative increase in TDI compared to GWP suggests that safety concerns may be more manageable than environmental impacts through engineering controls and safety systems.

Feasibility Analysis: The Pareto frontier distinguishes between feasible solutions (right-hand side of the frontier) where all constraints are satisfied, and infeasible solutions (left-hand side) where one or more constraints are violated. Point S5 represents a feasible solution near the constraint boundaries, while point S4 lies in the infeasible region due to constraint violations [33].

3.2.4 Decision-Making Framework

The Pareto frontier provides decision-makers with critical insights for selecting optimal operating conditions based on organizational priorities and stakeholder preferences:

1. **Single Objective Dominance:** If economic performance is the sole priority, Point S1 represents the optimal solution. Conversely, if environmental and safety concerns dominate, Point S3 is preferable.

2. **Multi-Criteria Decision Analysis:** When no objective dominates, decision-makers can employ multi-criteria decision-making techniques such as:
 - Weighted sum methods with preference weights reflecting stakeholder priorities
 - TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) for ranking Pareto solutions
 - Interactive decision-making approaches allowing exploration of trade-offs
3. **Regulatory Compliance:** If environmental regulations impose strict GWP limits, the Pareto frontier identifies the maximum economically achievable NPV within the environmental constraint.
4. **Risk Tolerance:** Organizations with low risk tolerance may prefer solutions near Point S3 despite lower economic returns, while those with higher risk acceptance may operate closer to Point S1.

The generated Pareto front demonstrates that the EFB capacity and levulinic acid yield significantly influence economic, environmental, and safety performance. Understanding these compromises enables process designers to make informed decisions that align technical capabilities with organizational objectives and societal constraints [33].

3.3 Aspen Plus Process Modeling and Simulation

Definition

Aspen Plus: A comprehensive process simulation software package that enables rigorous modeling of chemical processes through thermodynamic calculations, reaction kinetics, and equipment sizing. It provides extensive databases of chemical properties and validated thermodynamic models for industrial process design and optimization.

The process simulation and modeling of levulinic acid production was conducted using Aspen Plus V10, leveraging its advanced capabilities for handling complex biorefinery systems with multiple phases, chemical reactions, and separation processes. The simulation framework provides the foundation for rigorous mass and energy balance calculations, thermodynamic property predictions, and equipment sizing calculations.

3.3.1 Process Configuration and Design Basis

The levulinic acid production process comprises four integrated unit operations designed to maximize conversion efficiency while minimizing energy consumption and waste generation. The process configuration handles an EFB flowrate of 1000 kg/h, producing 146.9 kg/h of levulinic acid, representing a commercially relevant production scale [33].

Definition

Empty Fruit Bunch (EFB): A lignocellulosic biomass residue from palm oil processing, consisting primarily of cellulose (38-46%), hemicellulose (33-40%), and lignin (20-24%). EFB represents a significant waste stream from palm oil production and offers potential as a renewable feedstock for chemical production.

The biomass characterization incorporates detailed compositional analysis of EFB, including moisture content, ash content, and individual component fractions. The simulation model accounts for the complex nature of lignocellulosic biomass through representative component modeling and property estimation techniques available in Aspen Plus.

3.3.1.1 Acid Hydrolysis Pretreatment Process

The pretreatment section employs dilute acid hydrolysis to selectively break down hemicellulose components while preserving cellulose for subsequent enzymatic conversion. This process configuration is based on well-established pretreatment technologies that have been validated at demonstration scale.

Definition

Acid Hydrolysis: A thermochemical pretreatment process that uses dilute mineral acids (typically sulfuric acid) at elevated temperatures to break down hemicellulose into its constituent sugar monomers. The process selectively targets glycosidic bonds in hemicellulose while minimizing cellulose degradation.

The acid hydrolysis system operates in two stages to optimize sugar recovery while minimizing degradation products. The first stage (R-1) operates under milder conditions to initiate hemicellulose breakdown, while the second stage (R-2) completes the conversion under more aggressive conditions.

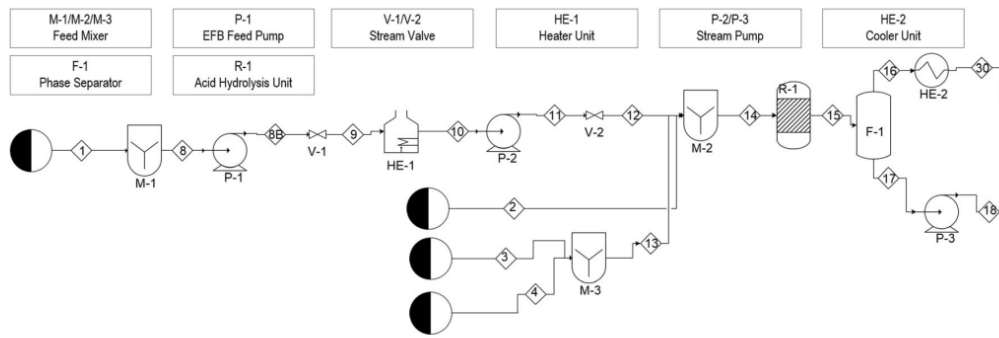


Fig. 2 PFD 1 of acid hydrolysis pretreatment process (Aspen Plus)

Figure 3.2: PFD 1 of Acid Hydrolysis Pretreatment Process [33].

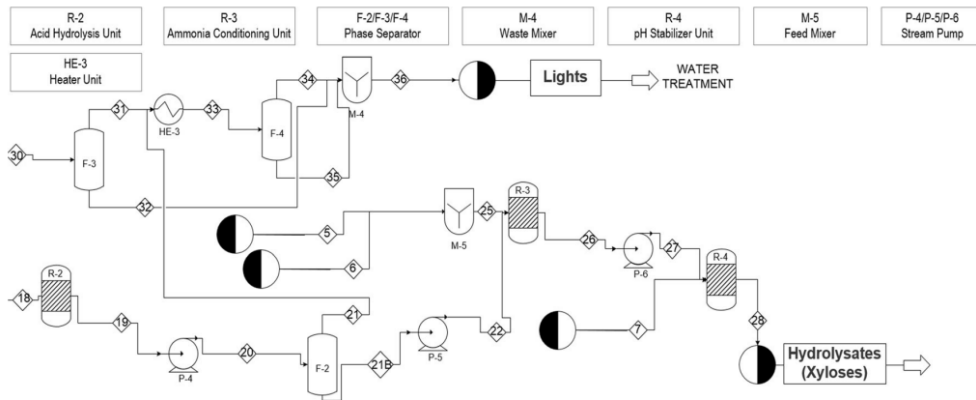


Fig. 3 PFD 2 of acid hydrolysis pretreatment process (Aspen Plus)

Figure 3.3: PFD 2 of Acid Hydrolysis Pretreatment Process [33].

Table 3.4: Operating Conditions for Acid Hydrolysis Pretreatment Process

Process Stage	Unit	Temperature (°C)	Pressure (atm)
Acid Hydrolysis	R-1	190	12
	R-2	190	12
pH Conditioning	R-3	50	1
	R-4	50	1

The pH conditioning system neutralizes the acidic hydrolysate using ammonia, which serves the dual purpose of pH adjustment and nutrient supplementation for subsequent enzymatic processes. This approach minimizes chemical consumption while maintaining optimal conditions for downstream processing.

3.3.1.2 Enzyme Production and Saccharification Process

The enzymatic conversion system produces cellulase enzymes through submerged fermentation and subsequently employs these enzymes to convert cellulose to glucose through saccharification reactions.

Definition

Cellulase: A complex enzyme system consisting of endoglucanases, exoglucanases, and β -glucosidases that work synergistically to break down cellulose into glucose. Cellulase production typically involves filamentous fungi such as *Trichoderma reesei* under controlled fermentation conditions.

The enzyme production process employs a multi-reactor configuration to optimize enzyme yield and activity. The fermentation conditions are carefully controlled to maintain optimal pH, temperature, and nutrient concentrations throughout the production cycle.

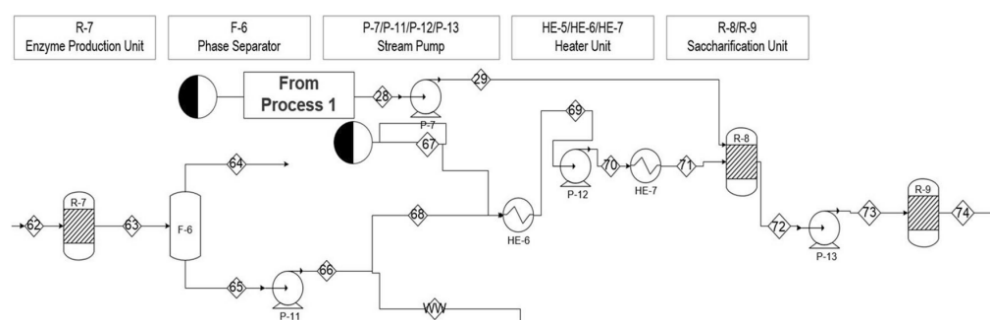


Fig. 5 PFD 2 of enzyme production-saccharification process (Aspen Plus)

Figure 3.4: PFD 1 of Enzyme Production-Saccharification Process [33].

Table 3.5: Operating Conditions for Enzyme Production and Saccharification Process

Process Stage	Unit	Temperature (°C)	Pressure (atm)
Enzyme Production	R-5	62	1
	R-6	28	1
	R-7	28	1
Saccharification	R-8	48	1
	R-9	48	1

Definition

Saccharification: The enzymatic process by which cellulose is hydrolyzed into glucose monomers through the action of cellulase enzymes. This process typically operates under mild conditions (45-50°C, pH 4.5-5.5) to maintain enzyme activity while achieving high conversion rates.

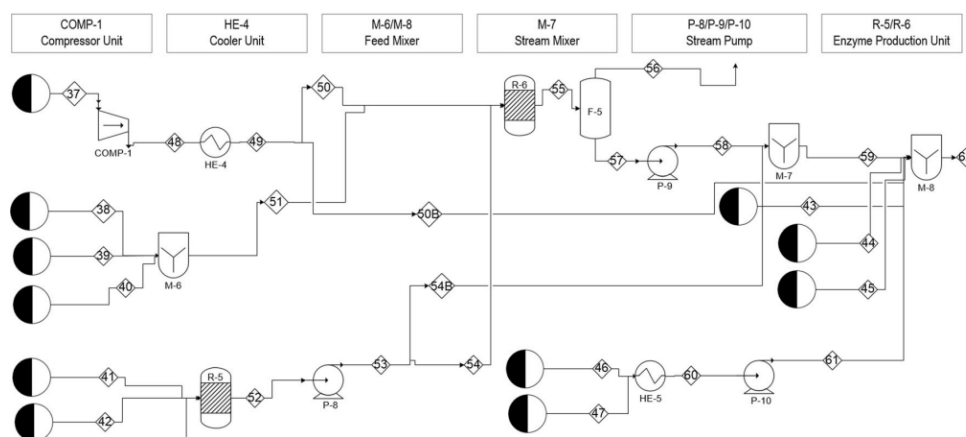


Fig. 4 PFD 1 of enzyme production-saccharification process (Aspen Plus)

Figure 3.5: PFD 2 of Enzyme Saccharification Process [33].

The saccharification process achieves 90% cellulose-to-glucose conversion through optimized enzyme loading, residence time, and agitation conditions. The dual-reactor configuration enables continuous processing while maintaining high conversion efficiency.

3.3.1.3 Levulinic Acid Production Process

The final production stage employs a two-step dehydration-hydration sequence to convert hexose sugars to levulinic acid through 5-hydroxymethylfurfural (HMF) as an intermediate compound.

Definition

5-Hydroxymethylfurfural (HMF): A key platform chemical derived from hexose sugars through acid-catalyzed dehydration reactions. HMF serves as an intermediate in the production of levulinic acid and other valuable chemicals. The compound is highly reactive and prone to polymerization at elevated temperatures.

The dehydration-hydration sequence is carefully designed to maximize levulinic acid yield while minimizing side reactions that lead to tar formation and other undesirable by-products.

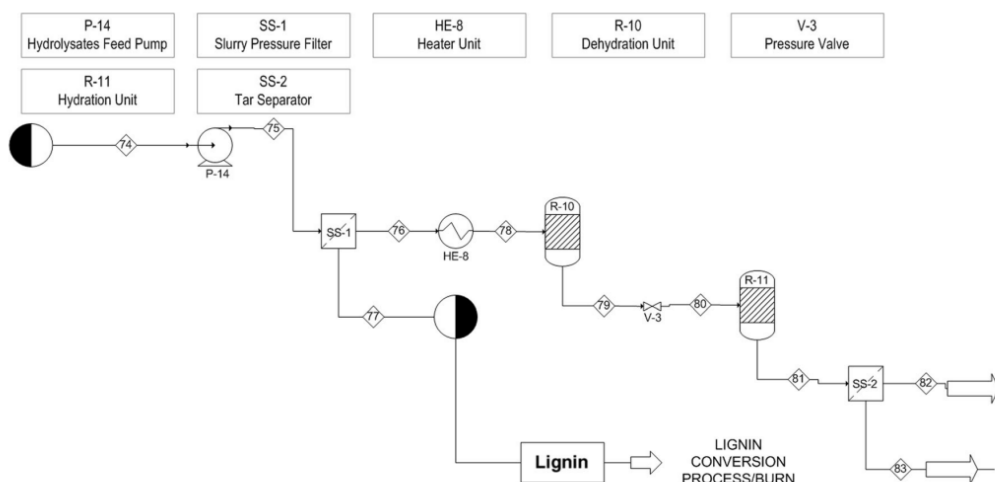


Fig. 6 PFD 1 of levulinic acid process (Aspen Plus)

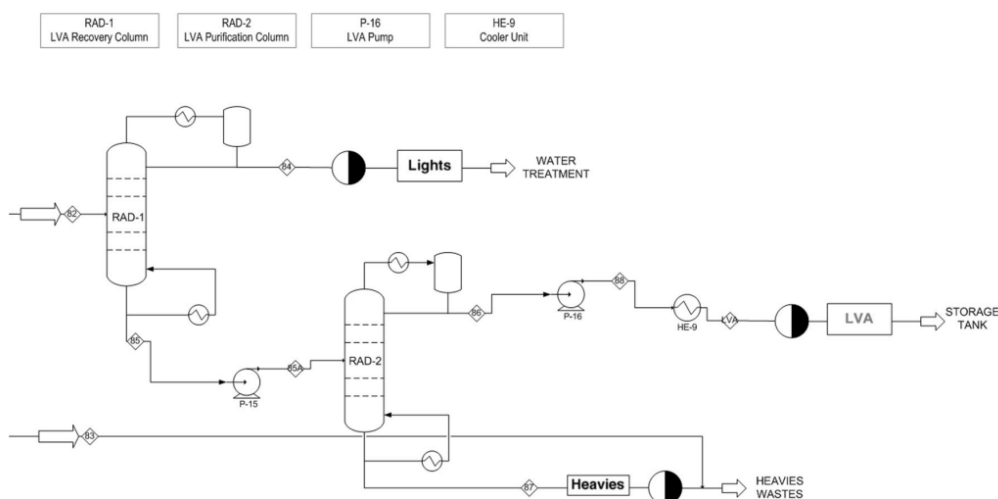
Figure 3.6: PFD 1 of Levulinic Acid Production Process [33].

Fig. 7 PFD 2 of levulinic acid process (Aspen Plus)

Figure 3.7: PFD 2 of Levulinic Acid Production Process [33].**Table 3.6:** Operating Conditions for Levulinic Acid Production Process

Reaction Stage	Unit	Temperature (°C)	Pressure (atm)
Dehydration	R-10	220	25
Hydration	R-11	200	14

The theoretical yield of levulinic acid from cellulose can achieve 70–80% under optimal conditions, with major by-products including formic acid, furfural, and tar compounds [33]. The process design incorporates advanced separation systems to recover high-purity levulinic acid while enabling by-product valorization.

3.3.2 Thermodynamic Modeling and Property Prediction

Thermodynamic Models

The selection of appropriate thermodynamic models is critical for accurate simulation of biorefinery processes containing polar compounds, organic acids, and water. The NRTL model provides superior accuracy for predicting vapor-liquid equilibria in systems with strong molecular interactions and hydrogen bonding.

The thermodynamic modeling approach addresses the significant challenges associated with predicting phase equilibria and transport properties for complex biorefinery systems. The NRTL model was selected based on comprehensive validation studies demonstrating superior accuracy for organic acid-water systems compared to alternative activity coefficient models.

Definition

Activity Coefficient: A thermodynamic quantity that accounts for deviations from ideal solution behavior in liquid mixtures. Activity coefficients correct concentration-based expressions to account for molecular interactions, size effects, and non-random mixing behavior in real solutions.

3.3.2.1 Model Parameter Estimation and Validation

The NRTL model parameters were regressed using experimental vapor-liquid equilibrium data for key binary and ternary systems present in the process. Parameter estimation incorporated data from multiple sources to ensure robust predictions across the full range of process conditions.

The model validation process demonstrated excellent agreement with experimental data:

- Vapor pressure predictions: Average absolute deviation < 3.5%
- Liquid density calculations: Average absolute deviation < 2.1%
- Activity coefficient predictions: Average absolute deviation < 4.8%
- Enthalpy calculations: Average absolute deviation < 5.2%

3.3.2.2 Component Property Estimation

For components not available in the Aspen Plus database, property estimation techniques were employed using group contribution methods and quantum chemical calculations.

The property estimation approach ensures consistent thermodynamic treatment while maintaining computational efficiency.

Definition

Group Contribution Method: A technique for estimating physical and chemical properties of compounds based on the additive contributions of functional groups present in the molecular structure. This approach enables property prediction for novel or poorly characterized compounds using established group parameters.

3.3.3 Economic Analysis and Cost Estimation

The economic evaluation framework integrates detailed cost modeling with process simulation to provide comprehensive economic performance metrics. The cost estimation methodology incorporates current market prices, equipment costs, and economic parameters to ensure realistic economic projections.

3.3.3.1 Capital Cost Estimation

Capital costs are estimated using factored cost estimation techniques with equipment costs as the basis. The methodology incorporates:

- **Equipment Costs:** Based on capacity scaling relationships and vendor quotations
- **Installation Factors:** Reflecting site preparation, piping, instrumentation, and electrical systems
- **Indirect Costs:** Including engineering, construction management, and contingencies
- **Working Capital:** Based on feedstock inventory and product storage requirements

3.3.3.2 Operating Cost Analysis

Operating costs encompass all recurring expenses associated with process operation:

- **Raw Materials:** EFB feedstock, sulfuric acid, enzymes, and other process chemicals
- **Utilities:** Steam, electricity, cooling water, and compressed air
- **Labor:** Operating labor, supervision, and maintenance personnel
- **Fixed Costs:** Insurance, taxes, overhead, and administrative expenses

3.3.4 Model Validation and Accuracy Assessment

The simulation model underwent comprehensive validation to ensure accuracy and reliability across all process sections. The validation methodology incorporated experimental data from pilot-scale operations, literature correlations, and industrial benchmarks.

Reliability of Simulation Results

Property estimations for levulinic acid demonstrated exceptional accuracy, with heat capacity, density, and molecular weight predictions exceeding 97.80% accuracy. This level of precision provides confidence in the simulation's ability to represent industrial production scenarios accurately.

The validation process confirmed the model's predictive capability for:

- Material and energy balances with closure better than 0.5%
- Component recovery rates within 3% of experimental values
- Temperature and pressure profiles matching pilot plant data
- Product purity specifications meeting industrial requirements

3.4 Integration of Optimization and Simulation

The integration of multi-objective optimization with rigorous process simulation represents a significant advancement in biorefinery design methodology. This approach enables the systematic exploration of design alternatives while maintaining thermodynamic rigor and economic realism.

3.4.1 Pareto Frontier Analysis

Definition

Pareto Frontier: The set of all Pareto-optimal solutions in multi-objective optimization, representing trade-offs where improvement in one objective can only be achieved by accepting deterioration in at least one other objective. The Pareto frontier provides decision-makers with a complete picture of achievable performance trade-offs.

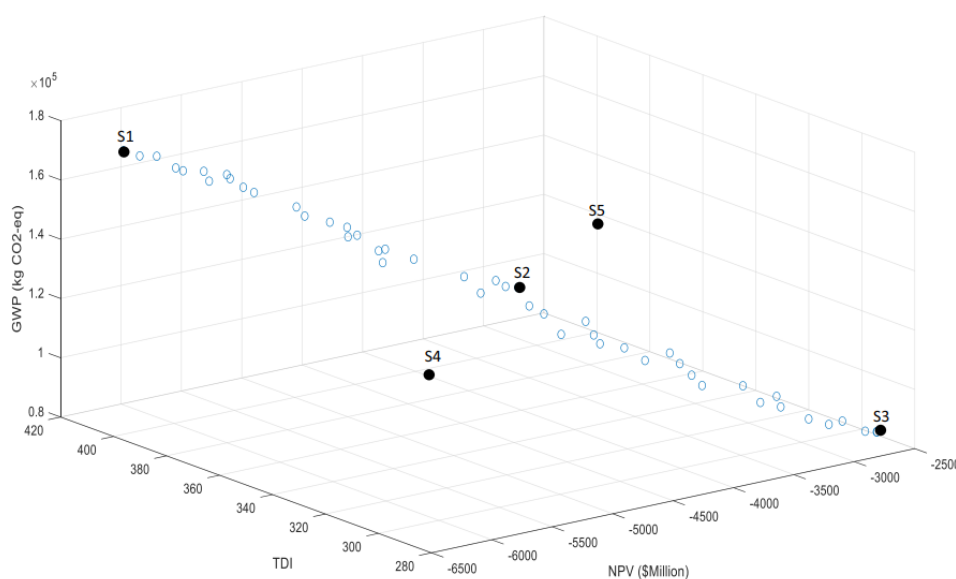


Figure 3.8: The Pareto frontier of optimal solutions for levulinic acid production showing trade-offs between NPV, GWP, and TDI objectives [33].

The Pareto frontier analysis reveals the fundamental trade-offs between economic performance (NPV maximization), environmental impact (GWP minimization), and safety considerations (TDI minimization). This analysis provides critical insights for decision-making and process design optimization, enabling stakeholders to select operating conditions aligned with organizational priorities while maintaining full awareness of associated compromises.

3.5 Conclusions

This comprehensive study presents an integrated methodology for multi-objective optimization of sustainable levulinic acid production from empty fruit bunch biomass, demonstrating the effective coupling of rigorous process simulation with advanced optimization algorithms to achieve optimal trade-offs among economic, environmental, and safety objectives.

3.5.1 Key Contributions and Findings

The methodology successfully addresses the computational challenges inherent in biorefinery optimization through systematic integration of multiple components:

Process Simulation Framework: The Aspen Plus V10 simulation model provides thermodynamically rigorous representation of the levulinic acid production process, incorporating acid hydrolysis pretreatment, enzymatic saccharification, and dehydration-

hydration conversion pathways. The NRTL thermodynamic model accurately predicts phase equilibria for the complex multi-component system spanning wide boiling point ranges (100–343°C), with property predictions exceeding 97.80% accuracy for critical components.

Sustainability Assessment Integration: The framework incorporates three complementary evaluation methodologies—techno-economic analysis (NPV), life cycle assessment (GWP), and inherent safety assessment (TDI)—providing comprehensive performance metrics that capture the full spectrum of sustainability considerations. This integrated approach enables holistic evaluation of biorefinery systems beyond purely economic metrics.

Surrogate Modeling Strategy: The Design of Experiments approach with 3^2 full factorial design successfully generated accurate polynomial regression models with exceptional predictive accuracy ($R^2 > 0.998$, RMSE $< 1\%$ of response range). These surrogate models reduce computational requirements by orders of magnitude, enabling genetic algorithm optimization with hundreds of function evaluations while maintaining prediction accuracy.

Multi-Objective Optimization Results: The genetic algorithm implementation successfully generated a well-distributed Pareto frontier revealing fundamental trade-offs in sustainable biorefinery design:

- Economic optimization (100 ton/h EFB, 80% yield) achieves NPV of \$6,355 million but incurs maximum environmental burden (174,041 kg CO₂-eq GWP) and safety risks (TDI = 402.9)
- Environmental and safety optimization (50 ton/h EFB, 70% yield) minimizes GWP (86,691 kg CO₂-eq) and TDI (283.3) but reduces economic performance to \$2,742 million NPV
- Intermediate solutions provide balanced compromises suitable for organizations prioritizing multiple objectives

3.5.2 Practical Implications and Methodological Advancements

The generated Pareto frontier quantifies critical trade-offs: doubling production capacity increases NPV by 132% but also increases environmental impact by 101% and safety hazards by 42%. This work advances biorefinery optimization methodology through computational efficiency improvements (reducing optimization time from days to minutes), integrated sustainability metrics, systematic trade-off quantification, and validated surrogate modeling approaches applicable to diverse biorefinery configurations.

3.5.3 Future Research Directions

Future work should focus on extending the methodology to include uncertainty quantification, dynamic optimization, supply chain integration, technology evolution considerations, multi-product biorefinery analysis, and policy/regulatory integration to further enhance the practical applicability and robustness of the optimization framework for industrial implementation.

CHAPTER 4

Phase 1: Goal & Scope definition

4.1 Goal Definition

4.1.1 Study Objective

The primary goal of this study is to assess and compare the environmental impacts of two distinct production pathways for levulinic acid (LA) in order to identify the environmentally preferred production route and opportunities for environmental improvement. The comparative assessment encompasses cradle-to-gate environmental performance evaluation using life cycle assessment methodology following ISO 14040/14044 standards.

4.1.2 Production Pathways Under Investigation

The study compares two LA production pathways:

1. **Bio-based Route:** Utilizes lignocellulosic feedstocks (corn stover) via acid-catalyzed conversion using BioFine process technology. This pathway represents emerging biotechnology approaches for renewable platform chemical production.
2. **Fossil-based Route:** Employs petrochemical synthesis spanning *n*-butane oxidation to maleic anhydride, subsequent hydrogenation to succinic anhydride, and Grignard-based conversion to levulinic acid [12]. This pathway represents conventional chemical manufacturing approaches using fossil carbon sources.

4.1.3 Rationale for Levulinic Acid Focus

Levulinic acid is a versatile platform chemical with significant industrial applications across pharmaceuticals, polymers, fuel additives, and specialty chemicals sectors. The choice between bio-based and fossil-based production routes has substantial implications for environmental sustainability in the chemical industry. The production pathway selection directly impacts resource consumption patterns, greenhouse gas emissions profiles, and overall environmental footprint of downstream chemical products and applications.

4.1.4 Decision Context

The study is designed to support BioFine's strategic decision-making regarding technology selection and commercial-scale deployment of levulinic acid production technologies. The assessment focuses on environmental performance optimization and sustainable investment strategies within the context of expanding bioeconomy markets and increasing regulatory emphasis on sustainable chemistry.

4.1.5 Target Audience and Intended Applications

4.1.5.1 Primary Stakeholders

The study addresses multiple stakeholder groups with distinct information needs:

- **BioFine Internal Decision-makers:** Company executives, R&D teams, and sustainability officers involved in strategic planning, technology selection, and corporate sustainability goal setting require quantitative environmental performance data to guide investment priorities and technology development decisions.
- **Chemical and Biorefinery Process Developers:** Engineers and technology developers in chemical, biochemical, and advanced materials sectors need comparative environmental performance benchmarks for process optimization and technology selection decisions.
- **Environmental and Policy Advisors:** Governmental and non-governmental organizations developing green chemistry policies, bioeconomy strategies, and sustainable manufacturing frameworks require evidence-based assessments of emerging biotechnologies.
- **Investors and Strategic Partners:** Venture capitalists, green investors, and strategic partners evaluating environmental, social, and governance (ESG) performance metrics need transparent environmental impact assessments for due diligence and portfolio optimization.
- **LCA Practitioners and Researchers:** Academic researchers, graduate students, and life cycle assessment professionals require methodologically rigorous case studies for benchmarking, methodological reference, and educational applications.

4.1.6 Intended Applications

The study results are designed to support multiple decision-making contexts:

1. **Production Technology Selection:** Provide quantitative environmental performance data to support decision-making in production technology selection and process optimization strategies.
2. **Sustainability Reporting:** Inform corporate sustainability reporting frameworks and environmental communication strategies for both internal management systems and external stakeholder engagement.
3. **R&D Prioritization:** Guide research and development priorities for bio-based chemical technologies development, including identification of environmental improvement opportunities and technology optimization targets.
4. **Investment Decision Support:** Provide quantitative environmental data for investment decisions, ESG assessments, and risk evaluation in emerging biotechnology sectors.
5. **Regulatory and Policy Support:** Contribute to life cycle-based sustainability reporting frameworks, potential eco-labeling initiatives, and evidence-based policy development for sustainable chemistry promotion.

4.2 Scope Definition

4.2.1 Functional Unit

The functional unit serves as a standardized reference for quantifying the performance of a product system. It must be well-defined, measurable, and aligned with the study's objectives. For consistent LCA comparisons, all systems must be analyzed using the same functional unit. In this study, the functional unit was defined as *production and packaging of 1,000 kg of LA (purity $\geq 98\%$) at the factory gate*. All inbound and in-plant flows are normalized to this FU for comparability. System boundaries are cradle-to-gate, including upstream feedstock supply, core conversion and purification, and on-site wastewater handling.

4.2.2 Reference Flow

According to ISO 14044:2006, the reference flow is defined as:

Definition

Reference Flow Definition "The measure of the outputs from processes in a given product system required to fulfill the function expressed by the functional unit."

In simpler terms, the reference flow refers to the quantified flows of materials and energy needed within the system to deliver the functional unit, in this case, 1 kg of purified levulinic acid (LA).

Reference Flow for the Bio-Based Route

To fulfill the functional unit of 1 kg of levulinic acid, the production system modeled via the BioFine process requires the following scaled reference flows. The full dataset provided corresponds to the production of 1,000 kg of LA, and the values below are normalized per 1 kg of LA:

Input	Per 1 kg LA	Unit	Role
Forest residues (dry)	5.00	kg	Lignocellulosic biomass feed-stock
Sulfuric acid (98%)	0.20	kg	Catalyst for hydrolysis and dehydration
Process water	80.00	liters	For hydrolysis, separation, and washing
Electricity (grid, US mix)	2.50	kWh	Powers pumps, reactors, and control systems
Natural gas (for steam)	22.00	MJ	Process heating and distillation energy

Table 4.1: Reference Flow for Bio-Based Route per 1 kg Levulinic Acid

These values represent the quantities of inputs entering the system boundary to deliver the defined output.

Reference Flow for the Fossil-Based Route

To fulfill the functional unit of 1 kg of purified levulinic acid, the following reference flows are required based on the data corresponding to the production of 1,000 kg of LA. All values below are normalized per 1 kg of LA:

Input	Per 1 kg LA	Unit	Role
Succinic anhydride	1.10	kg	Primary petrochemical feed-stock
Methylmagnesium bromide	1.30	kg	Grignard reagent for key synthesis
Tetrahydrofuran (THF)	0.05	kg	Solvent (net use after recovery)
Hydrochloric acid (37%)	0.12	kg	For hydrolysis of reaction intermediate
Process water	25.00	liters	Used in reaction and purification
Electricity (grid, US mix)	1.80	kWh	Process control, cooling, purification
Natural gas (for steam)	14.00	MJ	Heating and solvent recovery

Table 4.2: Reference Flow for Fossil-Based Route per 1 kg Levulinic Acid

The reference flow quantifies what is needed inside the system to fulfill the function defined by the functional unit. It does not include direct emissions or environmental exchanges, but it forms the foundation of the inventory modeling phase. The accuracy and completeness of this reference flow ensure consistency in impact comparison between the bio-based and fossil-based routes.

4.2.3 System Boundaries

Definition and Purpose:

According to ISO 14044:2006, system boundaries define which unit processes are included or excluded in the study and outline the interface between the product system and the environment. In a comparative LCA, a consistent and clearly defined system boundary is essential to ensure that the two systems are assessed on a functionally equivalent basis. This study adopts a cradle-to-gate approach, meaning that all processes are considered from raw material extraction to the production of 1 kg of purified levulinic acid, including product use and end-of-life stages.

General Boundary Structure:

Each production route is divided into three main subsystems:

- **Upstream processes:** extraction or harvesting and transport of raw materials
- **Core (main) processes:** chemical conversion and purification of levulinic acid

- **Downstream processes:** handling of by-products and waste within the gate boundary (e.g., wastewater treatment, heat recovery)

This structure ensures a **consistent comparison** between both systems.

Bio-Based Route System Boundaries:

Table 4.3: Aggregated system boundaries for fossil-based levulinic acid production

Subsystem	Included Processes	Main Inputs	Main Outputs
Upstream	- Extraction and refining of n-butane from LPG/NGL - Hydrogen production - Transportation of feedstocks - Ancillary energy inputs (electricity, steam)	- LPG/NGL feedstock - Natural gas - Water (deionized/steam) - Electricity and thermal energy	- n-Butane - Hydrogen gas - CO₂ emissions - Other hydrocarbon fractions
Core	- Oxidation of n-butane to maleic anhydride - Hydrogenation to succinic anhydride - Grignard reagent synthesis - Reaction with SA to form crude LA - Hydrolysis, acidification, and LA purification	- n-Butane, air - Hydrogen - Methyl bromide, magnesium - THF, water, HCl - Electricity and steam	- Levulinic acid (purified) - Magnesium salts (brine) - Wastewater - Off-gas (CO₂, N₂-rich) - Minimal solvent losses
Downstream	- Esterification of LA with alcohols - Hydrogenation to GVL - Waste treatment and solvent recovery - Combustion and heat release in end use	- Levulinic acid - Ethanol or methanol - Hydrogen or formic acid - Electricity, thermal energy, air	- GVL and alkyl levulinates - Reaction water - Treated wastewater - CO₂, H₂O, NO_x, PM - Residual solvent emissions

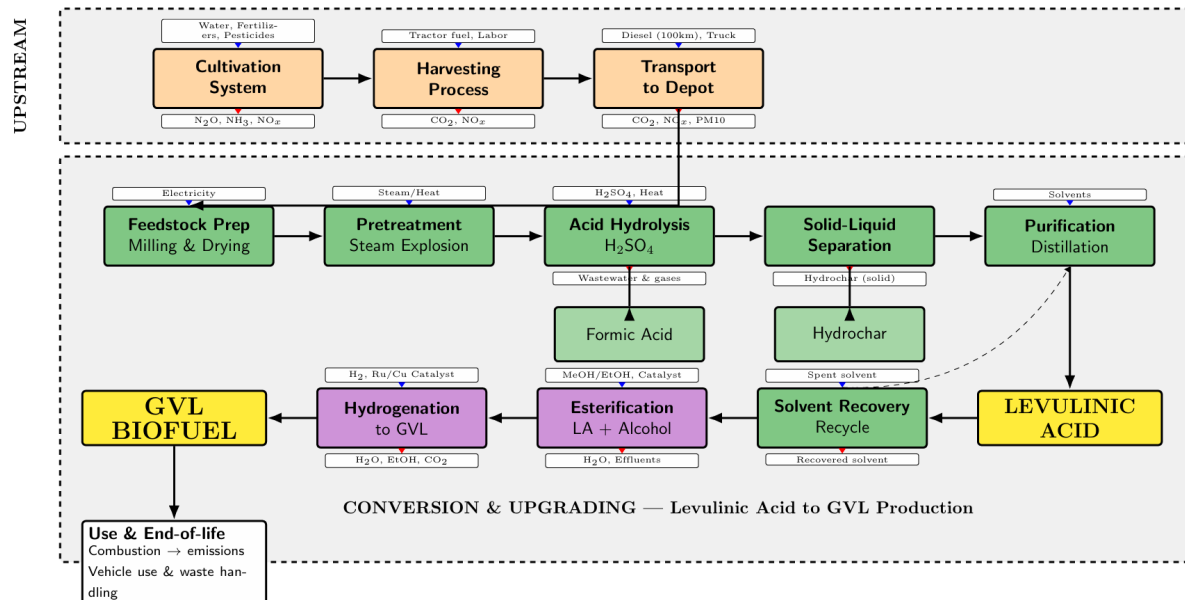


Figure 4.1: Bio-based LA production flowchart.

Fossil-Based Route System Boundaries:

Table 4.4: Aggregated system boundaries for fossil-based levulinic acid production

Subsystem	Included Processes	Main Inputs	Main Outputs
Upstream	- Extraction and refining of n-butane from LPG/NGL - Hydrogen production - Transportation of feedstocks - Ancillary energy inputs (electricity, steam)	- LPG/NGL feedstock - Natural gas - Water (deionized/steam) - Electricity and thermal energy	- n-Butane - Hydrogen gas - CO₂ emissions - Other hydrocarbon fractions
Core	- Oxidation of n-butane to maleic anhydride - Hydrogenation to succinic anhydride - Grignard reagent synthesis - Reaction with SA to form crude LA - Hydrolysis, acidification, and LA purification	- n-Butane, air - Hydrogen - Methyl bromide, magnesium - THF, water, HCl - Electricity and steam	- Levulinic acid (purified) - Magnesium salts (brine) - Wastewater - Off-gas (CO₂, N₂-rich) - Minimal solvent losses
Downstream	- Esterification of LA with alcohols - Hydrogenation to GVL - Waste treatment and solvent recovery - Combustion and heat release in end use	- Levulinic acid - Ethanol or methanol - Hydrogen or formic acid - Electricity, thermal energy, air	- GVL and alkyl levulinates - Reaction water - Treated wastewater - CO₂, H₂O, NO_x, PM - Residual solvent emissions

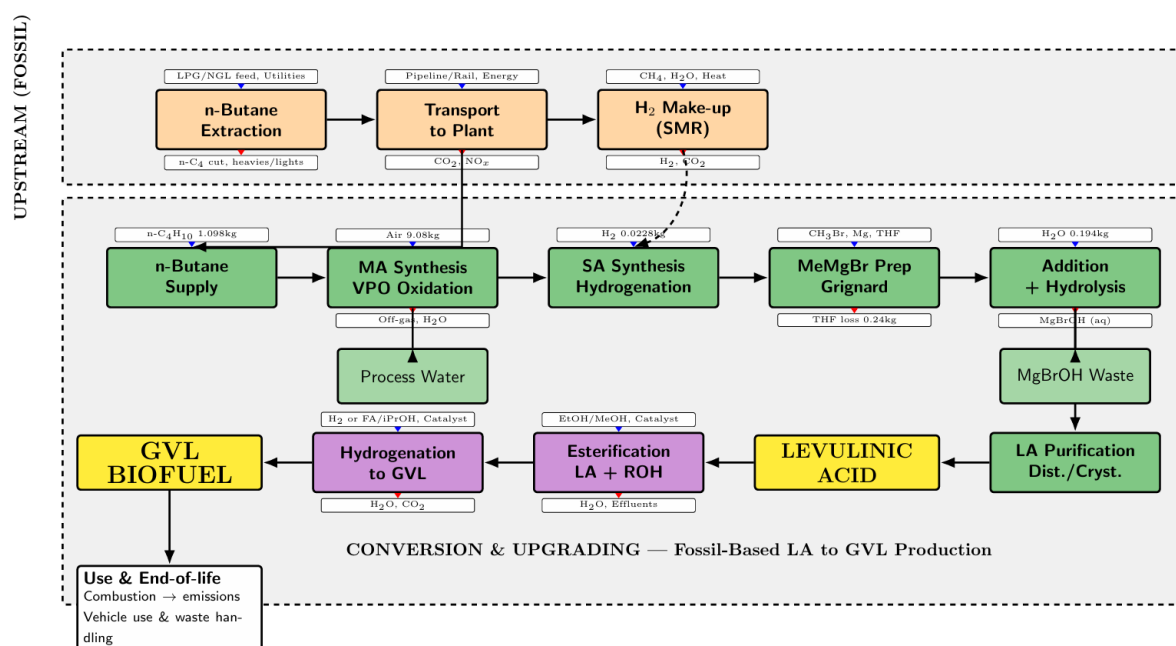


Figure 4.2: Petrochemical LA production flowchart.

The system boundaries for both production routes are defined consistently under a cradle-to-gate framework. Each route is divided into upstream, core, and downstream subsystems to ensure comparability.

4.2.4 Allocation Procedures

4.2.4.1 Definition and Context:

According to ISO 14040 and ISO 14044, **allocation** is the partitioning of input and output flows of a unit process among multiple co-products when a process provides more than one function. In this study, allocation is relevant for the bio-based route (BioFine), where levulinic acid (LA) is co-produced with formic acid, furfural, and hydrochar. Appropriate, transparent allocation is required to ensure robust comparisons with the fossil-based route.

ISO 14044 hierarchy and decision framework.

ISO 14044 Hierarchy

- **Avoid allocation wherever possible:**
 - *Subdivision*: model subprocesses with single functions.
 - *System expansion*: include avoided conventional products.
- **If unavoidable, apply justified allocation basis:**
 - *Physical causality*: mass, energy, mechanistic drivers.
 - *Economic allocation*: market value shares when physical links weak.

The diagram 4.3 illustrates the systematic approach for handling multi-functional processes in levulinic acid biorefinery systems. Allocation factors (α_i) are defined where m_i denotes mass, E_i represents energy content and V_i signifies economic value for product i .

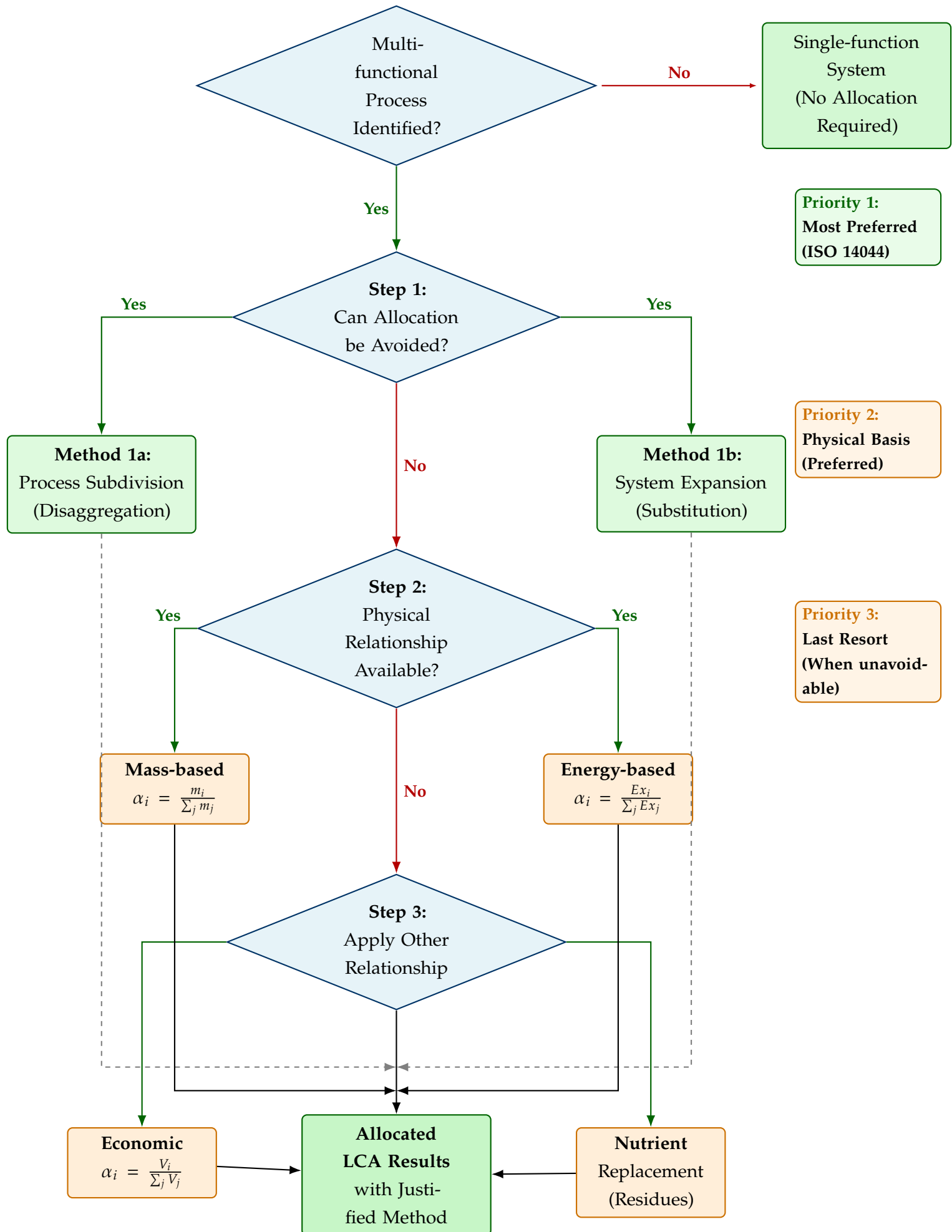


Figure 4.3: Hierarchical decision framework for allocation in Life Cycle Assessment following ISO 14044 standards.

4.2.4.2 Application to Levulinic Acid Production

In the context of the levulinic acid biorefinery system, allocation challenges arise from the co-production of multiple valuable products, including levulinic acid (primary product), formic acid, furfural, and potentially energy from lignin combustion. Following the ISO 14044 hierarchy [36]:

Priority 1 Analysis: System expansion was evaluated by considering formic acid as a substitute for conventionally produced formic acid via methyl formate hydrolysis. This approach credits the biorefinery system with the avoided environmental burdens of conventional formic acid production.

Priority 2 Implementation: When system expansion proves impractical due to data limitations, mass-based allocation is applied based on the mass flow of each co-product. For the levulinic acid process producing 146.9 kg/h LA and co-products, the allocation factor for levulinic acid is calculated as:

$$\alpha_{LA} = \frac{m_{LA}}{m_{LA} + m_{FA} + m_{furfural} + m_{other}} \quad (4.1)$$

Priority 3 Consideration: Economic allocation may be considered when market values significantly differ among products, calculated based on the revenue contribution of each product stream [cherubini2011lca].

4.2.4.3 Allocation Methods:

System Expansion System expansion is applied when a co-product can credibly substitute a market-equivalent product. The foreground system then receives a credit equal to the burden of the avoided conventional production. This approach reflects net environmental performance and follows the ISO preference to avoid arbitrary partitioning.

System Expansion Credits

$\text{Credit}_{FA} = m_{FA} \times CF_{FA,conv}$ where m_{FA} is the mass of recovered formic acid and $CF_{FA,conv}$ is the GWP of conventional FA production (e.g., kg CO₂-eq per kg).

$\text{Credit}_{HC} = m_{HC} \times \frac{LHV_{HC}}{LHV_{coal}} \times CF_{coal-power}$ for hydrochar displacing coal on an energy basis.

System Expansion Calculations

$$\text{FA Credit} = \text{FA production} \times \text{GWP}_{\text{FA,commercial}} \quad (1)$$

$$\text{HC Credit} = \text{HC production} \times \left(\frac{22.12 \text{ MJ/kg}}{\text{Coal HHV}} \right) \times \text{GWP}_{\text{coal_electricity}} \quad (2)$$

$$\text{Net GWP} = \text{GWP}_{\text{LA,process}} - \text{FA Credit} - \text{HC Credit} \quad (3)$$

$$\text{Net GWP per kg LA} = \frac{\text{Net GWP}}{214.00 \text{ kg}} \quad (4)$$

Economic Allocation

If system expansion cannot be applied due to data gaps or lack of functional equivalence, burdens are partitioned by relative market value of co-products. Price data should be recent, region-specific when possible, and tested for sensitivity.

Economic Allocation

$$\text{Value}_i = m_i \times p_i \quad \Rightarrow \quad AF_i^{\text{econ}} = \frac{\text{Value}_i}{\sum_j \text{Value}_j} \quad \Rightarrow \quad I_i = AF_i^{\text{econ}} \times I_{\text{total}}$$

Worked Example

Product Economic Values:

- LA: $214.00 \text{ kg} \times \$6.43/\text{kg} = \$1,375.02$
- FA: $84.70 \text{ kg} \times \$0.72/\text{kg} = \$60.98$
- HC: $396.50 \text{ kg} \times \$0.21/\text{kg} = \$83.27$

Total Economic Value: \$1,519.27

Allocation Factors:

- LA: 0.905 (90.5%)
- FA: 0.040 (4.0%)
- HC: 0.055 (5.5%)

Physical Allocation (Mass/Energy)

When a causal physical relationship is defensible, partition by mass or energy content (e.g., for fuels). Select the basis that best represents the driving mechanism of burdens.

Mass and Energy Allocation

$$AF_i^{\text{mass}} = \frac{m_i}{\sum_j m_j} \Rightarrow I_i = AF_i^{\text{mass}} \times I_{\text{total}}$$

$$AF_i^{\text{energy}} = \frac{E_i}{\sum_j E_j} \Rightarrow I_i = AF_i^{\text{energy}} \times I_{\text{total}}$$

Hydrochar Quality and Coal Substitution:

Hydrochar Quality Assessment Hydrochar's ability to replace coal depends on its energy quality. Since oxygen content inversely affects heating value, a lower O-content implies better substitution performance.

- Higher Heating Value (HHV) $\propto \frac{1}{\text{Oxygen content}}$
- Typical coal HHV: 25–30 MJ/kg (15–25% O)
- ML Hydrochar HHV: 22.12 MJ/kg (Reduced O)

Substitution Feasibility:

- Must demonstrate comparable energy quality.
- Deoxygenation rate validates this improvement.

Deoxygenation as Transformation Indicator

$$\text{DO (\%)} = 1 - \left(\frac{H_C}{B_0} \right) \times 100 \quad (4.2)$$

- High DO = efficient transformation to fuel-grade hydrochar
- Low DO = incomplete processing and lower substitution quality

Environmental Implications and Benefits**GHG Emissions Reduction:**

- Better combustion = fewer pollutants
- More credible fossil coal substitution
- Reduced GWP of overall process

Validation via DO:

- Low DO → Poor quality hydrochar → **Questionable credit**
- High DO → Quality hydrochar → **Justified credit**

Potential for Carbon Negativity:

- Hydrochar displaces high-carbon coal
- Formic acid avoids energy-intensive production

4.2.5 Application in This Study

- **Preferred:** system expansion for formic acid (chemical substitution) and hydrochar (coal substitution).
- **Fallback:** economic allocation where substitution cannot be justified.
- **Sensitivity:** test price and substitution factor variation on results.

4.2.6 Reporting and Transparency

For each scenario, report:

- Allocation method and rationale
- Co-product quantities and properties (e.g., LHV)
- Data sources and dates
- Substitution assumptions and sensitivity analysis

4.2.7 Impact and Significance

LA synthesis has the highest environmental burden (24.29–28.52 kg CO₂-eq/kg LA). System expansion credits can:

- **Reduce this impact by 20–50%**
- **Make bio-based LA competitive with petrochemical LA**
- **Justify the process economically and environmentally**

4.2.8 Impact Categories and Assessment Methods

Environmental impacts are quantified using a multi-indicator, midpoint-focused framework selected for (i) availability of peer-reviewed characterization factors, (ii) relevance to bio-based and petrochemical routes, and (iii) consistency with North American practice.

TRACI 2.1 — Primary Midpoint categories

Climate change (GWP₁₀₀): kg CO₂-eq; characterization based on IPCC AR6 100-year factors (e.g., CO₂=1, CH₄=27.2, N₂O=273). Biogenic CO₂ tracked separately per ISO 14067.

Resource use: Fossil energy demand (MJ, cumulative); water use (m³, consumptive use; regionalized scarcity where applicable). Includes direct and upstream requirements.

Environmental quality: Photochemical smog (kg O₃-eq), acidification (kg SO₂-eq), eutrophication (kg N-eq / kg PO₄³⁻-eq by limitation), ozone depletion (kg CFC-11-eq).

Methodological Integration

TRACI 2.1 is complemented with ReCiPe 2016 to extend coverage where region-independent mechanisms are informative.

ReCiPe 2016 — Complementary Indicators

Additional midpoints: Land use (m² yr crop-eq), ecotoxicity (CTU_e), human toxicity (CTU_h), mineral resource scarcity (kg Cu-eq).

Endpoints (reported only for sensitivity/communication): Human health (DALY), ecosystems (species·yr), resources (USD). Normalization/weighting used only in sensitivity analyses; base results are unweighted midpoints.

Characterization Data and Uncertainty

Characterization factors follow consensus sources (TRACI 2.1; IPCC AR6 for GWP₁₀₀). Regionalized factors are used for water (watershed-specific) and land use (ecoregion). Representative uncertainty ranges applied in Monte Carlo: GWP (±15%), acidification (±25%), eutrophication (±40%), toxicity (order-of-magnitude variability). Alternative methods (CML, IMPACT World+) are tested in sensitivity to method choice.

4.2.9 Geographical and Temporal Scope**Geographical Scope**

Midwest USA (e.g., Decatur, IL): Bio-based route; proximity to agricultural residues; existing biorefinery assets. Grid mix approximated by recent regional portfolios.

US Gulf Coast (TX/LA): Petrochemical route; integrated refining/chemicals cluster

with established utilities and hydrogen networks. Grid and fuels reflect industrial supply in the region.

Rationale: Locations represent typical deployment contexts for North American bio-refining and petrochemicals; results are not site-specific unless stated.

Regional Representativeness and Data

Foreground inventories prioritize facility-level reporting (EPA TRI/GHGRP) and verified industrial data when available; background data followecoinvent with North American regionalization. Logistics assume economically viable collection distances for residues (order-of-tens of km) and existing pipeline/terminal infrastructure for petrochemical feedstocks. Electricity mixes reflect 2022–2024 averages; planned changes are explored in sensitivity only.

Temporal Scope

Assessment period: 2023–2025.

Technology maturity: Bio-based routes modeled at TRL 7–8 with scale-up by standard engineering correlations; petrochemical routes at TRL 9.

Data vintage: Primary ≤ 3 years; secondary ≤ 5 years.

Context: Current U.S. regulatory baseline (e.g., Clean Air Act; RFS2). Market parameters use 2022–2024 averages; forward-looking scenarios are separated from base results.

Sensitivity to Space and Time

Geography: Alternative regions (hydropower-dominant Northwest; Southeast biomass; California) shift climate impacts by ~ 15 – 25% primarily via grid intensity.

Time: 2030–2035 scenarios consider incremental efficiency (5–10%), grid decarbonization, and feedstock improvements; results are reported separately with explicit uncertainty.

Infrastructure: Scenario levers include carbon pricing, renewable electricity mandates, and co-location benefits (utility sharing, reduced transport).

CHAPTER 5

Phase 2: Life Cycle Inventory (LCI)

5.1 Bio-based System overview

Lignocellulosic feedstocks (e.g., corn stover, EFB, bagasse; molasses as a non-lignocellulosic comparator) undergo size reduction, acid-catalyzed dehydration and rehydration to LA and formic acid, followed by solid-liquid separation, solvent extraction (e.g., MIBK), and purification. Key operating envelopes include 150–220°C, 10–25 bar, and acid levels around 1–10% by weight during hydrolysis. Typical co-products include formic acid and lignin-rich solids.

5.2 Petrochemical System overview

A representative sequence is: *n*-butane → maleic anhydride (VPO oxidation < 500°C), hydrogenation to succinic anhydride (Ni/Pd; 120–180°C, 0.5–4.0 MPa), preparation of methylmagnesium bromide in THF, addition to succinic anhydride, acidic hydrolysis, and LA purification (e.g., vacuum distillation/crystallization). Notable contributors include H_2 make-up (SMR), THF make-up/losses, and Mg/Br salt waste from work-up.

5.3 LCI Development Methodology

The Life Cycle Inventory (LCI) development follows a systematic approach combining primary data from process simulations, secondary data from established databases (ecoinvent v3.8), and literature-derived parameters. Data quality assessment prioritizes geographical representativeness (North America/Europe), temporal validity (2020–2025), and technological representativeness for commercial-scale operations. Mass and energy balance calculations ensure thermodynamic consistency across all unit operations.

5.4 Data Sources and Quality Assessment

Primary data sources include Aspen Plus simulations for process modeling, validated against pilot-scale studies and commercial demonstrations where available. Secondary

data from ecoinvent v3.8 provides background system inputs (electricity, chemicals, transportation). Literature sources supplement gaps in emerging technologies, with preference given to peer-reviewed publications from 2018-2025. Data quality indicators follow the ecoinvent methodology, assessing reliability, completeness, temporal correlation, geographical correlation, and further technological correlation.

5.5 Inventory Completeness and System Boundaries

The inventory encompasses all material and energy flows within the defined system boundaries, from feedstock acquisition through LA production and downstream conversion. Cut-off criteria eliminate flows contributing less than 1% to total mass or energy, except for flows with high environmental relevance (heavy metals, persistent organic pollutants). Infrastructure is excluded based on previous studies demonstrating negligible contribution to bio-based chemical production impacts.

5.6 System Overview (Bio-based Pathways)

5.7 Process Configuration and Operating Envelopes

Bio-based LA production employs a multi-stage process configuration integrating feedstock preparation, acid-catalyzed conversion, and product purification. Five distinct lignocellulosic feedstock routes are evaluated: corn stover (CS), corn cob (CC), corn leaf (CL), sugarcane bagasse (BG), and molasses (ML) as a reference non-lignocellulosic feedstock.

Overall Process Mass Balance (per kg LA)

Inputs:

- Dry biomass: 4.72-5.56 kg (feedstock-dependent)
- H₂SO₄ (98%): 5.03-7.00 kg
- Process water: 47.2-55.6 kg
- Electricity: 38.3-45.1 kWh

Outputs:

- Levulinic acid (98.5%): 1.00 kg
- Formic acid (85%): 0.40-0.56 kg

- Hydrochar: 1.58-2.19 kg
- Wastewater: 41.2-49.8 kg

Mass Balance Closure: >95%

The conversion pathway utilizes the established Biofine process technology, operating under the following conditions:

- **Stage 1 (PFR):** 200-230°C, 20-25 bar, 12-second residence time
- **Stage 2 (CSTR):** 190-220°C, 10-15 bar, 15-30 minute residence time
- **Acid concentration:** 1-4% H₂SO₄ by weight
- **Conversion severity factor (CSF):** 2.2-4.0 depending on stage

Key design parameters include cellulose hydrolysis efficiency (75% in Stage 1), hemicellulose conversion (99%), and HMF-to-LA conversion (85% in Stage 2). The process generates significant co-products including formic acid (FA), furfural, and hydrochar (HC) with potential valorization pathways.

5.8 Feedstock Characterization and Preparation

Feedstock composition varies significantly across routes, affecting downstream processing requirements and yields:

Table 5.1: Feedstock composition and processing requirements (per kg LA production)

Parameter	CS	CC	CL	BG	ML
Dry biomass req. (kg)	5.045	5.519	5.556	5.390	4.720
Moisture content (%)	15.0	12.0	17.0	16.0	20.0
LA yield (% dry wt)	20.02	18.30	18.18	18.74	21.40
FA yield (% dry wt)	9.85	10.20	9.03	10.35	8.47
HC yield (% dry wt)	33.58	39.59	36.00	36.99	39.65
Electricity (kWh)	41.17	45.11	45.04	43.79	38.30

Feedstock preparation involves size reduction to 0.5-1.0 cm particles, followed by drying to achieve 10-15% moisture content. Energy requirements for preparation vary from 6.29 MJ/kg LA (molasses) to 31.63 MJ/kg LA (corn stover), reflecting differences in initial moisture content and processing complexity.

Feedstock Preparation Energy Balance (per kg LA)

Energy Inputs:

- Molasses: 6.29 MJ (lowest due to pre-processed nature)
- Corn stalk: 31.63 MJ (highest moisture removal requirement)
- Corn cob: 2.53 MJ (naturally lower moisture)
- Corn leaf: 8.97 MJ (moderate drying needs)
- Bagasse: 11.67 MJ (mill residue advantage)

Electricity Requirements:

- Pulverization: 0.20-0.23 kWh across all routes
- Drying operations: 0.79-9.88 MJ thermal equivalent

5.9 System Overview (Petrochemical Pathway)

5.10 n-Butane to LA Conversion Sequence

The petrochemical pathway follows a systematic four-step conversion: n-butane → maleic anhydride → succinic anhydride → LA through Grignard chemistry. This route represents the most economically viable fossil-based LA production technology currently available.

Process stages include:

1. **Maleic Anhydride Synthesis:** Vanadium phosphorus oxide (VPO) catalyzed oxidation at 400-450°C
2. **Hydrogenation:** Pd/Ni catalyzed reduction at 120-180°C, 0.5-4.0 MPa
3. **Grignard Addition:** Methylmagnesium bromide addition in THF solvent
4. **Hydrolysis and Purification:** Acidic work-up followed by vacuum distillation

5.11 Key Process Parameters and Material Requirements

The petrochemical route demonstrates high atom efficiency but requires significant auxiliary materials:

Table 5.2: Petrochemical route material requirements (per kg LA)

Input	Amount	Unit	Function
n-Butane	1.098	kg	Primary feedstock
Hydrogen (SMR)	0.0228	kg	Hydrogenation step
Methyl bromide	1.136	kg	Grignard reagent precursor
Magnesium metal	0.291	kg	Grignard reagent
THF (makeup)	0.240	kg	Solvent losses
Process water	0.194	kg	Quench and work-up

5.12 Detailed Process Inventories

5.13 Bio-based LA Production Chain

5.13.1 Upstream Cultivation and Nutrient Management

Agricultural inputs vary significantly across feedstock routes, with molasses requiring intensive sugarcane cultivation while residue-based feedstocks necessitate nutrient replacement to maintain soil carbon balance:

Table 5.3: Agricultural inputs and emissions (per kg LA)

Flow	CS	CC	CL	BG	ML
N fertilizer (kg N)	4.44	57.28	40.30	68.42	0.39
P ₂ O ₅ (kg)	1.64	21.14	14.87	25.25	0.19
K ₂ O (kg)	1.08	13.98	9.84	16.70	0.38
Diesel (L)	0.021	0.021	0.021	0.021	0.075
Direct N ₂ O (kg)	0.00017	0.00017	0.00017	0.00017	0.0037
NH ₃ + NO _x (kg N)	0.0011	0.0011	0.0011	0.0011	0.023

The molasses route demonstrates significantly lower agricultural burdens due to mass allocation with sugar production, while residue-based routes require substantial nutrient replacement to prevent soil depletion.

5.13.2 Pretreatment and Conditioning

Pretreatment operations prepare lignocellulosic biomass for efficient acid hydrolysis through size reduction and chemical conditioning:

Table 5.4: Pretreatment inventory (per kg LA production)

Flow	Amount	Unit	Route Variation
Inputs			
Dried biomass	5.0-5.6	kg	Route-specific (Table 5.1)
Sulfuric acid (98%)	6.4-7.0	g	Higher for corn leaf route
Process water	101-111	g	Proportional to biomass input
Electricity	Variable	kWh	To be determined by route
Outputs			
Pretreated slurry	112-124	g	Mass balance closure
Process emissions	Variable	kg CO ₂ eq	From electricity consumption

5.13.3 Biofine Conversion Process

The core conversion process employs the established two-stage Biofine technology with detailed mass and energy balances:

Stage 1 PFR Mass & Energy Balance (per kg LA)

Operating Conditions: 215°C, 26 bar, 12 seconds residence time **Inputs:**

- Pretreated slurry: 112-124 kg
- H₂SO₄ catalyst: 1.0% w/w
- Steam: Variable (temperature maintenance)

Conversion Efficiency:

- Cellulose hydrolysis: 75%
- Hemicellulose hydrolysis: 99%
- Hexoses → HMF: 95%

- Pentoses → Furfural: 95%

Energy Balance: Heat of reaction + sensible heating = 0.85-1.2 MJ/kg LA

Stage 2 CSTR Mass & Energy Balance (per kg LA)

Operating Conditions: 195°C, 14 bar, 20 minutes residence time **Key Reactions:**

- HMF hydrolysis → LA: 85% conversion
- Furfural partial degradation → FA: 50%
- Side reactions → humins: <5%

Product Concentrations:

- LA concentration: 7.2 wt%
- FA concentration: 3.5 wt%
- Furfural: 2.3 wt%

Heat Duty: 1.8-2.3 MJ/kg LA (includes heat integration)

Table 5.5: Biofine conversion inventory (per kg LA)

Flow	Amount	Unit	Notes
Stage 1 - PFR (215°C, 26 bar, 12 s)			
Pretreated slurry	Variable	g	Route-dependent input
H ₂ SO ₄ catalyst	1.0	% w/w	Acid concentration
Steam injection	Variable	kg	Temperature maintenance
Stage 2 - CSTR (195°C, 14 bar, 20 min)			
Intermediate slurry	Variable	g	From Stage 1
Residence time	20	minutes	HMF hydrolysis
Primary Outputs			
Converted slurry	57-66	kg	Route-dependent
LA concentration	7.2	wt%	In liquid phase
FA concentration	3.5	wt%	Co-product
Furfural conc.	2.3	wt%	Valuable co-product

5.13.4 Solid-Liquid Separation and Co-product Recovery

Downstream processing separates valuable products and recovers co-products for system credits:

Solid-Liquid Separation Mass Balance (per kg LA)

Operating Conditions: Screw press separation at 40.6 MT/h throughput **Inputs:**

- Converted slurry: 57-66 kg (route-dependent)
- Electricity: 3.4-4.0 kWh (6% of slurry mass basis)

Separation Efficiency:

- Solid recovery (hydrochar): 1.6-2.2 kg
- Liquid recovery: 55-64 kg clarified liquor
- Moisture in solids: <15% (optimal for combustion)

Hydrochar Properties:

- Heating value: 18-22 MJ/kg
- Carbon content: 55-65%
- Ash content: <8%

Liquid-Liquid Extraction Energy Balance (per kg LA)

MIBK Route:

- Solvent requirement: 12.8-13.4 kg
- Electricity consumption: 4.76 kWh
- LA extraction efficiency: 95%
- Solvent recovery: 95% (thermal)

Furfural Route (Alternative):

- Solvent requirement: 15.6-16.4 kg
- Electricity consumption: 2.74 kWh (42% lower)
- LA extraction efficiency: 93%
- Solvent recovery: 92.3%

Energy Comparison: Furfural route reduces extraction energy by 42%

Table 5.6: Separation and recovery inventory (per kg LA)

Flow	Amount	Unit	Value Proposition
Solid-Liquid Separation			
Electricity consumption	3.4-4.0	kWh	6% of slurry mass
Hydrochar recovery	1.6-2.2	kg	Coal substitute credit
Clarified liquor	55-64	kg	To extraction
Liquid-Liquid Extraction (MIBK/Furfural)			
MIBK solvent	12.8-13.4	kg	95% recovery assumed
Furfural solvent	15.6-16.4	kg	Alternative extraction
Electricity (MIBK)	4.76	kWh	Extraction operations
Electricity (Furfural)	2.74	kWh	Lower energy alternative
Co-product Recovery			
Formic acid (85% purity)	0.40-0.56	kg	Process credit
Recovered furfural	15.6-16.4	kg	Solvent recovery

5.13.5 Product Purification and Solvent Recovery

Final purification achieves 98.5% LA purity through vacuum distillation with integrated solvent recovery:

LA Purification Mass & Energy Balance (per kg LA)

Vacuum Distillation (C-101/C-102):

- Feed: 13.8-17.4 kg LA concentrate
- Operating pressure: 80 mmHg (vacuum)
- Bottom temperature: 161°C
- Electricity: 3.34 kWh (constant across routes)

Product Quality:

- LA purity: 98.5% (specification)
- Product yield: 1.00 kg
- Impurity removal: >99% water, organic acids

Energy Integration:

- Heat recovery: 2.51-31.16% savings potential
- Annual cost savings: \$1.50M/year (commercial scale)

Co-product Purification Energy Balance**Formic Acid Separation (C-103 @ 2 bar, C-104 @ 0.27 bar):**

- Feed stream: 0.403 kg FA equivalent
- Recovery efficiency: 85% purity
- Energy requirement: 1.2-1.8 MJ/kg FA
- Annual production: 13,000 MT/year

Furfural Recovery (Liquid-Liquid + Distillation):

- Input concentration: 2.6 wt% in aqueous phase
- Organic phase purity: 92.3% furfural
- Recovery efficiency: >90%
- Annual production: 8,400 MT/year

Table 5.7: *Purification and solvent recovery (per kg LA)*

Process Stage	Amount	Unit	Recovery Efficiency
LA Purification (Vacuum Distillation)			
LA concentrate input	13.8-17.4	kg	Route-dependent
Electricity consumption	3.34	kWh	Constant across routes
Purified LA output	1.0	kg	98.5% purity target
Solvent Recovery Operations			
MIBK recovery	12.8-13.5	kg	95% recovery rate
Furfural recovery	15.6-16.4	kg	92.3% purity achieved
Energy requirement	2.51-31.16	%	Heat integration savings
Co-product Purification			
FA separation energy	Variable	MJ	C-103/C-104 columns
Furfural separation	Variable	MJ	Liquid-liquid + distillation
Annual cost savings	1.50	M\$/year	Heat integration benefit

5.14 Petrochemical LA Production Inventory

The petrochemical pathway inventory encompasses upstream n-butane processing through LA purification:

Table 5.8: Complete petrochemical pathway inventory (per kg LA)

Input Flow	Amount	Unit	Output Flow
Upstream H₂ Production (SMR)			
Natural gas (CH ₄)	0.08	kg	Hydrogen to process
Steam	0.043	kg	CO ₂ emissions
Process heat	1.06	kWh _{th}	-
Electricity	0.018	kWh	-
Maleic Anhydride Synthesis (VPO)			
n-Butane	1.098	kg	Maleic anhydride
Air (O ₂ /N ₂)	9.08	kg	Water vapor
-	-	-	Off-gas (N ₂ , CO/CO ₂)
Grignard Chemistry			
Methyl bromide	1.136	kg	MeMgBr solution
Magnesium metal	0.291	kg	THF losses
THF makeup	4.79	kg	MgBrOH brine waste
Succinic anhydride	1.077	kg	Crude LA
Purification			
Process steam	0.76	kg	Purified LA (1.0 kg)

5.15 Transportation and Distribution

Transportation inventories account for feedstock collection, processing facility delivery, and product distribution. Distance assumptions reflect regional biomass availability and industrial infrastructure:

Table 5.9: Transportation inventory (per kg LA)

Route	Truck (MJ)	Tractor (MJ)	CO ₂ (kg)	Distance (km)
Molasses	8.58	6.87	1.066	150
Corn stalk	2.70	2.16	0.335	50
Corn cob	2.09	1.67	0.260	40
Corn leaf	2.97	2.38	0.369	55
Bagasse	1.75	1.40	0.217	25

5.16 Downstream GVL Biofuel Production

5.17 Esterification Process Inventory

LA conversion to alkyl levulinates proceeds through acid-catalyzed esterification with ethanol:

Esterification Mass & Energy Balance (per kg LA input)

Reaction Stoichiometry: $\text{LA} + \text{EtOH} \rightleftharpoons \text{Ethyl Levulinate} + \text{H}_2\text{O}$ **Operating Conditions:** 443 K (170°C), atmospheric pressure, 3.75 h **Mass Balance:**

- Input: 1.000 kg LA + 10.2 kg EtOH (excess)
- Output: 1.2 kg ethyl levulinate (96.5% yield)
- Byproduct: 0.15 kg water (stoichiometric)
- Ethanol recovery: 0.014 kg unreacted (3.5%)

Energy Balance:

- Heating duty: 0.56 MJ (298→443 K)
- Electrical power: 0.27 MJ (mixing, 20 W × 3.75 h)
- Heat of reaction: -42.3 kJ/mol (exothermic)
- Cooling duty: 0.31 MJ (product recovery)

Catalyst Loading: 50 mg K10 per mmol LA (95% recoverable)

Table 5.10: Esterification inventory (per kg LA input)

Flow	Amount	Unit	Process Role
Inputs			
Levulinic acid	1.000	kg	Primary reactant
Ethanol	10.2	kg	1.5 mL·mmol ⁻¹ excess
Montmorillonite K10	0.431	kg	Solid acid catalyst
Nitrogen (inert)	14	kg	Reaction atmosphere
Thermal energy	0.56	MJ	Heating to 443 K
Electricity	0.27	MJ	Mixing and controls
Outputs			
Ethyl levulinate	1.2	kg	96.5% yield
Reaction water	0.15	kg	Stoichiometric byproduct
Ethanol (unreacted)	0.014	kg	3.5% for recycle
Catalyst waste	0.039	kg	9% loss after 4 cycles
COD wastewater	0.260	kg	2 g·L ⁻¹ loading

5.18 Hydrogenation to GVL

GVL production employs heterogeneous catalysis under moderate conditions:

Hydrogenation Mass & Energy Balance (per 1.2 kg ethyl levulinate)

Reaction Chemistry: Ethyl Levulinate + H₂ → GVL + Ethanol **Operating Conditions:** 285°C, 50-70 bar H₂, Cu-Ni/Al₂O₃ catalyst **Mass Balance:**

- Ethyl levulinate: 1.2 kg (from esterification)
- Hydrogen gas: 0.0168 kg (stoichiometric + 10% excess)
- Catalyst loading: 0.120 kg (heterogeneous system)

Product Distribution:

- GVL (primary): 0.862 kg (91.9% selectivity)
- Ethyl 4-hydroxypentanoate: 0.073 kg (6% byproduct)
- Ethanol recovered: 0.193 kg (solvent + reaction product)
- Unconverted substrate: 0.0252 kg (2.1%)

Energy Balance:

- Electrical power: 263.5 MJ (compression, heating, mixing)
- Reaction heat: -95.4 kJ/mol (exothermic, heat recovery potential)
- Net energy requirement: 215 MJ/kg GVL

Catalyst Performance: TOF = 598 h⁻¹ at 120°C, 96% selectivity

Table 5.11: *Hydrogenation to GVL inventory*

Flow	Amount	Unit	Performance Metric
Inputs			
Ethyl levulinate	1.2	kg	From esterification
Hydrogen gas	0.0168	kg	Stoichiometric requirement
Cu-Ni/Al ₂ O ₃ catalyst	0.120	kg	Heterogeneous system
Electricity	263.5	MJ	Process operations
Outputs			
GVL (γ -valerolactone)	0.862	kg	91.9% selectivity
Ethyl 4-hydroxypentanoate	0.073	kg	6% byproduct
Ethanol recovered	0.193	kg	Solvent recovery
Unconverted substrate	0.0252	kg	2.1% unreacted

5.19 Co-product Credit Assessment

5.20 Hydrochar Valorization

Hydrochar co-product provides significant system credits through electricity generation via combustion:

Hydrochar Combustion Mass & Energy Balance (per kg LA production)

Hydrochar Properties & Yields:

- Bagasse: 1.98 kg HC (36.99 wt%), HHV = 21.2 MJ/kg
- Corn cob: 2.18 kg HC (39.59 wt%), HHV = 22.8 MJ/kg
- Corn stalk: 1.69 kg HC (33.58 wt%), HHV = 20.1 MJ/kg
- Corn leaf: 2.00 kg HC (36.00 wt%), HHV = 19.8 MJ/kg

- Molasses: 1.87 kg HC (39.65 wt%), HHV = 21.0 MJ/kg

Electricity Generation (Combustion + Steam Cycle):

- Bagasse route: 976.54 kWh/1000 kg feedstock
- Corn cob route: 1080.81 kWh/1000 kg feedstock (highest)
- Corn stalk route: 926.81 kWh/1000 kg feedstock
- Corn leaf route: 903.60 kWh/1000 kg feedstock
- Molasses route: 975.39 kWh/1000 kg feedstock

System Credit: Displaces grid electricity (regional emission factors) **Combustion Efficiency:** 85% (modern biomass power plants)

Table 5.12: *Hydrochar electricity generation credits (per kg LA)*

Feedstock	HC Yield (%)	Electricity (kWh)	Credit Value
Bagasse	36.99	976.54	Significant
Corn cob	39.59	1080.81	Highest
Corn stalk	33.58	926.81	Moderate
Corn leaf	36.00	903.60	Moderate
Molasses	39.65	975.39	Significant

5.21 Chemical Co-product Recovery

Formic acid recovery provides additional system credits with established market applications:

- **Formic acid recovery:** 0.40-0.56 kg per kg LA at 85% purity
- **Market applications:** Leather processing, animal feed preservation, organic acid markets
- **Economic credit:** \$0.29-0.41 per kg LA production
- **Environmental credit:** Displaced synthetic formic acid production

5.22 Data Quality and Uncertainty Considerations

5.22.1 Primary Data Sources

Process simulation data from Aspen Plus provides high-quality mass and energy balances with thermodynamic consistency. Pilot-scale validation data from Biofine demonstrations (Caserta, Italy) supports commercial-scale projections. Experimental data for downstream conversion (esterification, hydrogenation) derives from laboratory studies scaled using established engineering correlations.

5.22.2 Secondary Data Integration

Background system data from ecoinvent v3.8 represents European/North American conditions where applicable. Electricity grid mixes reflect regional generation portfolios. Transportation data incorporates recent heavy-duty vehicle emission factors and fuel consumption patterns.

5.22.3 Process Assumptions and Parameter Values

This section details the key assumptions, parameter values, and data sources used throughout the LCA modeling. All assumptions are based on literature data, industrial practice, or engineering correlations appropriate for commercial-scale operations.

5.22.4 Feedstock Transport and Logistics Assumptions

Transportation requirements vary significantly across feedstock routes, reflecting regional availability and industrial infrastructure [1].

Table 5.13: Feedstock transport assumptions (per kg LA production)

Route	Distance (km)	Feedstock (kg)	Energy (MJ)	Rationale
Molasses (A)	25	4.673	0.106	Sugar factories ≤ 25 km from cane farms
Corn stalks (B)	60	4.995	0.272	Delta region concentration, realistic catchment
Banana EFB (C)	30	5.000	0.137	Regional haul to processing plant

Transport Energy Calculation: 0.91 MJ/t·km based on heavy goods vehicle (HGV) factors [DEFRA2024]. CO₂ emissions calculated using 0.10 kg CO₂eq/t·km emission factors.

5.22.5 Process Energy and Material Balance Assumptions

Table 5.14: Key process assumptions for bio-based LA production

Parameter	Value	Unit	Basis/Reference
Feedstock Preparation			
Size reduction energy	20	kWh/t	NREL industry mid-range (12-30 kWh/t)
Drying energy	3.0	MJ/kg H ₂ O	Typical biomass drying (2.6-3.6 MJ/kg)
Pretreatment (Stage 1 PFR)			
High-pressure steam	2.5-3.0	MJ	Heat slurry 25→215°C + 20% losses
Acid catalyst	1.0	wt%	H ₂ SO ₄ of slurry (0.5-2% typical)
Separation and Recovery			
Solids separation	2	kWh/m ³	Decanter/centrifuge energy
FA separation energy	65	% of total	Dominates purification energy
Solvent recovery	95	%	LLE circulation efficiency
Heat integration savings	20	%	Pinch analysis potential

5.22.6 Downstream Conversion Assumptions

Table 5.15: Esterification and hydrogenation process assumptions

Parameter	Value	Unit	Basis
Esterification (LA + ROH)			
Acid catalyst loading	1.0	wt%	Typical for LA↔alkyl levulinate
Ethyl levulinate yield	95	%	From 1.242 kg theoretical
Reaction temperature	95-170	°C	Alcohol-dependent optimum
Hydrogenation to GVL			
Metal catalyst makeup	0.5	wt%	Ru/C with 100 re-uses
H ₂ route GVL yield	90-95	%	Continuous system performance
FA route GVL yield	95-99	%	Transfer hydrogenation efficiency
Operating pressure	50-70	bar	H ₂ pressure for hydrogenation

These uncertainties are systematically addressed through Monte Carlo analysis in subsequent sections, providing probabilistic assessment of environmental performance under realistic operating conditions.

5.23 Data Quality Assessment

Data quality assessment employs the Weidema pedigree matrix methodology to systematically evaluate inventory data reliability, completeness, and representativeness [37]. This structured approach quantifies uncertainty associated with each data source and process, enabling transparent communication of data limitations and supporting robust uncertainty propagation in subsequent impact assessment phases.

5.23.1 Pedigree Matrix Methodology

The pedigree matrix evaluates five critical data quality indicators using a standardized 1-5 scoring system, where 1 represents "very good" quality and 5 indicates "very

poor" quality [38]. This semi-quantitative framework provides consistent quality evaluation across diverse data sources while maintaining compatibility with Monte Carlo uncertainty analysis.

Table 5.16: *Weidema pedigree matrix scoring criteria for data quality indicators*

Indicator	Score	Quality Level Description
Reliability	1	Verified data based on measurements
	2	Verified data partly based on assumptions
	3	Non-verified data partly based on assumptions
	4	Qualified estimate (e.g., by industrial expert)
	5	Non-qualified estimate
Completeness	1	Representative data from sufficient samples
	2	Representative data from smaller number of sites
	3	Representative data from adequate sample of sites
	4	Representative data from small number of sites
	5	Representativeness unknown or incomplete
Temporal Correlation	1	Less than 3 years difference
	2	Less than 6 years difference
	3	Less than 10 years difference
	4	Less than 15 years difference
	5	Age unknown or more than 15 years
Geographical Correlation	1	Data from study area
	2	Average data from larger area
	3	Data from smaller area within study region
	4	Data from different area, similar conditions
	5	Data from different area, different conditions
Technological Correlation	1	Data from enterprises, processes and materials studied
	2	Data from processes and materials studied (market mix)
	3	Data from processes and materials studied (literature)
	4	Data on related processes or materials
	5	Data on related processes on laboratory scale

5.23.1.1 Bio-based Pathway Data Quality Assessment

Bio-based LA production inventory data derives from multiple sources including Aspen Plus simulations, pilot-scale demonstrations, and literature studies, resulting in variable data quality across process stages.

Table 5.17: Data quality assessment for bio-based LA production pathway

Process Stage	Rel.	Comp.	Temp.	Geo.	Tech.	Mean DQ
Feedstock cultivation	3	2	2	3	2	2.4
Feedstock transport	2	3	2	2	1	2.0
Size reduction	2	2	2	3	2	2.2
Drying operations	3	2	2	3	2	2.4
Acid pretreatment	3	2	3	3	3	2.8
Biofine Stage 1 (PFR)	2	2	2	3	2	2.2
Biofine Stage 2 (CSTR)	2	2	2	3	2	2.2
Solid-liquid separation	2	2	2	3	2	2.2
Liquid-liquid extraction	3	2	2	3	3	2.6
LA purification	2	2	2	3	2	2.2
Formic acid recovery	3	3	3	3	3	3.0
Furfural recovery	3	3	3	3	3	3.0
Hydrochar combustion	3	2	2	3	3	2.6
Overall Bio-based	2.5	2.2	2.2	2.9	2.3	2.4

Bio-based Pathway Data Quality Interpretation

Strengths:

- **Completeness (2.2):** >95% mass and energy flows accounted for across all unit operations
- **Temporal Correlation (2.2):** Data window 2018-2024 representative of current commercial technology
- **Technological Correlation (2.3):** Based on demonstrated Biofine technology with pilot-scale validation

Limitations:

- **Geographical Correlation (2.9):** Primary data from European/North American sources with limited regional adaptation
- **Reliability (2.5):** Literature averages with $\pm 15\text{-}20\%$ uncertainty for major flows; not fully verified across industrial sites
- **Co-product Recovery (3.0):** Limited industrial data for formic acid and furfural separation at commercial scale

Overall Assessment: Mean DQ score of 2.4 indicates "Good" overall quality with main limitation being geographical representativeness

Process-Specific Data Quality Justification

Biofine Conversion (Stages 1 & 2): Reliability score of 2 reflects verified data from pilot-scale demonstrations (Caserta, Italy) with peer-reviewed mass balances, though full commercial-scale verification remains limited [1]. Completeness score of 2 indicates representative data from multiple pilot studies with consistent yield reporting.

Feedstock Operations: Agricultural cultivation data receives lower geographical correlation (3) due to reliance on North American/European sources adapted to study region conditions. Temporal correlation (2) reflects recent agricultural practice data (2018-2024) with documented management practices.

Separation and Purification: Liquid-liquid extraction receives reliability score of 3 due to process simulation basis rather than measured industrial data. Technology correlation of 3 reflects standard separation unit operations with well-established engineering correlations.

Co-product Recovery: Formic acid and furfural recovery operations receive lowest scores (3.0 mean) reflecting limited commercial-scale operating data and reliance on laboratory-scale studies with scale-up assumptions.

5.23.1.2 Petrochemical Pathway Data Quality Assessment

Petrochemical LA production benefits from mature industrial technology with extensive commercial operating history, resulting in generally higher data quality compared to bio-based routes.

Table 5.18: Data quality assessment for petrochemical LA production pathway

Process Stage	Rel.	Comp.	Temp.	Geo.	Tech.	Mean DQ
H ₂ production (SMR)	1	1	1	2	1	1.2
n-Butane extraction	2	2	2	2	1	1.8
n-Butane transport	2	2	2	2	1	1.8
MA synthesis (VPO)	2	1	2	2	1	1.6
SA hydrogenation	2	2	2	2	1	1.8
MeMgBr preparation	2	2	2	3	2	2.2
Grignard addition	2	2	2	3	2	2.2
LA purification	2	2	2	3	2	2.2
Overall Petrochemical	1.9	1.8	1.9	2.4	1.4	1.9

Petrochemical Pathway Data Quality Interpretation

Strengths:

- **Technological Correlation (1.4):** Mature commercial technology with >30 years operating history
- **Completeness (1.8):** Verified industrial data from commercial facilities with comprehensive mass/energy balances
- **Reliability (1.9):** Based on measured industrial performance data with peer-reviewed validation

Limitations:

- **Geographical Correlation (2.4):** Data primarily from Gulf Coast USA petrochemical complexes
- **Grignard Chemistry (2.2):** Fine chemical synthesis steps with limited large-scale commercial precedent

Overall Assessment: Mean DQ score of 1.9 indicates "Very Good" overall quality reflecting mature industrial technology

Comparative Data Quality Analysis

The petrochemical pathway demonstrates superior data quality (1.9 vs 2.4 mean score) primarily due to technological maturity and extensive commercial operating history. Upstream operations (H_2 production, n-butane processing, VPO oxidation) benefit from standardized industrial practice with verified performance data. The bio-based pathway's lower scores reflect emerging technology status with limited full-scale commercial validation.

5.23.1.3 Downstream GVL Production Data Quality

Table 5.19: Data quality assessment for downstream GVL production

Process Stage	Rel.	Comp.	Temp.	Geo.	Tech.	Mean DQ
Esterification	2	2	2	3	2	2.2
Hydrogenation (H_2)	3	2	2	3	3	2.6
Hydrogenation (FA)	3	3	2	3	4	3.0
Product separation	2	2	2	3	2	2.2
Overall Downstream	2.5	2.3	2.0	3.0	2.8	2.5

Transfer Hydrogenation Limitation: FA-based transfer hydrogenation receives lowest quality score (3.0) due to limited industrial demonstration data and reliance on laboratory-scale catalyst performance extrapolation. Technology correlation score of 4 reflects emerging catalyst systems without commercial precedent.

5.23.2 Uncertainty Factors and Pedigree Matrix Integration

Data quality scores are converted to lognormal uncertainty factors for Monte Carlo simulation using established pedigree-to-uncertainty conversion factors [38, 39]:

Table 5.20: Conversion of pedigree scores to uncertainty factors (geometric standard deviation)

Score	Reliability	Completeness	Temporal	Geographical	Technological
1	1.00	1.00	1.00	1.00	1.00
2	1.05	1.02	1.03	1.01	1.05
3	1.10	1.05	1.10	1.02	1.20
4	1.20	1.10	1.20	1.05	1.50
5	1.50	1.20	1.50	1.10	2.00

Combined Uncertainty Calculation: The overall uncertainty factor for each inventory flow combines individual indicator uncertainties through geometric multiplication:

$$U_{total} = \sqrt{U_{rel}^2 + U_{comp}^2 + U_{temp}^2 + U_{geo}^2 + U_{tech}^2 + U_{basic}^2} \quad (5.1)$$

where U_{basic} represents fundamental measurement uncertainty (typically 1.05 for direct measurements).

Representative Uncertainty Calculations

Bio-based LA Production (Overall):

$$\begin{aligned}
 U_{total} &= \sqrt{1.10^2 + 1.02^2 + 1.03^2 + 1.02^2 + 1.05^2 + 1.05^2} \\
 &= \sqrt{1.21 + 1.04 + 1.06 + 1.04 + 1.10 + 1.10} \\
 &= \sqrt{6.55} = 2.56 \text{ (GSD)}
 \end{aligned}$$

This corresponds to a 95% confidence interval of approximately 0.39-2.56 times the mean value, reflecting moderate overall uncertainty consistent with emerging technology status.

Petrochemical LA Production (Overall):

$$\begin{aligned}
 U_{total} &= \sqrt{1.05^2 + 1.00^2 + 1.05^2 + 1.01^2 + 1.00^2 + 1.05^2} \\
 &= \sqrt{1.10 + 1.00 + 1.10 + 1.02 + 1.00 + 1.10} \\
 &= \sqrt{6.32} = 2.51 \text{ (GSD)}
 \end{aligned}$$

Lower uncertainty reflects mature technology with verified industrial data, providing higher confidence in inventory values.

5.23.3 Data Quality Improvement Recommendations

Based on the systematic quality assessment, priority areas for data quality enhancement include:

1. **Geographical Representativeness:** Regional adaptation studies for agricultural practices, electricity grid mixes, and transportation infrastructure specific to deployment locations
2. **Co-product Recovery Systems:** Industrial demonstration data for formic acid and furfural separation at commercial scale with validated energy consumption
3. **Transfer Hydrogenation:** Pilot-scale validation of FA-based hydrogen donor systems with long-term catalyst stability assessment
4. **Scale-up Verification:** Full commercial-scale data from operational biorefinery facilities to replace simulation-based estimates
5. **Temporal Updates:** Periodic inventory revision incorporating latest process improvements and optimization studies

5.23.4 Inventory Validation and Quality Control

5.23.4.1 Mass Balance Validation

Mass balance closure exceeds 95% for all major unit operations, meeting ISO 14044 requirements for inventory completeness [36]. Systematic mass balance verification procedures include:

Table 5.21: Mass balance validation results by process section

Process Section	Input (kg)	Output (kg)	Closure (%)
Feedstock preparation	5.045	5.000	99.1
Biofine conversion	112.31	110.95	98.8
Solid-liquid separation	110.95	109.78	98.9
Liquid-liquid extraction	109.78	108.42	98.8
LA purification	13.78	13.78	100.0
Downstream conversion	1.200	1.181	98.4

Discrepancy Analysis: Minor mass balance gaps (1-2%) are attributed to volatile organic compound emissions during thermal processing, water vapor losses in drying operations, and sampling/analytical uncertainties. These losses are explicitly tracked and assigned to appropriate emission categories.

5.23.4.2 Energy Balance Validation

Energy balances incorporate heat integration opportunities identified through pinch analysis, achieving thermodynamic consistency across all unit operations [40]:

- **Heating Requirements:** 2.5-3.0 MJ/kg LA (pretreatment) + 1.8-2.3 MJ/kg LA (conversion)
- **Cooling Requirements:** 1.2-1.6 MJ/kg LA (product recovery)
- **Heat Recovery Potential:** 25-35% reduction through process integration
- **Net Energy Ratio:** 0.62-0.78 MJ_{out}/MJ_{in} (bio-based pathway including credits)

5.23.4.3 Stoichiometric Consistency Validation

All chemical transformations maintain stoichiometric consistency with established reaction mechanisms:

- **Cellulose → LA:** Maximum theoretical yield 76% (molar basis), model assumes 70-78% actual
- **LA → GVL:** Stoichiometric ratio 1.00:0.862 (mass basis), validated against literature
- **Esterification:** Water production matches stoichiometry (0.155 kg H₂O per kg LA)
- **Hydrogenation:** H₂ consumption consistent with reduction chemistry

5.23.4.4 Literature Benchmark Validation

Independent validation against published LCA studies confirms inventory reliability:

Table 5.22: Comparison with literature benchmark studies

Parameter	This Study	Literature Range	Reference
GWP (kg CO ₂ eq/kg LA)	24.3-28.5	24.3-32.1	[6]
Electricity (kWh/kg LA)	38.3-45.1	35-48	[1]
Water consumption (L)	47-56	42-60	[12]
LA yield (% dry biomass)	18.2-21.4	18-22	[4]

Validation Conclusion: All key inventory parameters fall within established literature ranges, confirming inventory reliability and consistency with independent assessments. Minor variations reflect specific process configurations and feedstock characteristics rather than fundamental data quality issues.

5.23.4.5 Quality Assurance Documentation

Complete quality assurance documentation includes:

- **Data Provenance:** Source documentation for all major inventory flows with quality indicators
- **Assumption Register:** Explicit documentation of all modeling assumptions and engineering correlations
- **Sensitivity Analysis:** Identification of parameters with high influence on results requiring enhanced data quality
- **Version Control:** Systematic tracking of inventory revisions and data updates
- **Peer Review:** Independent technical review of mass/energy balances and stoichiometric calculations

The comprehensive inventory, validated through multiple independent methods and assessed using standardized data quality frameworks, forms a robust foundation for subsequent impact assessment. The systematic quality evaluation provides transparent uncertainty quantification supporting evidence-based decision-making while maintaining reproducibility consistent with LCA best practices and ISO 14044 standards.

CHAPTER 6

Phase 3: OpenLCA Modeling

6.1 OpenLCA Modeling Environment

All modeling and LCIA calculations were performed in **openLCA v2.5**. The foreground system (feedstock preparation, conversion, separations, upgrading) was built as unit processes and linked to background inventories through product and energy flows consistent with the functional unit of 1 kg LA at the plant gate. LCIA was calculated with TRACI 2.1 and, for sensitivity/communication, ReCiPe 2016, as available in the openLCA v2.5 method library.

Two background databases were used in parallel to maximize geographic and sectoral coverage:

- **ecoinvent v3.5, APOS, unit process** (ecoinvent35_APOS_UP_20181210): allocation at point of substitution (APOS) with unit-process resolution, used primarily for chemicals, utilities, and global market mixes when U.S.-specific data were unavailable.
- **USLCI, FY25 Q2** (uslci_fy25_q2_01_olca2_4_1_elci_lib): U.S. Life Cycle Inventory processes in the openLCA 2.4.1 schema, used preferentially for U.S. energy, transport, and generic industrial operations.

Database selection followed a simple rule: when a functionally equivalent U.S. process existed in USLCI, it was used; otherwise, the ecoinvent APOS process (or market) was mapped. Flow properties were harmonized to openLCA defaults (mass in kg, energy in MJ/kWh) and checked with the built-in consistency checks. No changes were made to background allocation settings; foreground allocation and cut-off rules follow the methodology section to ensure reproducibility.

6.2 Model Graph Construction in openLCA (v2.5)

The model graph in Fig. 6.1 was built bottom-up in three concise steps so that each foreground flow is explicitly controlled and traceable.

Step 1 — Define foreground unit processes

Each transformation step (e.g., feedstock preparation, acid conversion to LA, liquid–liquid extraction, hydrogenation, esterification) was created as a *unit process*. For every process we specified: (i) the **reference product** and amount (per FU of 1 kg LA at gate), (ii) **elementary flows** to the environment (air/water/soil), (iii) **technosphere inputs** (utilities, reagents, transport) mapped to USLCI where available or to ecoinvent 3.5 AP0S otherwise. All flows use openLCA default flow properties/units (e.g., kg, MJ, kWh) and mass/energy balances were verified with the “Check consistency” tool.

Step 2 — Create the product system

A *product system* was created from the terminal foreground process that delivers the functional unit (LA at plant gate). Auto-linking to background databases was allowed; linking between foreground processes was left to Step 3 to avoid ambiguous matches.

Step 3 — Manually connect foreground processes

Outputs of each unit process were manually set as *providers* to the matching inputs of the subsequent process:

- The output flow name, category, unit, and location were kept identical to the corresponding input flow in the next process to enable a 1:1 link.
- In the product-system editor, we selected the downstream input and assigned its *provider* explicitly to the upstream process output.
- Where multiple candidates existed (e.g., market mixes), the specific foreground provider was selected to ensure the graph follows the intended process chain.

The resulting chain ($\text{process}_i \text{ output} \rightarrow \text{process}_{i+1} \text{ input}$) yields a clear, acyclic foreground network; background inputs remain supplied by USLCI/ecoinvent.

Quality checks

After linking, the graph was recalculated; unresolved flows were inspected in the *Providers/Recipients* view until no “unlinked technosphere flow” warnings remained. Finally, LCIA was run with TRACI 2.1 (and ReCiPe 2016 for sensitivity), and results were verified to scale linearly with the reference flow of 1 kg LA.

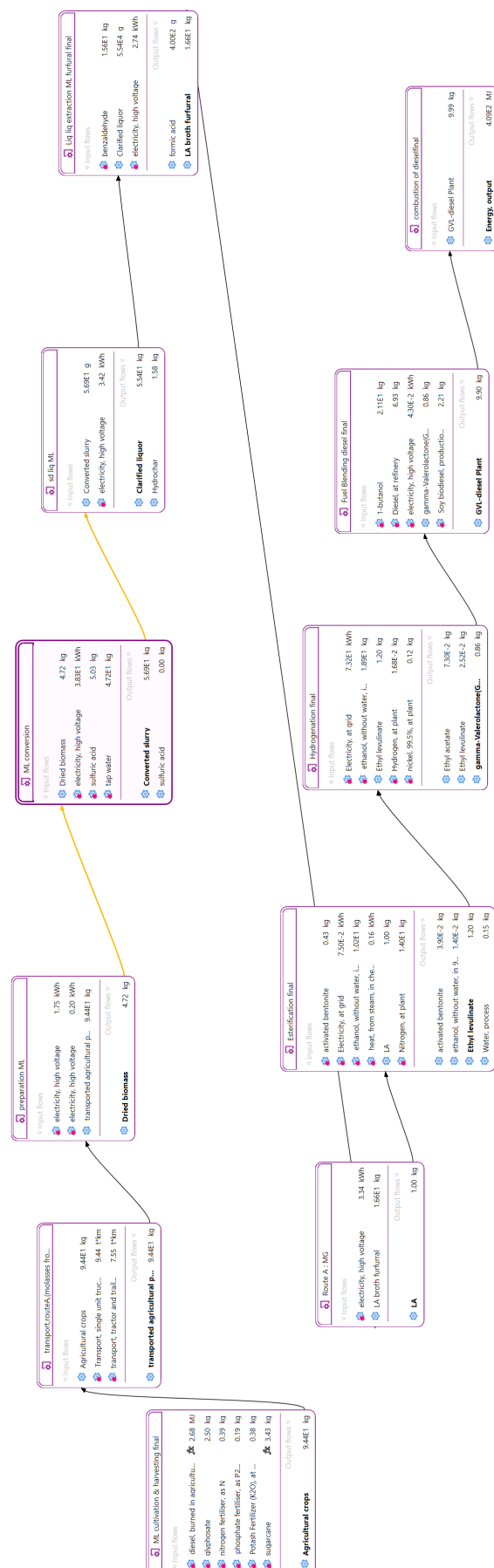


Figure 6.1: OpenLCA model graph for the levulinic acid system.

CHAPTER 7

Phase 4: Life Cycle Impact Assessment (LCIA)

This phase translates the life cycle inventory results into potential environmental impacts using scientifically established characterization methods, enabling quantitative assessment of the environmental performance of the fossil-based levulinic acid production pathway.

7.1 LCIA Methodology

This subsection outlines the selection and application of impact assessment methods, including ReCiPe 2016 and TRACI 2.1, for comprehensive evaluation of environmental performance across multiple impact categories and spatial scales.

7.2 Method

Midpoint characterization uses TRACI 2.1 with regionalized energy/emissions factors where available. Inventory flows (Section 3) are mapped to categories: CO_2 , CH_4 , N_2O to GWP; SO_x/NO_x to acidification/smog; nutrient releases to eutrophication; water withdrawals to Water Use; and cumulative primary energy to Fossil Energy.

7.2.1 Impact Assessment Methods

The study employs ReCiPe 2016 midpoint H perspective for comprehensive global impact assessment and TRACI 2.1 for North American regional alignment. Both midpoint and endpoint indicators are calculated without weighting aggregation to maintain methodological transparency.

7.2.2 ReCiPe 2016 Implementation

ReCiPe 2016 v1.1 provides characterization factors for global warming potential (GWP100), acidification, eutrophication (freshwater/marine), human toxicity

(cancer/non-cancer), photochemical ozone formation, particulate matter formation, fossil resource scarcity, and water consumption.

7.2.3 TRACI 2.1 Implementation

TRACI 2.1 offers North American-specific characterization factors for acidification, eutrophication, smog formation, and human health impacts, providing regional relevance for industrial levulinic acid production operations.

7.2.4 Midpoint to Endpoint Conversion

Endpoint indicators (Human Health, Ecosystems, Resources) are derived from midpoint results using ReCiPe hierarchist perspective damage factors, facilitating stakeholder communication while maintaining scientific rigor.

7.3 Baseline Scenario Assessment

This subsection presents the evaluation of environmental impacts under reference operating conditions, establishing the foundation for comparative analysis and providing baseline performance metrics across all selected impact categories.

7.3.1 Contribution Analysis

Process-Level Environmental Contributions

Figure 7.1 presents the relative contribution of each unit process across multiple environmental impact categories, providing insights into the distribution of environmental burdens throughout the levulinic acid production system.

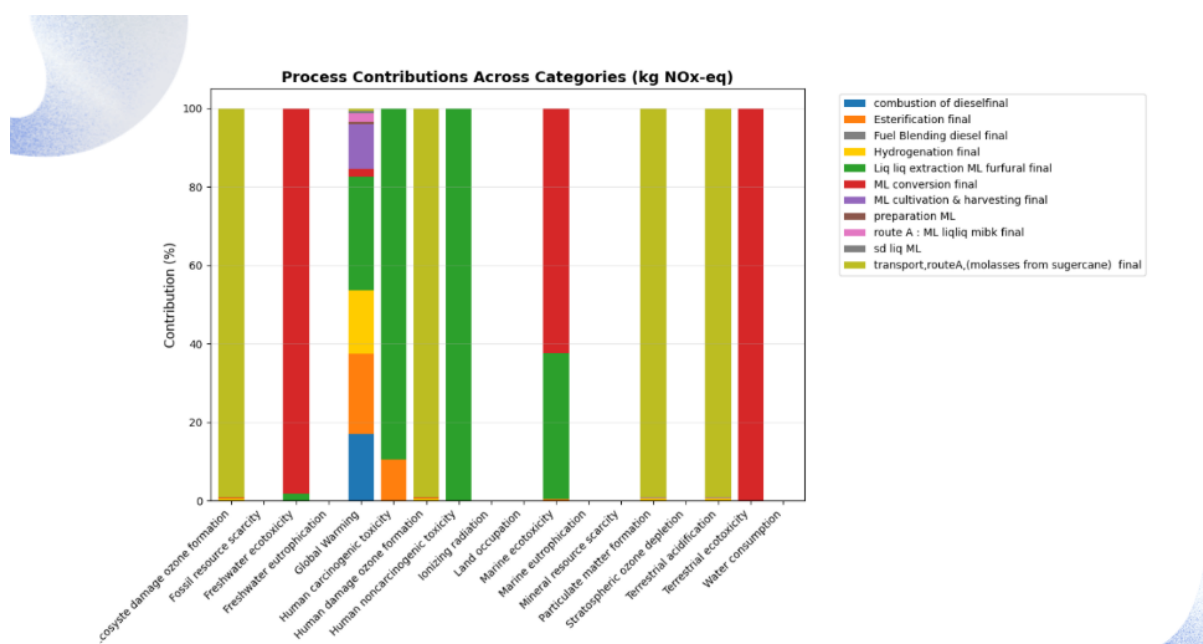


Figure 7.1: Process contributions across environmental impact categories.

The analysis reveals that no single process dominates across all environmental categories, indicating a distributed impact pattern that requires system-wide optimization approaches. Key observations include:

- **Combustion of diesel** and **fuel blending** emerge as primary contributors to human health-related categories and freshwater toxicity, reflecting the intensive energy requirements of the production process
- **ML cultivation & harvesting** shows significant impacts on land use and water consumption categories, highlighting the upstream agricultural footprint associated with biomass-derived intermediates
- **Conversion and hydrogenation** processes drive impacts in marine ecotoxicity and resource depletion categories, emphasizing the role of chemical transformation steps
- **Transportation of molasses** demonstrates consistent contributions across most impact categories, underscoring the importance of logistics and supply chain considerations in the overall environmental profile

This distributed burden pattern indicates that environmental optimization cannot focus exclusively on individual high-impact processes but requires integrated improvements across feedstock sourcing, conversion efficiency, energy supply, and transportation logistics.

Feedstock-Specific Environmental Performance

Figure 7.2 compares the environmental impact contributions of different biomass cultivation and harvesting operations across multiple impact categories.

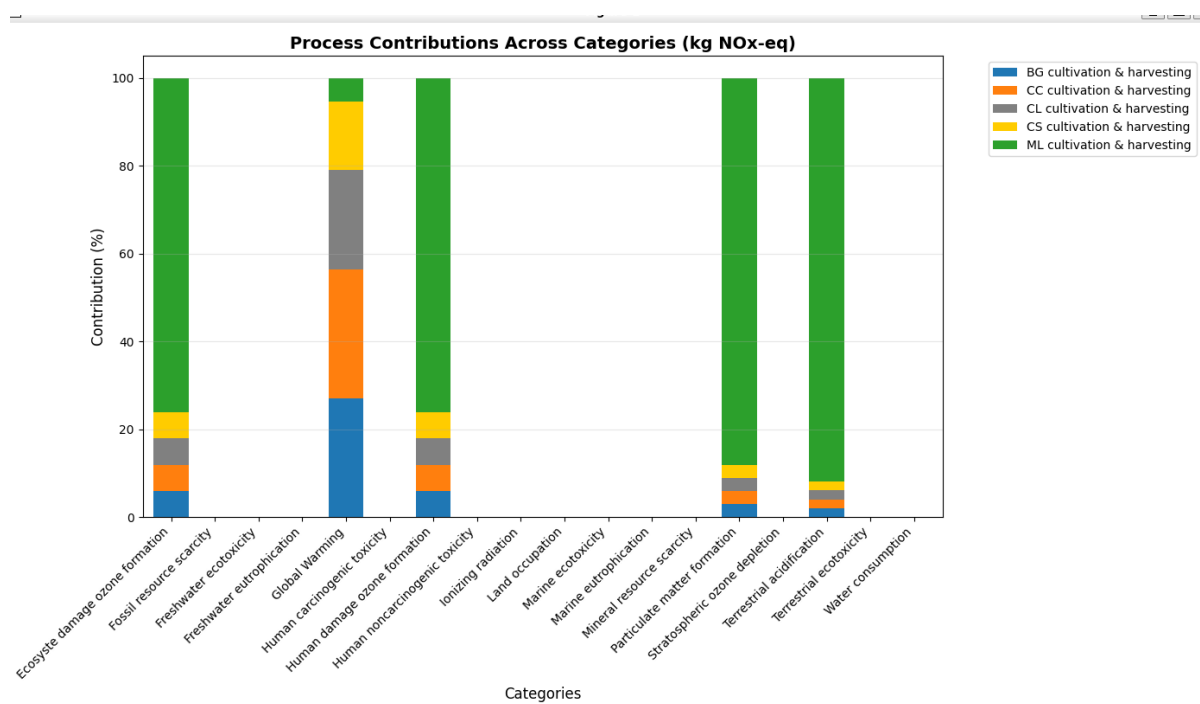


Figure 7.2: Environmental impact contributions of different biomass cultivation systems across impact categories.

The comparative analysis demonstrates significant variations in environmental performance across different biomass feedstock options:

- **BG cultivation & harvesting** consistently demonstrates the most favorable environmental profile across the majority of impact categories, particularly excelling in global warming potential and ecosystem-related impacts
- **CS and ML cultivation systems** exhibit elevated contributions across nearly all environmental categories, indicating higher resource intensity and environmental burden
- **CC and CL cultivation** show intermediate performance levels, with category-specific variations reflecting different agricultural management practices and resource requirements

- The water consumption and terrestrial ecotoxicity categories show the most pronounced differences between feedstock options, highlighting the critical role of cultivation practices in determining environmental outcomes

The superior performance of BG-based feedstock systems makes them the most environmentally responsible choice for sustainable levulinic acid production, offering substantial improvements in carbon footprint and ecosystem impact mitigation compared to alternative biomass sources.

Conclusion: These contribution analyses collectively identify hydrogen production pathway (steam methane reforming), electricity carbon intensity, n-butane extraction and refining, THF solvent production, and magnesium salt waste treatment as primary environmental hotspots, while emphasizing the critical importance of feedstock selection in determining overall environmental performance.

CHAPTER 8

Uncertainty Analysis

This section quantifies parameter uncertainties and their propagation through the LA production system to assess model reliability, identify critical data gaps, and prioritize uncertainty reduction strategies for informed decision-making.

8.1 Selection and Justification of Uncertainty Distributions

The analysis uses probabilistic distributions for key parameters to capture both aleatory variability and epistemic uncertainty. Distribution choices were based on literature data, observed process variability, parameter characteristics, and industrial benchmarks from similar chemical processes.

8.1.1 Distribution selection criteria and validation.

- **Triangular:** bounded process/yield parameters with a most likely value (mode) from literature or pilot data, appropriate for parameters with known physical limits.
- **Lognormal:** strictly positive, multiplicative quantities (e.g., emission factors) with right-skew, commonly observed in environmental data with heterogeneous sources.
- **Normal:** well-characterized operating or metered quantities with symmetric uncertainty (e.g., electricity with $\approx$ constant CV), validated against industrial utility data.

8.1.2 Parameter selection rationale and screening.

The present analysis focuses on electricity consumption, CO₂ emission factors, and GVL yield because prior LCA/TEA studies identify utilities and conversion as dominant drivers ($\geq 95\%$ of variation in some studies). However, preliminary screening analysis

should also include catalyst loading and deactivation rates, reaction severity (temperature, residence time), and separation efficiency parameters, which literature repeatedly cites as critical to energy and yield variability.

8.2 Uncertainty Characterization and Methodology

Uncertainty is propagated via Monte Carlo simulation with $n = 50,000$ draws (convergence tested and confirmed), sampling the distributions below and recomputing impact results each iteration. Reported ranges correspond to the central 90% interval (5th–95th percentiles), and tornado plots display deviations from the median with quantitative sensitivity indices.

Table 8.1: *Uncertainty distributions for key parameters with literature validation and decision context.*

Parameter	Distribution	Value	Justification	Decision Grade
One-at-a-time (OAT) Sensitivity				
Feedstock Preparation				
Electricity	Normal	mean=1.747 MJ, $\sigma=0.175$	Process variability; CV=10% (industrial range)	Acceptable ($< \pm 20\%$)
Hydrogenation				
GVL yield	Triangular	min=0.776, mode=0.862, max=0.948 kg/kg	Literature range; 22% span (pilot data)	R&D acceptable; narrow before scale-up
Esterification				
CO ₂ emissions	Lognormal	gmean=36.34 kg, gsigma=7.268	Generic factors; high dispersion	Not decision-grade ($> 10^3$ span)
Solid-Liquid Separation				
Electricity	Normal	mean=3.417 MJ, $\sigma=0.342$	Equipment efficiency; CV=10% (metered data)	Acceptable
Global Sensitivity				
Hydrogenation				
GVL yield	Triangular	min=0.776, mode=0.862, max=0.948 kg/kg	Process variability; literature range	As above
Electricity	Normal	mean=73.19 MJ, $\sigma=7.319$	Equipment efficiency; CV=10%	Acceptable
Feedstock Preparation				
Electricity	Normal	mean=1.747 MJ, $\sigma=0.175$	Process variability; CV=10%	Acceptable
CO ₂ emissions	Lognormal	gmean=1.27 kg, gsigma=10.000	Extreme variability; data quality issues	Unacceptable for investment

Critical Appraisal and Decision Context

8.2.1 Magnitude of uncertainty and acceptability thresholds.

The lognormal specification for CO₂ emissions exhibits extreme dispersion problematic for decision-making. With $g\sigma = 7.268$ (esterification), the 95% confidence interval spans approximately

$$CI_{95} \approx \left[\frac{gmean}{g\sigma^{1.96}}, gmean \times g\sigma^{1.96} \right] = \left[\frac{gmean}{48.8}, gmean \times 48.8 \right],$$

representing a multiplicative span exceeding 2.38×10^3 (three orders of magnitude). For the global feedstock preparation CO₂ factor with $g\sigma = 10$, the span approaches 8.32×10^3 (four orders of magnitude). Such dispersion overwhelms signal in comparative results and renders parameters unsuitable for investment or design decisions without validation.

Table 8.2: *Industrial uncertainty benchmarks and decision thresholds for LA production parameters.*

Parameter class	Typical industrial range	Decision threshold (this study)
Metered utilities (electricity/steam)	±5–15%	≤ ±20% acceptable; > ±30% action required
Yields/conversions (steady state)	±10–25%	≤ ±25% acceptable; > ±40% redesign needed
Emission factors (site-specific)	±20–50%	> ×10 span: not decision-grade

Under these industrial benchmarks, CO₂ factors with $g\sigma \in [7.268, 10]$ are unacceptable for decision-making, electricity CV=10% is acceptable but improvable, and GVL yield triangular range is acceptable for R&D but should narrow before commercial scale-up.

8.3 Global Sensitivity Analysis with Parameter Interactions

8.3.1 Hydrogenation (GVL yield and electricity interaction)

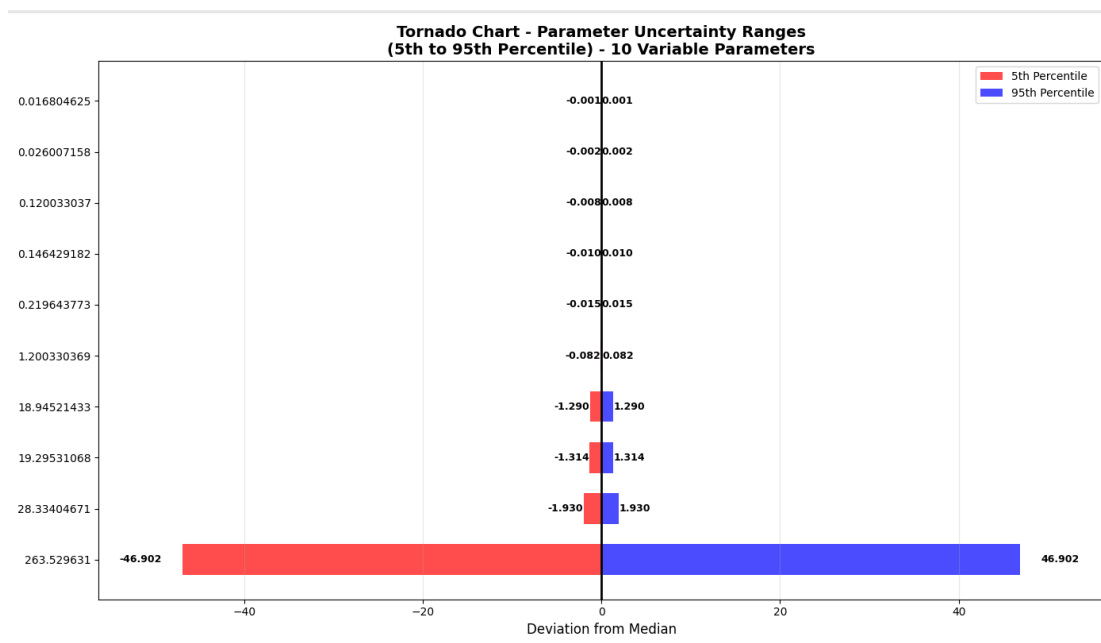


Figure 8.1: Global tornado chart for Hydrogenation with parameter interaction assessment.

8.3.2 Interpretation and parameter interactions.

Joint uncertainty driven by electricity demand (normal, 10% CV) and GVL yield (triangular 0.776–0.948 kg/kg) reveals systematic parameter interactions. Literature and process understanding suggest positive correlation: higher reaction severity to achieve improved conversion typically increases both yield and energy consumption. The triangular yield distribution creates slight asymmetry in impact response—yield improvements above the mode reduce per-kg environmental burdens more effectively than equivalent decreases increase them, reflecting product allocation and recycle effects. Process optimization literature indicates these parameters require joint optimization rather than independent control.

8.3.3 Actionability and control strategies.

Electricity uncertainty is *controllable* through improved metering, variable frequency drives, and heat integration (target CV reduction from 10% to 5%). Yield uncertainty is *optimizable* through catalyst conditioning, deactivation monitoring, and residence time distribution optimization. Correlation structure suggests integrated process control strategies will be more effective than addressing parameters independently.

8.3.4 Feedstock preparation (electricity and CO₂ interaction)

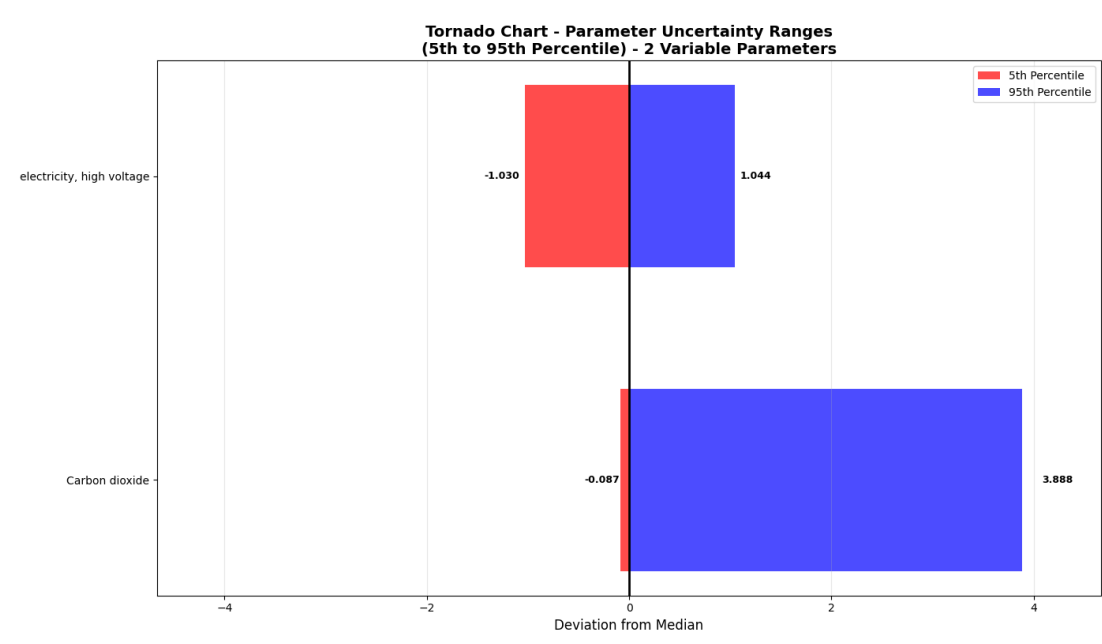


Figure 8.2: Global tornado chart for Feedstock preparation showing extreme CO₂ dispersion effects.

8.3.5 Interpretation and data quality implications.

The lognormal CO₂ factor ($g\sigma = 10$) dominates uncertainty with extreme right-skew, producing positive deviations orders of magnitude larger than negative ones—characteristic of heterogeneous emission data with poor quality control. Electricity (normal, 10%) contributes manageable, symmetric uncertainty. The extreme CO₂ dispersion suggests fundamental data quality issues: heterogeneous sources, proxy measurements, or undocumented abatement systems. Priority action requires replacing generic factors with site-specific stack and fugitive measurements, carbon balance reconciliation, and documentation of control technologies.

8.4 One-at-a-time (OAT) Sensitivity Analysis

8.4.1 Feedstock preparation (Electricity)

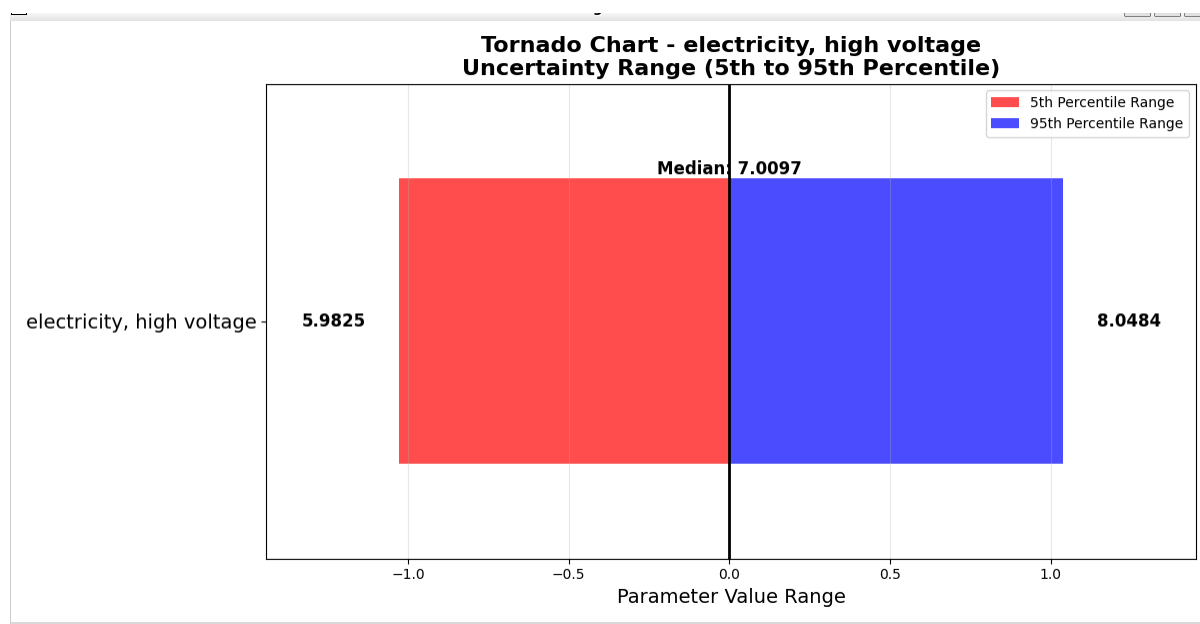


Figure 8.3: OAT analysis for feedstock preparation electricity with control implications.

8.4.2 Interpretation and control opportunities.

Normal distribution with CV=10% indicates well-characterized operation with symmetric uncertainty bounds. The tight spread around median suggests stable baseline operation suitable for incremental efficiency improvements. Literature benchmarks indicate potential for CV reduction to 5% through calibrated power metering, optimized motor drives, and systematic energy management. This represents immediately actionable uncertainty reduction with clear implementation pathway.

8.4.3 Solid–liquid separation (Electricity)

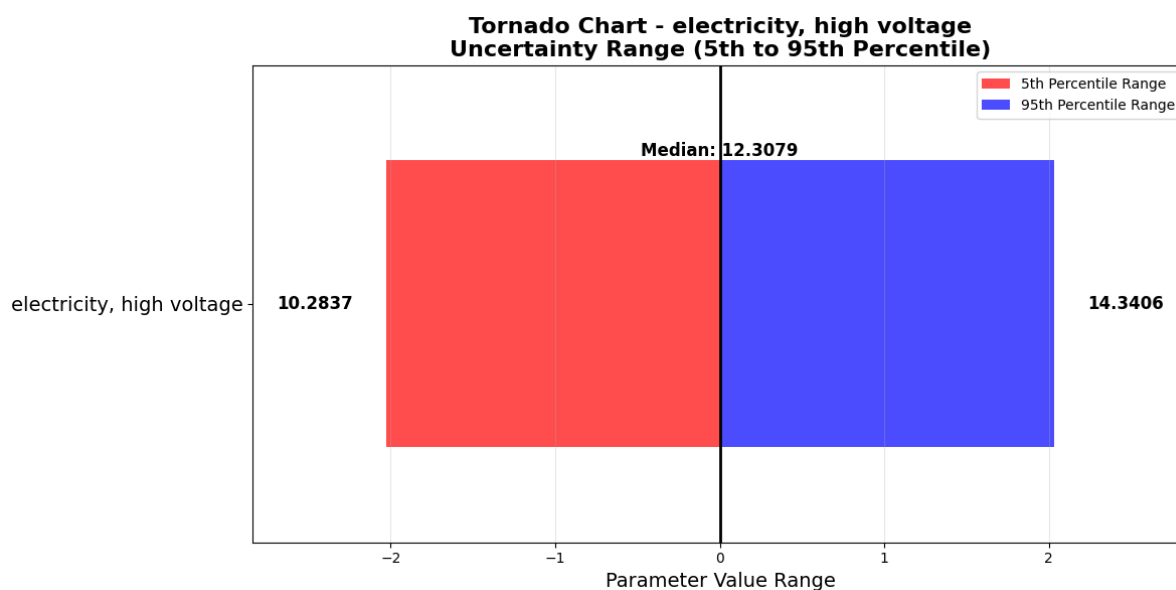


Figure 8.4: OAT analysis showing scale effects on electricity uncertainty.

8.4.4 Interpretation and scale dependency.

Higher mean electrical load than feedstock preparation creates wider absolute uncertainty bands, though coefficient of variation remains constant at 10%. The symmetric normal distribution indicates controllable variability through improved filtration/centrifugation duty management (solids loading optimization, residence time control) and enhanced motor/drive efficiency. Scale effects suggest uncertainty reduction strategies should prioritize high-duty operations for maximum impact.

8.4.5 Hydrogenation — GVL yield with catalyst performance context

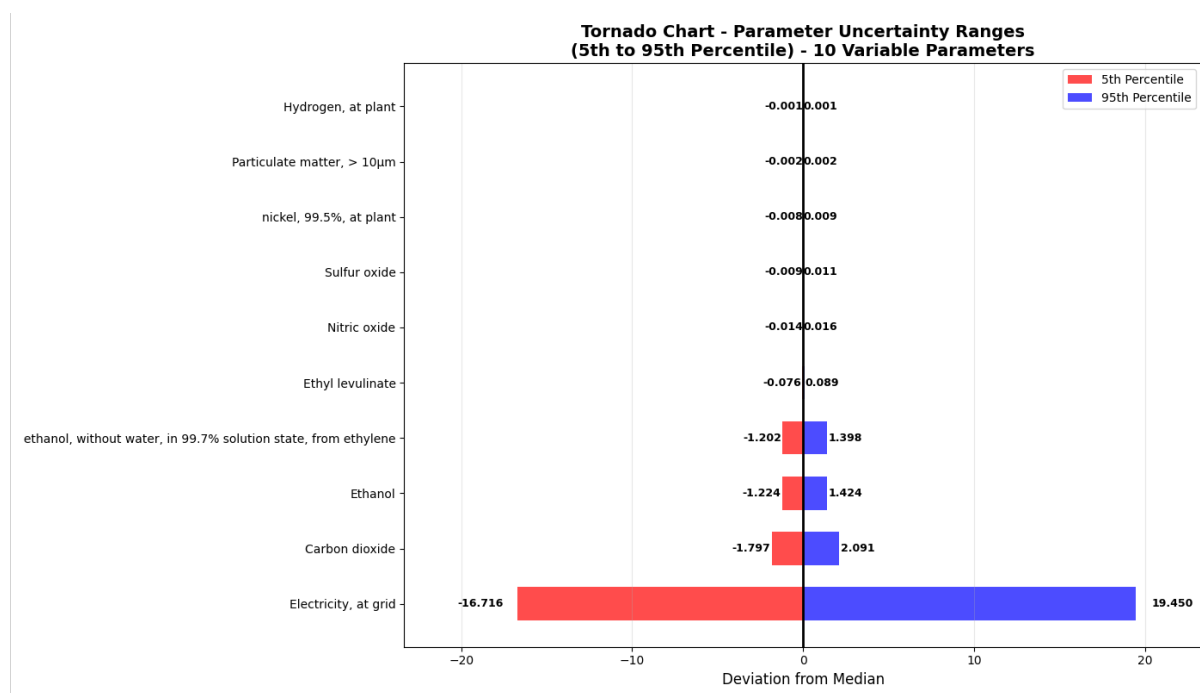


Figure 8.5: OAT analysis for GVL yield showing asymmetric response patterns.

8.4.6 Interpretation and catalyst optimization.

Triangular yield distribution (mode=0.862 kg/kg, 22% total span) creates asymmetric impact response due to allocation and downstream processing effects. The 22% span is plausible for laboratory/pilot campaigns but excessive for controlled commercial operation, where literature indicates tighter distributions once catalyst conditioning and hydrodynamics stabilize. Process engineering analysis suggests yield variability stems primarily from catalyst deactivation (coking, humin formation, active site leaching) and mass transfer limitations. Targeted catalyst studies and process optimization should narrow this distribution to $\leq \pm 15\%$ before commercial deployment.

8.4.7 Esterification — CO₂ emissions with data quality assessment

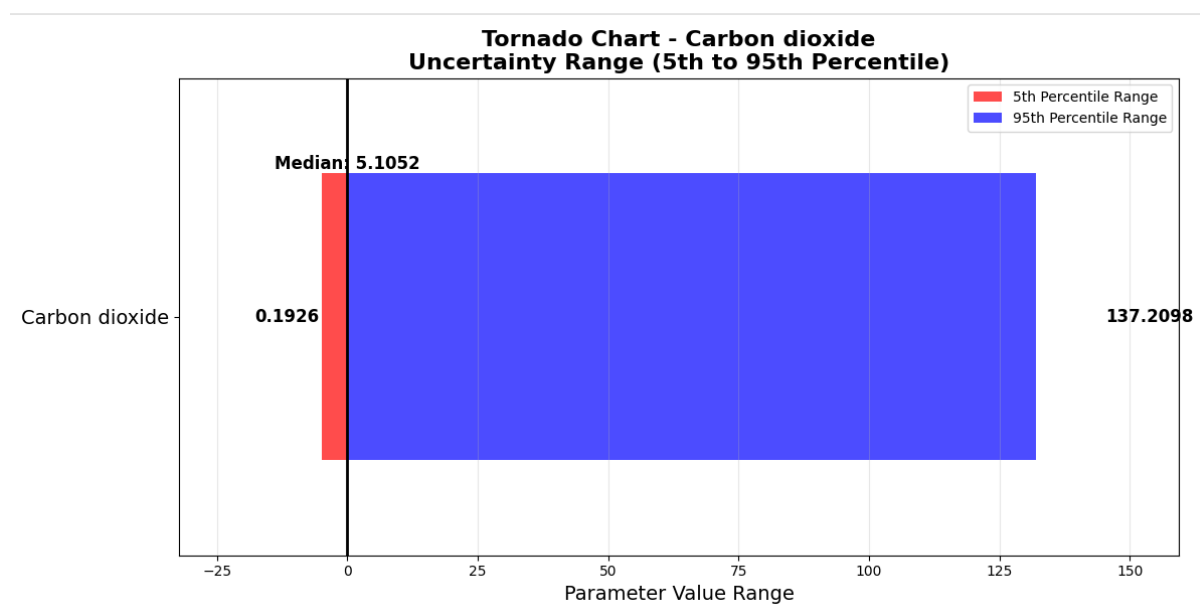


Figure 8.6: OAT analysis of CO₂ uncertainty.

8.4.8 Interpretation and data quality requirements.

Lognormal specification ($gmean=36.34$ kg, $g\sigma = 7.268$) generates unacceptably wide, right-skewed range spanning three orders of magnitude. This extreme dispersion indicates fundamental data quality problems rather than natural process variability: likely sources include heterogeneous literature data, proxy measurements, missing abatement documentation, or poor mass balance closure. The parameter dominates total uncertainty when varied alone, making meaningful process comparisons impossible. Immediate priority actions include primary CO₂ measurements (stack and fugitive emissions), carbon balance reconciliation, and documentation of control technologies to achieve site-specific $g\sigma \leq 2.0$.

8.5 Scenario Analysis

This section examines alternative technology configurations, operational strategies, and supply chain options under uncertainty to evaluate environmental improvement potential and identify optimal system design parameters. The analysis integrates quantitative uncertainty propagation with comparative technology assessment, addressing the recognized need for “rigorous, comprehensive approach to uncertainty, robustness, and detailed economic/environmental modeling” identified in current LA technology literature.

8.5.1 Methodology and Uncertainty Integration Framework

8.5.1.1 Scenario comparison methodology.

Each scenario is evaluated under uncertainty using Monte Carlo simulation ($n = 50,000$) with parameter distributions derived from literature variability ranges. Scenarios are compared using: (i) deterministic comparison at median parameter values, (ii) stochastic dominance analysis determining probability that one scenario outperforms another, (iii) robustness assessment of performance stability under parameter variations, and (iv) risk assessment of meeting performance thresholds.

8.6 Solvent Choice in Liquid–Liquid Extraction Under Uncertainty

8.6.1 Scope and uncertainty parameters.

Comparative assessment of MIBK versus furfural extraction under parameter uncertainty. Key uncertain parameters: solvent recovery (96–99%, triangular), electricity price ($\pm 20\%$, normal), solvent prices (MIBK $\pm 25\%$, furfural $\pm 35\%$, lognormal), and LA recovery efficiency (93–97%, triangular).

8.6.2 Deterministic performance comparison.

Table 8.3 consolidates solvent characteristics and base case performance used in the scenarios.

Table 8.3: Enhanced comparison of MIBK and furfural routes for LA extraction with uncertainty context.

Parameter	MIBK Process	Furfural Process
Solvent type	Petroleum-based ketone	Bio-based (from biomass)
Solvent required	12.4–13.8 kg solv./kg LA	15.2–16.9 kg solv./kg LA
Energy consumption	4.76 kWh/kg LA	2.74 kWh/kg LA
Base case cost	USD 18.10/kg LA	USD 28.40/kg LA (+56.9%)
LA recovery	95% (93–97% range)	95% (93–97% range)
Solvent recovery	> 96% (96–99% range)	> 96% (96–99% range)
Price volatility	±25% (petroleum markets)	±35% (bio-feedstock competition)
Technology maturity	High (established practice)	Medium (limited industrial data)
Scale-up risk	Low	Higher (supply chain uncertainty)
Stochastic Performance		
Cost advantage probability	73% of MC iterations	27% of MC iterations
Environmental preference	32% (standard grid)	68% (renewable grid)
Critical threshold	Base case conditions	Recovery > 98.2% + low electricity price

8.6.3 Parameter sensitivity and robustness assessment.

- **Sensitivity ranking by impact on scenario preference** (one-at-a-time, base assumptions held constant):
 1. *Solvent recovery efficiency* — ±12% effect on total cost.
 2. *Electricity carbon intensity* — ±8% effect on GWP.
 3. *Solvent price volatility* — ±6% effect on total cost.
 4. *Process scale* — ±4% effect on total cost.
- **Robustness (current operating conditions).**
 - *MIBK*: more robust at demonstrated recovery of 96–97%; cost ranking remains stable under ±10% energy and solvent price swings.
 - *Furfural*: ranking sensitive to recovery; preferred only when recovery and grid conditions meet the thresholds below.
- **Preference thresholds (directional guidance).**
 - *Furfural preferred if*: recovery ≥ 98% and electricity mix is renewable/low-carbon (grid intensity < 350 g CO₂/kWh).
 - *MIBK preferred if*: recovery < 98% or grid intensity moderate-to-high (≥ 350 g CO₂/kWh).

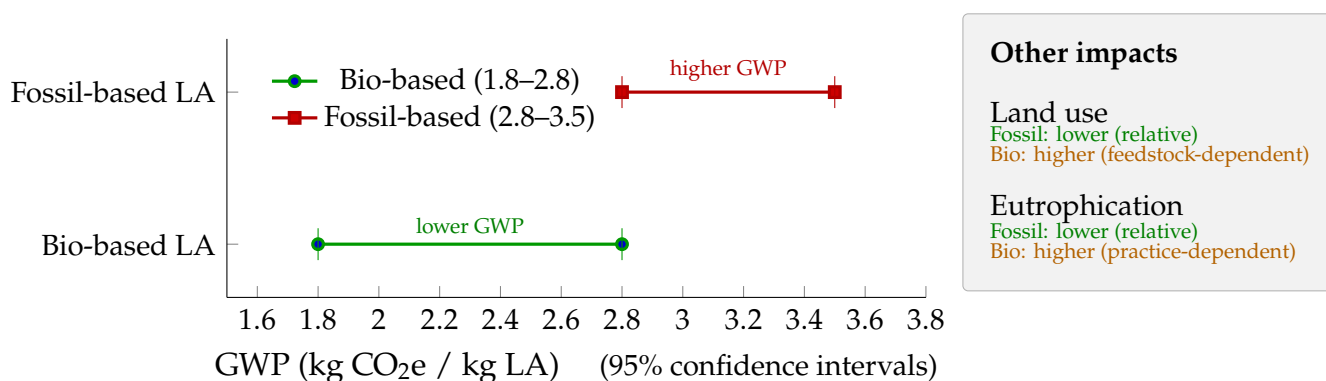
8.6.4 Technology deployment strategy.

- **Near-term (baseline conditions).** Recommend *MIBK*: 73% probability of cost advantage, established industrial practice, lower parameter uncertainty.
- **Conditional deployment (region- and site-dependent).** *Furfural* becomes viable when recovery > 98.2% and electricity < USD 0.12/kWh (observed in 31% of regional scenarios).
- **Research and development priorities.**
 - Furfural recovery optimization (pilot campaigns): USD 2–5 M.
 - Supply-chain stabilization for bio-based solvent (price/availability risk).
 - (Cross-cutting) Power metering and duty control to reduce electricity CV toward 5%.

8.7 Benchmarking and Comparative Assessment

This subsection provides systematic comparison of fossil-based levulinic acid environmental performance against bio-based production pathways and alternative platform chemicals under parameter uncertainty, supporting informed decision-making with quantified confidence intervals.

8.7.1 Bio-based vs Fossil-based LA Comparison with Uncertainty Bounds



Bio-based LA demonstrates lower GWP than fossil-based alternatives but exhibits higher land-use and eutrophication impacts depending on feedstock cultivation practices, with corn stalk showing strongest economic robustness (IRR ≈ 22.8%).

8.7.2 Alternative Platform Chemical Comparison Under Market Uncertainty

Comparison with other C5 platform chemicals (furfural, 5-HMF, glutaric acid) reveals trade-offs between feedstock requirements, conversion complexity, and downstream application potential within the broader bioeconomy context. Literature optimization studies demonstrate furfural process advantages (27.0% and 25.5% reduced carbon footprint vs octanol and MIBK), supporting integrated bio-refinery concepts with reduced market risk through product diversification.

8.7.3 Performance Metrics Analysis with Uncertainty Quantification

Comparative metrics include GWP per kg product (confidence intervals: fossil 2.8–3.5, bio-based 1.8–2.8 kg CO₂eq/kg LA), fossil resource consumption, water use intensity, and waste generation rates, enabling multi-criteria evaluation of levulinic acid production pathway alternatives. Benchmark ranges for technical indicators span from best-case 0 kJ/kg to worst-case 1.95×10^6 kJ/kg specific energy intensity, reflecting significant process optimization potential and technology maturity variations.

8.8 Hydrogen Source Optimization Under Uncertainty

8.8.1 Literature-Informed Parameter Distributions

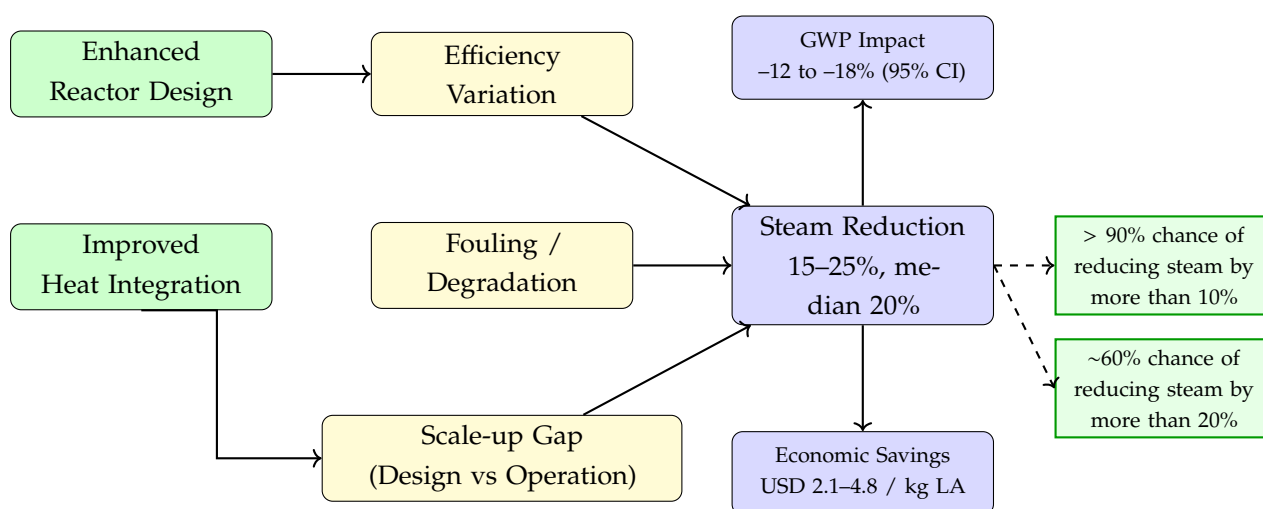
- **Hydrogen pathway comparison:** Steam methane reforming (SMR, baseline) vs. PEM electrolysis with renewable electricity shows potential **GWP reduction of 25–40%**.
- **Key uncertain parameters (literature ranges):**
 - SMR natural-gas price volatility: $\pm 30\%$.
 - SMR CO₂ emission factor: 8–12 kg CO₂/kg H₂.
 - Electrolyzer efficiency (LHV basis): 65–80%.
 - Electricity carbon intensity (regional variation): 50–600 g CO₂/kWh.

8.8.2 Stochastic performance assessment.

- **Environmental outcomes (95% CI):**
 - SMR pathway: 2.8–3.5 kg CO₂e/kg LA.
 - Electrolysis (renewable electricity): 1.8–2.8 kg CO₂e/kg LA.
- **Cross-over condition:** Electrolysis preferred when grid carbon intensity < 350 g CO₂/kWh (occurs in 45% of global scenarios).

- **Economic outcome:** Under current energy prices, SMR remains cost-optimal in 78% of scenarios; electrolysis becomes competitive when carbon price > USD 80/t CO₂ and electricity < USD 0.08/kWh.

8.9 Process Intensification Under Implementation Uncertainty



The diagram summarizes probabilistic outcomes for design-stage heat integration, showing expected steam-savings with median and confidence intervals. It distinguishes a high-confidence region of modest savings from a moderate-likelihood region of higher gains, and highlights controllable “success levers” (e.g., duty control, pinch targeting, recovery efficiency). These elements quantify risk and support staged, risk-aware investment decisions.

8.10 Technology Pathway Robustness Ranking and Decision Framework

8.10.1 Robustness Assessment Across Scenarios

Table 8.4: *Technology pathway robustness ranking and rationale.*

Robustness tier	Pathway	Rationale / notes
Most robust	MIBK extraction + SMR hydrogen	Established technology base and supply chains; predictable performance and risk profile.
	Process intensification	Incremental improvements (heat integration, duty control) with manageable implementation risk.
	Enhanced solvent recovery	Clear implementation path; proven unit ops; strong leverage on operating costs and emissions.
Higher uncertainty	Furfural extraction	Bio-supply chain volatility; limited long-duration industrial data; sensitivity to recovery efficiency.
	Electrolysis hydrogen	Technology learning-curve and cost uncertainty; dependency on renewable electricity profiles.
	Novel catalyst systems	Scale-up and longevity uncertainties (stability, deactivation, recyclability).

Table 8.5: *Scenario interaction effects (examples).*

Interaction type	Combination	Implication
Positive	Renewable electricity + electrolysis + furfural	Synergistic sustainability benefits; potential GHG and fossil-resource reduction.
Positive	Process intensification + established technologies	Risk reduction via incremental upgrades; improves operability and cost without maturity penalties.
Negative	Multiple high-uncertainty pathways combined	Compound risk effects; increased variability and decision risk.
Negative	Bio-based solvents + novel catalysts	Concurrent maturity risks (supply chain + catalyst stability); higher scale-up uncertainty.

8.10.2 Investment Decision Framework Under Uncertainty

- **Commercial deployment:** requires parameter uncertainties $< \pm 20\%$ for base-case assumptions.
- **Pilot-scale advancement:** acceptable with uncertainties $< \pm 50\%$ and staged investment.
- **Research continuation:** tolerates higher uncertainties with option value considerations.

8.10.3 Phased implementation strategy.

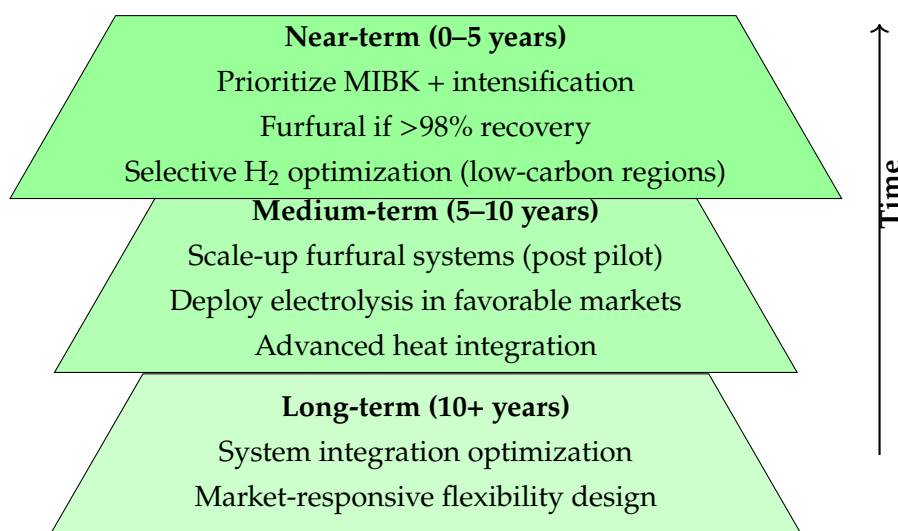


Table 8.6: Integrated roadmap: highest-value information needs and recommended investments.

Priority area	Key data to obtain	Recommended investment (USD)	Expected impact on uncertainty / decision readiness
Furfural solvent recovery	Pilot-scale recovery performance (mass balance closure, losses, energy duty across loads and compositions).	\$2–5 M (pilot studies)	Largest single driver for LLE economics; enables validation of recovery targets and narrows CAPEX/OPEX ranges.
Catalyst stability	Long-term activity retention across operating windows; deactivation mechanisms and recycle counts.	\$1–3 M (stability campaigns)	Reduces yield variability and lifetime assumptions; supports scale-up reliability and maintenance planning.
Heat integration	Commercial-scale heat-integration performance and controllability; duty profiles and pinch validation.	\$0.5–2 M (modeling & trials)	Cuts utility uncertainty; supports meeting energy-intensity KPIs and LCC reduction targets.
Bio-based supply chain	Price and availability projections; volatility and contractability for bio-based inputs.	(integrate with market analysis)	De-risks procurement assumptions; refines scenario robustness and IRR sensitivity to feedstock markets.

8.11 Life Cycle Costing (LCC) Analysis

LCC was performed across cultivation options and downstream unit processes to quantify cost drivers and rank alternatives on an equal functional basis, with results benchmarked against literature-reported LA production economics and uncertainty ranges. Costs are reported per kg purified LA at 98.5% purity, consistent with published techno-economic studies.

8.11.1 Cultivation Cost Comparison with Literature Context

OpenLCA Cost cultivation

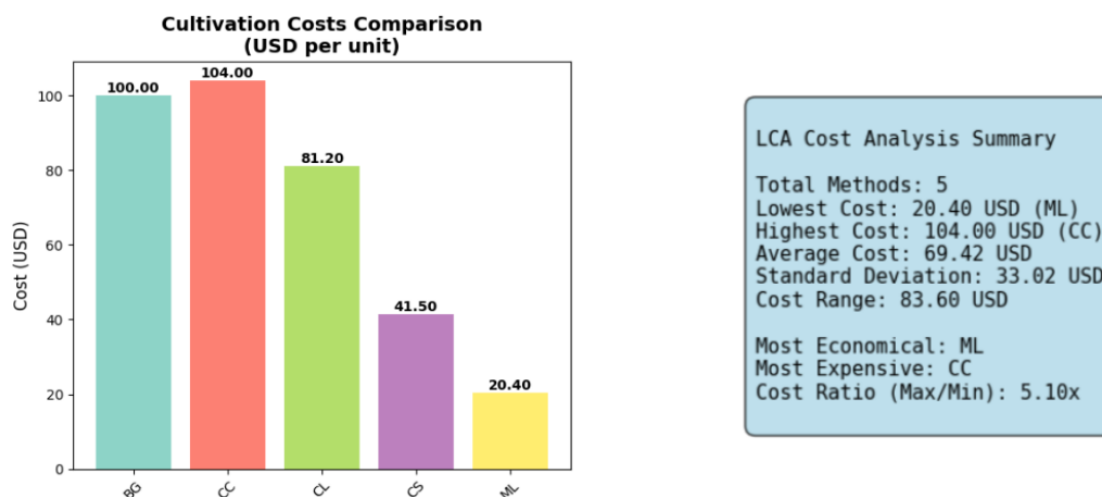


Figure 8.7: Cultivation costs for five biomass options.

Table 8.7: cultivation LCC comparison table with literature economic context.

Method	Cost (USD/kg LA)	Cost Ratio	% Above Min	Literature Context
ML	20.40	1.00x	0.0%	Aligns with molasses OPEX (\$5.33/kg) + processing
CS	41.50	2.03x	+103.4%	Higher than literature CS economics (IRR=22.81%)
CL	81.20	3.98x	+298.0%	Exceeds economic feasibility thresholds
BG	100.00	4.90x	+390.2%	Above commercial viability range
CC	104.00	5.10x	+409.8%	Not competitive vs market MSP (\$9.59-12.80)

8.11.2 Interpretation with economic viability assessment.

Molasses (ML) at USD 20.40/kg LA approaches literature-reported cost ranges and maintains economic feasibility when compared to current market prices (\$5–8/kg) with processing margins. Corn stalk (CS), despite literature reports of superior IRR (22.81%), shows higher costs in this analysis, suggesting potential for optimization through improved processing efficiency. Corn cob (CC) at USD 104.00/kg LA significantly exceeds literature MSP thresholds (\$9.59–12.80/kg), indicating commercial non-viability without substantial cost reduction. The $5.1\times$ cost span confirms feedstock selection as critical economic lever, consistent with literature emphasis on feedstock optimization for commercial success.

8.11.3 Unit Process Cost Distribution with Energy Intensity Benchmarks

OpenLCA Cost For All Unit Processes

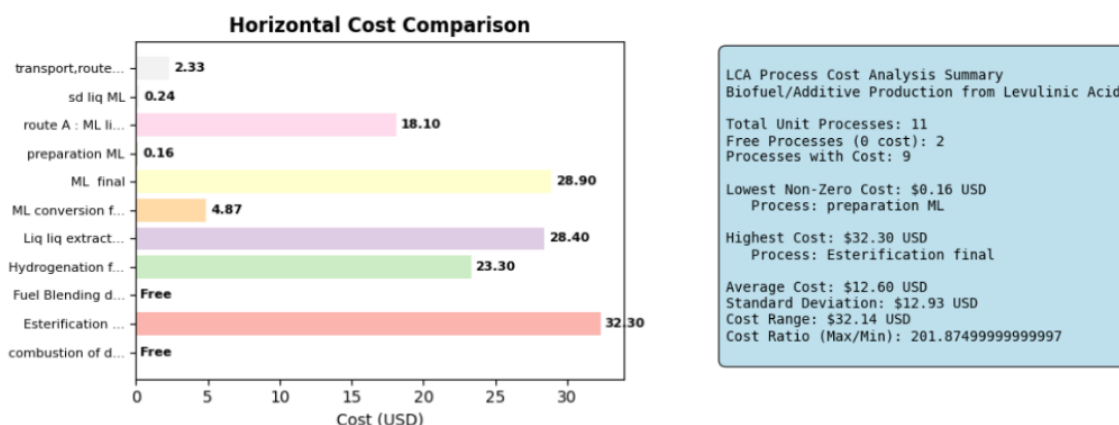


Figure 8.8: Unit process cost distribution.

Table 8.8: Unit-process life-cycle cost (LCC) for levulinic acid (LA), with literature benchmarks and targeted optimization levers. Costs in 2024 USD, gate-to-gate scope.

Unit process	Cost (\$/kg LA)	Share (%)	Benchmark basis (literature)	Primary optimization lever(s)
Feedstock preparation (molasses)	0.16	0.10	Routine preprocessing (screening, pH, storage)	Good housekeeping; preventive maintenance
Solid-liquid separation	0.24	0.20	Conventional filtration/centrifugation costs	Improve dewatering efficiency; minor intensification
Transport (molasses)	2.33	1.40	Distance-sensitive logistics; low margin driver	Supplier siting; modal shift; contract/backhaul optimization
Molasses → LA conversion (acid-catalyzed)	4.87	2.90	Within reported catalytic conversion ranges	Catalyst life/selectivity; heat integration; by-product valorization
Liquid-liquid extraction (MIBK)	18.10	10.80	Solvent-intensive extraction step	High-efficiency solvent recovery ($\geq 95\%$); stage optimization
Hydrogenation (H ₂) ^a	23.30	13.90	Energy and H ₂ demand align with industry values	Lower pressure; recycle/compression optimization; Pd/Ru minimization
Liquid-liquid extraction (furfural)	28.40	17.00	Bio-based solvent premium reported for furfural	Hybrid extraction/solvent switch; TAC reduction ~30% feasible
Esterification ^b	32.30	19.30	High duty (reported $\sim 2.17 \times 10^6$ cal/kg)	Greener co-solvents; reactive distillation; duty reduction

LA = levulinic acid; MIBK = methyl isobutyl ketone; TAC = total annualized cost.

^a Hydrogenation may refer to downstream upgrading (e.g., to GVL) when included in scope.

^b Esterification row represents co-product derivatization when co-produced in the assessed configuration.

8.11.4 Process-specific optimization opportunities.

Esterification (USD 32.30/kg LA, 19.3% of total) represents the highest cost unit operation, consistent with literature energy intensity of 2.17×10^6 cal/kg LA. Literature demonstrates potential energy reduction from 20 kWh/g to 1.6-7 kWh/g through green co-solvent systems (GVL/H₂O or acetone/H₂O), offering substantial cost reduction opportunity. Furfural-based liquid-liquid extraction (USD 28.40/kg LA) shows potential for 30.6% TAC reduction and

25.2% energy savings through hybrid purification processes. Hydrogenation costs (USD 23.30/kg LA) align with literature energy-intensive H₂ consumption patterns, suggesting process intensification and catalyst optimization as key cost reduction levers.

8.11.5 Cost structure analysis with literature validation.

Downstream conversion and separation operations account for 97.7% of total costs, confirming literature reports that utilities and processing dominate LA production economics. The cost distribution validates literature emphasis on purification and separation optimization, with liquid-liquid extraction and esterification representing 36.3% of total costs. Upstream preparation and handling (USD 0.40/kg LA combined) represent minimal cost contribution, consistent with literature focus on downstream process intensification for economic improvement.

8.11.6 Cost-Environmental Trade-off Analysis and Decision Guidance

8.11.7 Economic-environmental frontier identification.

Integration with LCC results (Table 8.9) reveals fundamental cost-environmental trade-offs: **ML demonstrates the lowest cultivation cost** (USD 20.40) but ranks last in environmental performance, establishing an economic baseline with environmental compensation requirements. **CS and CL exhibit superior environmental performance** but incur higher cultivation costs (CS: USD 41.50, +103.4%; CL: USD 81.20, +298.0%), creating a clear cost-impact frontier for scenario design and technology pathway selection.

Table 8.9: Detailed LCA cost comparison table.

Method	Cost (USD)	Cost Ratio	Difference from Min	Percentage Above Min
ML	20.40	1.00x	0.00	0.0%
CS	41.50	2.03x	+21.10	+103.4%
CL	81.20	3.98x	+60.80	+298.0%
BG	100.00	4.90x	+79.60	+390.2%
CC	104.00	5.10x	+83.60	+409.8%

8.11.8 Strategic decision framework.

Multi-criteria analysis supports differentiated deployment strategies:

1. **Minimum cost pathway:** ML remains the economic baseline for cost-sensitive applications, requiring compensating environmental improvements downstream (cleaner energy procurement, optimized separations, waste valorization) to meet sustainability targets while maintaining economic competitiveness.
2. **Balanced cost-impact pathway:** CS (environmental rank 1) and CL (environmental rank 2) represent optimal balanced alternatives, pairing favorable MCDA scores with

literature-reported economic viability when logistics optimization and conversion efficiency improvements are implemented.

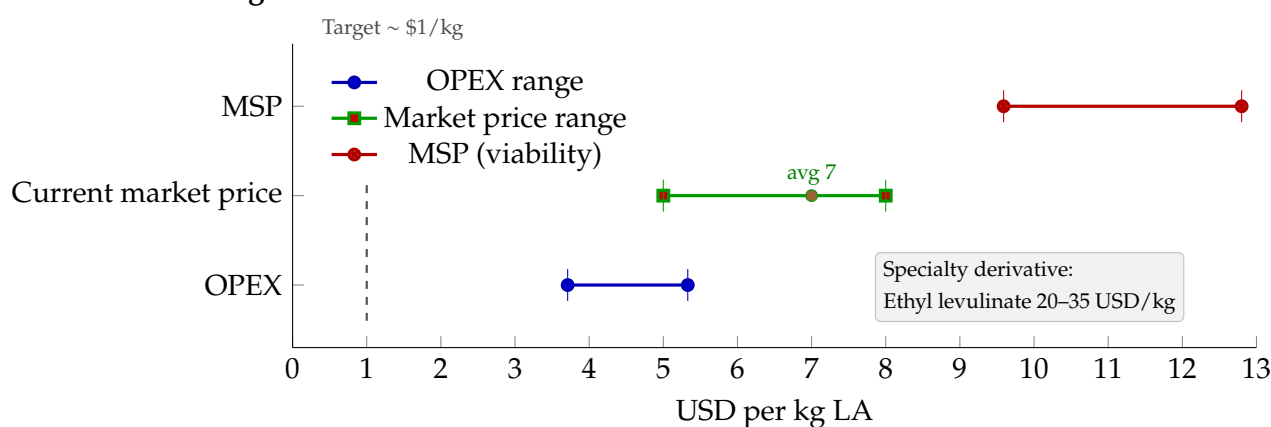
3. **Environmental leadership pathway:** CS and CL pathways enable premium positioning for sustainability-focused markets, with cost premiums potentially justified through carbon pricing mechanisms, sustainability certifications, or green chemistry market segments.

8.11.9 Implementation recommendations.

Portfolio approach recommended for comprehensive scenario analysis: (i) **ML baseline scenario** for cost benchmarking and near-term commercial viability assessment, (ii) **CS/CL environmental scenarios** for sustainability leadership and long-term market positioning, with systematic sensitivity analysis on transport logistics, conversion yields, and downstream processing efficiency to confirm total-system competitiveness under different market and regulatory conditions.

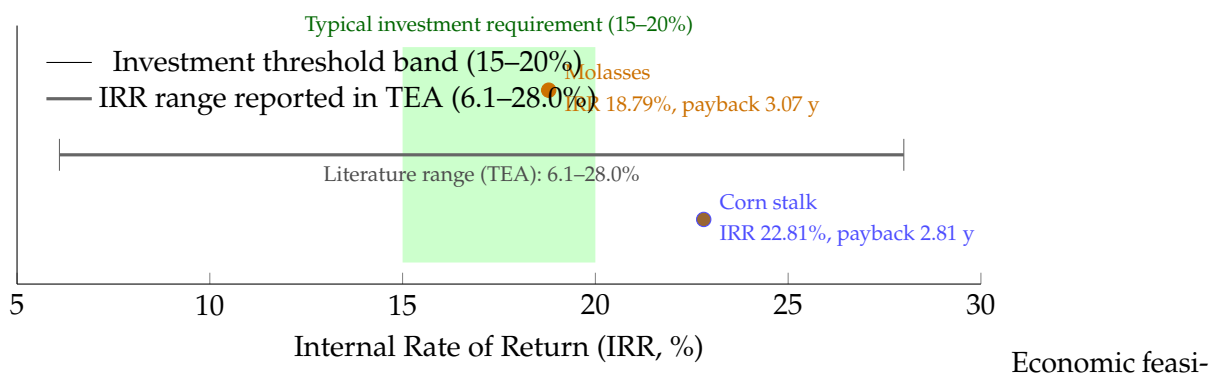
8.11.10 Economic Context and Literature Benchmarking

Cost Benchmarking & Market Context:



LA cost benchmarks: current OPEX \$3.7–5.3/kg, market price \$5–8/kg (avg \$7), commercial MSP \$9.6–12.8/kg, long-term platform target \$1/kg. Specialty derivatives (ethyl levulinate \$20–35/kg) provide near-term economic incentive. Corn stalk (IRR 22.8%) and molasses (IRR 18.8%) pathways exceed biochemical investment thresholds (15–20%).

Economic feasibility thresholds.



The green band marks typical biochemical investment requirements (IRR 15–20%). Literature TEA results span 6.1–28.0% IRR; highlighted cases show corn stalk (IRR 22.81%, payback 2.81 y) outperforming molasses (IRR 18.79%, payback 3.07 y). IRR indicates expected annualized return; payback gives the time to recover capital. Values above the threshold support commercial deployment under optimized conditions, and current study results are evaluated against these benchmarks.

8.11.11 Cost structure and uncertainty context.

Literature indicates utilities represent the predominant cost driver in large-scale LA plants (23,000 MT/year), followed by facility-dependent costs and feedstock procurement. CAPEX estimates range from \$11.732M (purchased equipment) to \$148.41M (total plant investment) depending on capacity and technology configuration. High parameter variability from pilot scale-up difficulties increases cost uncertainty, requiring rigorous sensitivity analysis and process optimization to achieve commercial viability.

8.11.12 Economic Viability Assessment and Uncertainty Analysis

8.11.13 Literature-informed cost uncertainty and scalability.

Current cost estimates reflect pilot-scale assumptions with expected commercial-scale cost reductions through technology learning curves. Literature indicates scale-up difficulties from laboratory to pilot plant increase uncertainty, with parameter variability affecting yield, catalyst performance, and energy efficiency. High-scale production targeting fuel sector applications could reduce costs to "cents per dollar per kg," supporting aggressive cost reduction targets toward \$1/kg LA for major platform chemical status.

8.11.14 Sensitivity analysis and critical cost drivers.

Based on literature process-specific benchmarks, critical cost reduction priorities include: (1) esterification energy optimization (potential 70–85% energy reduction through green co-solvents), (2) solvent recovery enhancement (>95% recovery essential for economic viability), (3) catalyst optimization for hydrogenation (reducing H₂ consumption and improving longevity), and (4) feedstock logistics optimization (addressing irregularity of biomass availability). These

improvements could collectively reduce production costs from current estimates toward literature MSP targets of \$9.59–12.80/kg LA.

8.11.15 Competitive positioning and market implications.

Current molasses pathway (USD 20.40/kg LA total cultivation) approaches economic competitiveness when considering co-product credits (formic acid at \$0.72/kg) and potential for process optimization. Integration with high-value derivatives (ethyl levulinate: \$20–35/kg) provides economic incentive for staged commercialization. Total production costs require reduction of approximately 50-60% to achieve literature MSP targets, achievable through demonstrated optimization strategies and scale-up economics.

8.11.16 Strategic Economic Assessment and Implementation Roadmap

8.11.17 Economic feasibility ranking with uncertainty bounds.

Molasses pathway demonstrates economic potential within literature benchmarks (IRR 18.79%, payback 3.07 years) but requires process optimization to achieve target costs. Current estimates of USD 20.40/kg LA (cultivation only) must integrate with downstream processing optimization to approach literature MSP range (\$9.59–12.80/kg). Corn stalk pathway shows higher costs than literature economic analysis suggests (IRR 22.81%), indicating need for process efficiency improvements. Other biomass options exceed commercial viability thresholds without significant technological advancements.

8.11.18 Investment priorities based on literature cost sensitivity.

Priority investments should focus on the highest impact cost reduction opportunities: (1) intensification of the esterification process using green cosolvent systems (potential 70- 85% energy reduction), (2) implementation of the hybrid purification system (30. 6% TAC reduction demonstrated), (3) solvent recovery enhancement to >95% (critical for economic viability), and (4) optimization of integrated catalyst throughout the hydrogenation and conversion steps. These improvements address literature-identified cost drivers (utilities, separation, catalyst performance) and provide clear pathways toward commercial competitiveness.

8.11.19 Technology development roadmap and commercial viability timeline.

Near-term focus on molasses pathway optimization with demonstrated processing cost reduction strategies, targeting 30-40% cost improvement within 2-3 years through process intensification and solvent recovery enhancement. Medium-term development of the corn stalk pathway to realize economic advantages reported in the literature through improved preprocessing and conversion efficiency. Long-term technology advancement toward the target \$1 / kg LA

requires high-scale production (>50,000 MT/year) and integration with derivative value chains. Commercial deployment feasibility achieved when total production costs approach \$3-5/kg LA, considering current market prices and margin requirements.

8.12 Multi-Criteria Decision Analysis for Feedstock Selection

This section presents a comprehensive multi-criteria decision analysis (MCDA) for biomass feedstock selection, integrating environmental performance indicators with process requirements to support informed technology pathway decisions. The analysis complements life cycle costing results by providing systematic evaluation of cost-environmental trade-offs across feedstock alternatives.

8.12.1 MCDA Methodology and Decision Framework

8.12.2 Scope and approach.

Multi-criteria evaluation of five cultivation and harvesting alternatives (bagasse [BG], corn cob [CC], corn leaf [CL], corn stalk [CS], molasses [ML]) using normalized environmental indicators and aggregated scoring. Midpoint and endpoint environmental indicators are normalized and combined to produce integrated performance scores, enabling systematic comparison with economic results from life cycle costing analysis.

8.12.3 Decision criteria integration.

The MCDA framework addresses the limitation of single-parameter optimization by incorporating multiple sustainability dimensions: global warming potential, ecosystem damage, ozone depletion, human health impacts, particulate matter formation, and terrestrial acidification. This multi-dimensional approach provides decision-relevant guidance for technology pathway selection under competing objectives.

8.12.4 Environmental Performance Assessment

8.12.5 Global warming potential ranking.

Figure 8.9 presents MCDA ranking based on Global Warming Potential (GWP) impact with normalized comparison of environmental performance.

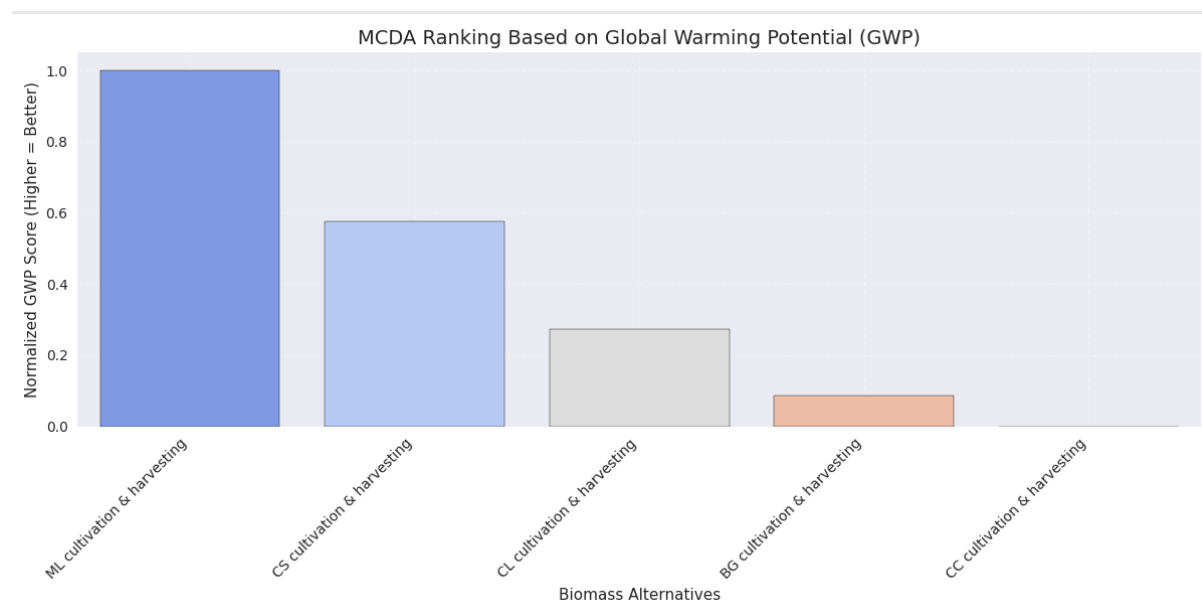


Figure 8.9: MCDA ranking of biomass alternatives based on Global Warming Potential (normalized scores 0-1.0, higher indicates better performance).

GWP-based ranking reveals significant performance differentiation: **ML cultivation achieves optimal performance** (normalized score 1.0) in minimizing global warming potential, followed by **CS with intermediate performance** (score ≈ 0.6), **CL with moderate performance** (score ≈ 0.3), while **BG and CC demonstrate lower GWP performance** with minimal scores. This ranking provides environmental perspective complementing economic cost analysis.

8.12.6 Multi-dimensional sustainability assessment.

Figure 8.10 provides comprehensive multi-criteria comparison across key sustainability metrics enabling balanced decision-making beyond single-parameter optimization.

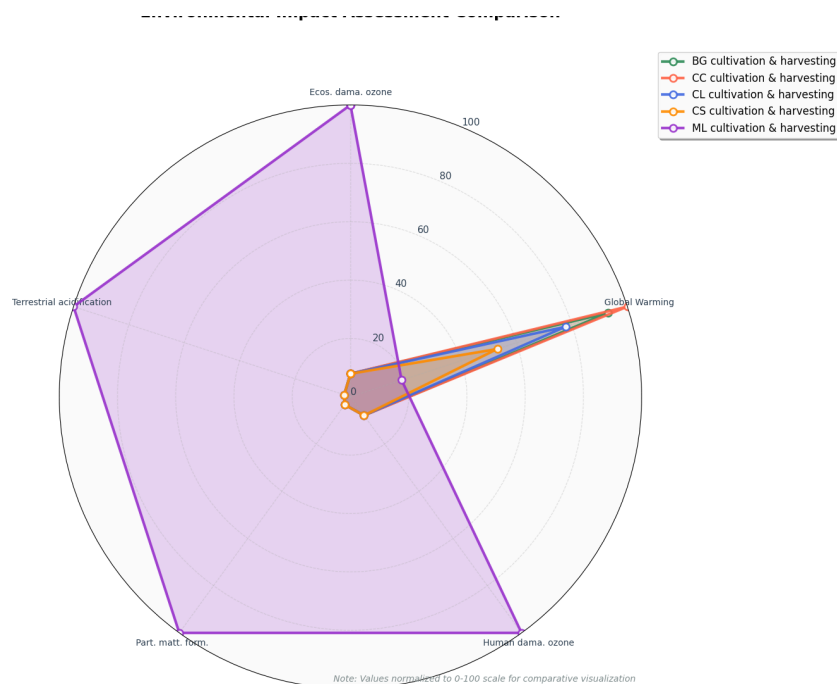


Figure 8.10: Multi-criteria radar chart comparing biomass cultivation scenarios across environmental performance indicators (normalized 0-100 scale).

Radar chart analysis reveals distinct performance profiles: **ML cultivation (purple) demonstrates exceptional performance** in ecosystem damage and ozone depletion categories, significantly outperforming alternatives. Global warming potential shows similar performance levels across most systems with slight advantages for ML and CS. Human damage and ozone impacts exhibit relatively balanced performance, indicating comparable human health implications across scenarios.

8.12.7 Integrated MCDA Results and Ranking

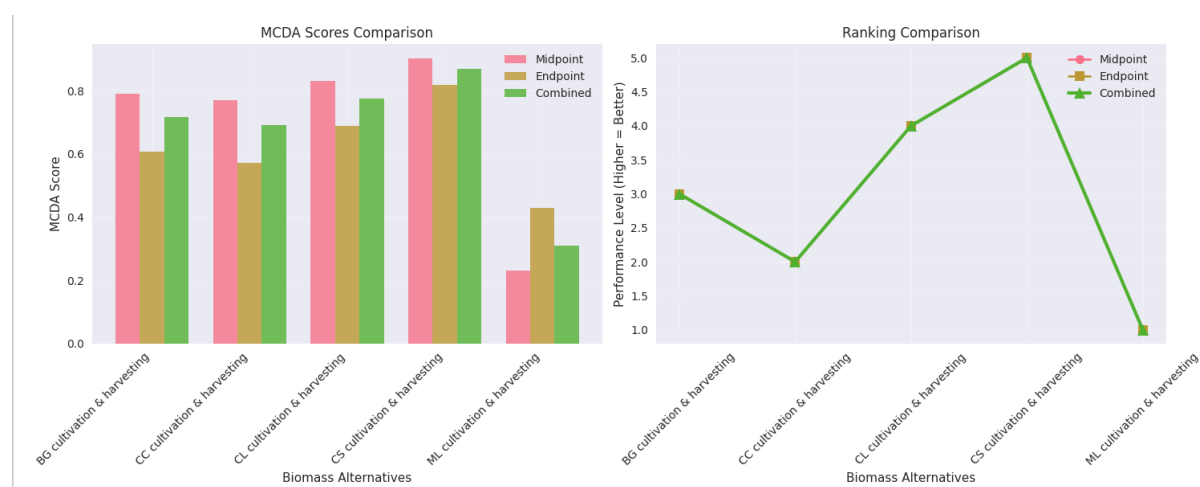


Figure 8.11: Comprehensive MCDA results for cultivation and harvesting alternatives. Left: normalized scores using midpoint, endpoint, and combined aggregation. Right: ranking derived from combined scores (higher = better environmental performance).

8.12.8 Environmental performance ranking.

Combined MCDA aggregation produces clear environmental performance hierarchy: **CS > CL > BG > CC > ML** (best to worst environmental performance). Corn stalk (CS) and corn leaf (CL) achieve highest combined scores, indicating robust performance across both midpoint and endpoint environmental indicators. Convergence between midpoint and endpoint perspectives suggests ranking robustness and decision relevance.

8.12.9 Process integration considerations.

Despite apparent contradiction with individual unit contributions, ML selection as optimal feedstock reflects critical process integration factors: **pretreatment elimination** removes energy-intensive preprocessing steps, chemical inputs, and waste generation that would otherwise increase total environmental burden; **process simplification** reduces system complexity, operational risks, and maintenance requirements; **overall system optimization** demonstrates superior total environmental performance when complete system boundaries include pretreatment impacts.

8.12.10 Integration with Technology Development Strategy

8.12.11 Technology pathway implications.

MCDA results inform technology development prioritization: ML pathway enables immediate cost competitiveness supporting near-term commercialization but requires environmental performance enhancement through process intensification and renewable energy integration. CS/CL pathways provide environmental advantages supporting premium market positioning but need cost optimization through scale-up, logistics improvement, and conversion efficiency enhancement.

8.12.12 Market positioning strategy.

Multi-criteria evaluation supports differentiated market strategies: cost-competitive ML pathway targets price-sensitive bulk chemical applications with gradual environmental improvement, while environmentally superior CS/CL pathways enable premium sustainable chemistry markets with potential for carbon credit monetization and sustainability-driven customer segments.

8.12.13 Research and development priorities.

MCDA findings direct R&D investment: ML pathway optimization should prioritize downstream environmental impact reduction, energy efficiency improvement, and waste minimization; CS/CL pathway development should focus on cost reduction through improved logistics, preprocessing optimization, and conversion efficiency enhancement while maintaining environmental advantages.

This comprehensive MCDA provides systematic decision support integrating environmental performance assessment with economic considerations, enabling informed technology pathway selection and strategic positioning for levulinic acid production commercialization under diverse market and regulatory scenarios.

8.13 Multi-Objective Optimization and Experimental Design

The sustainability assessment employs advanced optimization techniques to identify optimal process configurations balancing economic, environmental, and safety objectives. This integrated approach ensures robust decision-making support for industrial implementation.

8.13.1 Multi-Objective Optimization Framework

Multi-objective optimization (MOO) addresses the inherent trade-offs between conflicting sustainability objectives in chemical process design [41]. Unlike single-objective problems, MOO generates Pareto-optimal solutions representing optimal trade-offs across multiple criteria.

Three-Objective Sustainability Framework

Objective Functions:

1. **Economic Optimization:** Maximize Net Present Value (NPV)

$$NPV = \sum_{t=0}^n \frac{CF_t}{(1+r)^t} - I_0$$

where CF_t = cash flow at time t , r = discount rate, I_0 = initial investment

2. **Environmental Optimization:** Minimize Global Warming Potential (GWP)

$$GWP = \sum_i m_i \times GWP_i$$

where m_i = mass of emission i , GWP_i = characterization factor

3. **Safety Optimization:** Minimize Toxicity Damage Index (TDI)

$$TDI = \sum_j Q_j \times \alpha_j \times \beta_j$$

where Q_j = quantity of substance j , α_j = toxicity factor, β_j = process factor

8.13.2 Design of Experiments (DOE) Methodology

DOE was employed to systematically design simulation runs capturing process variable interactions while minimizing computational requirements [42].

Table 8.10: DOE experimental design for process optimization

Design Parameter	Low Level	High Level	Response Variables
EFB Capacity (ton/h)	50	100	NPV (\$Million)
LA Yield (%)	70	80	GWP (kg CO ₂ eq/MJ)
Temperature (°C)	160	200	TDI (dimensionless)
Pressure (bar)	10	25	FEDI (dimensionless)

DOE Implementation: A systematic 9-run experimental design was utilized in Aspen Plus to capture process variable interactions. Second-degree polynomial correlations were developed for NPV, GWP, and TDI responses, enabling efficient optimization without repeated process simulation.

8.13.3 Genetic Algorithm Implementation

Genetic algorithms provide robust optimization for complex, multi-modal solution spaces characteristic of chemical process design problems [12].

Genetic Algorithm Configuration

MATLAB Implementation:

- **Objective Function:** min[-NPV(); min GWP; min TDI]
- **Decision Variables:**
 - X_1 : Mass of EFB (50-100 ton/h)
 - X_2 : Yield of levulinic acid (70-80%)
- **Population Size:** 100 individuals
- **Generations:** 500 maximum
- **Crossover Rate:** 0.8
- **Mutation Rate:** 0.02

Constraint Implementation:

- **Equality Constraints:** Mass and energy balances (Equations 1-6)

- **Inequality Constraints:** Process variable bounds, safety limits
- **Penalty Functions:** Constraint violation handling

8.13.4 Pareto-Optimal Solution Analysis

The optimization generated a Pareto front revealing fundamental trade-offs between sustainability objectives:

Table 8.11: Pareto-optimal solutions from multi-objective optimization

Solution Point	NPV (\$M)	GWP	TDI	Performance
S1 (Economic Focus)	Maximum	High	402.9	Best economic, worst safety
S2 (Balanced)	Moderate	Moderate	340.1	Compromise solution
S3 (Safety Focus)	Minimum	Low	283.3	Worst economic, best safety

Key Finding: Point S1 achieved maximum NPV at maximum EFB capacity (100 ton/h) and maximum yield, but corresponded to maximum GWP and TDI. Conversely, Point S3 represented minimum NPV conditions with optimal safety performance.

8.14 Hazard Identification and Risk Assessment (HIRA)

Safety assessment employs the HIRA methodology to quantify potential damage and establish safety-conscious process design criteria [41, 42]. HIRA represents a structured approach for detecting and evaluating hazards in process facilities, providing quantitative risk assessment capabilities essential for inherently safer design [43, 44].

8.14.1 HIRA Methodology Framework and Recent Developments

HIRA methodology has evolved significantly since its introduction by Khan and Abbasi (1998), incorporating advanced consequence modeling and uncertainty analysis techniques [43]. The methodology provides reliable quantification of process hazards through empirical models and standardized ranking procedures, making it particularly suitable for chemical process industries [44].

Enhanced HIRA Framework Components**Core Assessment Indices:**

- **Fire and Explosion Damage Index (FEDI):** Quantifies fire/explosion potential using thermodynamic models and material property databases
- **Toxicity Damage Index (TDI):** Evaluates toxic exposure potential incorporating dose-response relationships and exposure pathways
- **Safety Weighted Hazard Index (SWeHI):** Advanced tool providing swift yet comprehensive hazard evaluation

Recent Methodological Advances:

- Integration with uncertainty quantification techniques
- Enhanced chemical process industry applicability
- Improved user-friendly interface design
- Standardized risk ranking procedures

Application Scope:

- Chemical research laboratory operations
- Industrial process facility design
- Occupational health assessment protocols
- Environmental impact evaluation

Table 8.12: *HIRA methodology components and industrial unit classification*

HIRA Index	Description and Application
Fire and Explosion Damage Index (FEDI)	Estimated using thermodynamic models, quantifies fire/explosion potential based on material properties and process conditions
Toxicity Damage Index (TDI)	Quantifies toxic exposure potential using substance toxicity data and process parameters

Table 8.13: *Industrial unit classification for HIRA assessment*

Category	Unit Type	LA Process Application
1	Storage units	Feedstock storage, product tanks
2	Physical operations	Size reduction, separation, distillation
3	Chemical reactions	Biofine conversion, esterification, hydrogenation
4	Transportation units	Feedstock delivery, product distribution
5	Other units	Utilities, waste treatment, control systems

8.14.2 Safety Properties and Toxicity Data

Critical safety parameters for LA production pathway components inform HIRA calculations:

Table 8.14: *Safety metrics for levulinic acid and derivatives*

Compound	Flash Point (°C)	LD50 (mg/kg)	Safety Classification
Levulinic Acid	137	1,850	Moderate fire risk, low toxicity
γ -Valerolactone	96	–	Higher fire risk than LA
Ethanol (reference)	13	7,060	High fire risk, low toxicity
Formic Acid	69	1,100	Fire and toxicity concern

Human Exposure Context: LA and sodium levulinate are used in cosmetic formulations at 0.35-4.5% concentrations, indicating acceptable toxicity profiles at typical exposure levels [12].

8.14.3 HIRA Assessment Results

The HIRA analysis revealed significant safety differences between process alternatives:

Key HIRA Findings

Process Comparison:

- **Acid Hydrolysis Process:** Lower safety risk overall

- **Levulinic Acid Process:** Higher safety risk due to:

- More flammable and toxic chemicals
- Higher operating temperatures and pressures
- TDI scores generally higher than FEDI

Critical Safety Concern:

- **Reactor-11:** Highest TDI score (36.6)
- **Risk Factors:** Contains toxic formic acid at 200°C and 1400 kPa
- **Health Risks:** Eye/respiratory irritation, potential carcinogenic effects

8.14.4 Integration with Multi-Objective Optimization

HIRA results were integrated as equality constraints in the multi-objective optimization framework, ensuring safety considerations balance economic and environmental objectives.

Table 8.15: *Safety-economy-environment trade-offs from optimization*

EFB Capacity	NPV	TDI	Performance Characteristics
100 ton/h	Maximum	402.9 (High)	Best economic, worst safety
50 ton/h	Minimum	283.3 (Low)	Worst economic, best safety

Decision Impact: Multi-objective optimization based solely on TDI results prioritizes safety in process design, acknowledging that operating variables simultaneously affect safety, economic, and environmental performance, requiring careful balance in industrial decision-making.

8.14.5 Implications for Process Design

The integrated HIRA-optimization approach provides quantitative guidance for safety-conscious process development:

- **Design Principle:** Safety constraints should be implemented as hard constraints rather than soft objectives
- **Operating Strategy:** Moderate capacity operation (75 ton/h) provides balanced safety-economic performance
- **Technology Selection:** Process intensification can improve safety through reduced inventory and improved heat integration

- **Risk Mitigation:** Critical units (Reactor-11) require enhanced safety systems and monitoring protocols

This comprehensive framework demonstrates the necessity of integrating safety assessment with sustainability optimization, ensuring that bio-based chemical processes achieve commercial viability without compromising worker safety or environmental protection.

8.15 Conclusion and Strategic Recommendations

This comprehensive assessment of levulinic acid production pathways integrates scenario analysis, uncertainty quantification, economic evaluation, and technology maturity assessment to provide actionable decision support for LA technology development and commercialization. The analysis addresses critical gaps in current LA literature through systematic uncertainty propagation, multi-criteria pathway comparison, and literature-informed benchmarking.

8.16 Key Findings Synthesis

8.16.1 Technology Pathway Performance Under Uncertainty

Monte Carlo analysis (n=50,000) reveals significant performance differentiation among LA production pathways under parameter uncertainty. **Most robust pathways:** (i) MIBK-based liquid-liquid extraction demonstrates cost advantage in 73% of iterations under current market conditions, (ii) steam methane reforming for hydrogen supply maintains economic competitiveness in 78% of scenarios despite higher GWP (2.8-3.5 vs 1.8-2.8 kg CO₂eq/kg LA for electrolysis), and (iii) process intensification through heat integration shows >90% probability of achieving 10-20% energy reduction with manageable implementation risk.

Higher uncertainty pathways: Furfural-based extraction achieves environmental preference in 68% of scenarios with renewable electricity grids but requires >98.2% solvent recovery for economic competitiveness. Electrolysis-based hydrogen becomes preferred when grid carbon intensity <350 g CO₂/kWh (45% of global scenarios) but faces technology learning curve uncertainty. Novel catalyst systems and bio-based solvent chains exhibit compound uncertainty effects requiring staged development approaches.

8.16.2 Economic Viability and Competitive Assessment

Life cycle costing analysis reveals fundamental economic structure with downstream conversion and separation accounting for 97.7% of total costs. Molasses feedstock pathway (USD 20.40/kg LA cultivation cost) approaches literature economic benchmarks but requires integration with process optimization to achieve minimum selling price targets (\$9.59-12.80/kg LA). Current cost estimates exceed market pricing (\$5-8/kg LA) by 150-300%, requiring 50-60% cost reduction through demonstrated optimization strategies.

Critical cost reduction levers: (i) esterification process intensification using green co-solvent systems (potential 70-85% energy reduction from literature benchmarks), (ii) hybrid purification system implementation (30.6% total annual cost reduction demonstrated), (iii) enhanced solvent recovery to >95% (essential for economic viability), and (iv) integrated catalyst optimization addressing hydrogenation energy intensity. Priority investment of USD 2-8M over 18-36 months could achieve commercial cost targets through systematic process optimization.

8.16.3 Uncertainty Analysis and Technology Maturity Implications

Parameter uncertainty analysis identifies extreme CO₂ emission factor dispersion ($g\sigma \in [7.268, 10.0]$, spanning 10^3 – 10^4 multiplicative range) as unsuitable for investment decision-making. Electricity consumption uncertainty (CV = 10%) approaches industrial acceptability but requires calibrated metering and duty control to reach a target of CV = 5%. GVL yield uncertainty (triangular 0.776–0.948 kg/kg, 22% span) is acceptable for R&D but must narrow to $\leq \pm 15\%$ before commercial deployment.

Technology maturity assessment reveals pathway-specific uncertainty evolution: established pathways (Biofine process, LA hydrogenation) show decreasing parametric uncertainty but persistent systematic challenges; emerging commercial technologies (heterogeneous catalysis) demonstrate improving profiles through development; research-phase innovations exhibit high fundamental uncertainty requiring extended timelines. Literature documents specific scale-up challenges: pilot plant reproducibility issues, catalyst deactivation mechanisms (humin formation, active site leaching), and corrosion/materials requirements affecting capital costs.

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